The Σ_1 Theory of Non-Structural Subtyping Is Decidable

Yaoyu Zhao ✉
EPFL, Switzerland

Abstract

Subtype constraints are inequalities $s \leq t$ between type expressions with unknowns. With a least type $\bot$ and a greatest type $\top$, the subtype order is non-structural: it relates trees of different shapes. For a binary covariant constructor, the first-order theory of this order is undecidable, already for sentences with the quantifier prefix $\exists\forall\exists$, whereas satisfiability of a conjunction of constraints is decidable in polynomial time and entailment between constraints was shown decidable only recently. We determine the Σ_1 level of the quantifier hierarchy: it is decidable whether a Boolean combination of existential (Σ_1) and universal (Π_1) sentences over subtype constraints is true. The result holds for the signature $\{\bot, f, \top\}$ with one constructor f of any arity and any variance, in particular for function types, over arbitrary, regular and finite trees, and for the signature $\{\bot, f\}$ with a covariant constructor. Satisfiability and entailment are special cases, and so is the entailment of a disjunction of k inequalities, to which every sentence reduces. A disjunction can be entailed although none of its disjuncts is, so the k failures of a countermodel have to be realized by one solution.

We reduce a countermodel to a *hole* of a finite automaton with k start states: a tuple of k words, one witness path for each failed inequality, such that no forbidden pair of states occurs where the rest of one word is a prefix of the rest of another word. The central technical theorem bounds the total length of a shortest hole. Its proof has two parts. Words that can be continued forever are replaced by ultimately periodic continuations whose mutual constraints form a system of difference constraints, and their periodic stretches are then compressed. The remaining tuples are shortened by deleting repeated blocks of a hierarchy of markers in all words at once. The bound is elementary: four iterated exponentials of a number that is polynomial in the size of the automaton for a fixed number of queries, and five iterated exponentials of a polynomial in the size of the sentence, while the problem is PSPACE-hard. The decision procedures, their correctness and the elementary bound are mechanized in Lean 4.

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Supplementary Material The Lean development: `https://github.com/noti0na1/deci-sigma1`

1 Introduction

Type inference with subtyping collects the typing requirements of a program as *subtype constraints* $s \leq t$ between type expressions with unknowns [21, 12, 2, 9, 32]. Types are finite or infinite trees. When the type language has a least type $\bot$ and a greatest type $\top$, the subtype order is *non-structural*: $\bot \leq t \leq \top$ holds for every tree t, so the order relates trees of different shapes. For instance, with a binary constructor f that is covariant in both arguments,

$$f(\bot,\, f(\bot, \bot)) \;\leq\; f(f(\top, \bot),\, \top),$$

although the two trees have no common shape.

Three questions about a *constraint* Γ, a finite set of such inequalities read as their conjunction, have been studied for this order. *Satisfiability* asks whether Γ has a solution; it is decidable in polynomial time [17, 28]. *Entailment* asks whether every solution of Γ

satisfies a further inequality $x \leq y$; it is the basis of constraint simplification [31, 43, 36, 33]. Its decidability, posed by Rehof in 1996 as Problem 16 of the TLCA list of open problems [35], was studied by Niehren and Priesnitz with automata [25, 26] and settled positively only recently [46]. Finally, the full *first-order theory* of the order is undecidable [39], and for a binary covariant constructor, with or without $\top$, this already holds for sentences with the quantifier prefix $\exists\forall\exists$ [45].

Satisfiability is an existential (Σ_1) sentence $\exists\bar{x}.\ \bigwedge\Gamma$ in which no negation occurs, and entailment is a universal (Π_1) sentence $\forall\bar{x}.\ \neg\bigwedge\Gamma \vee x \leq y$ of a special form: its quantifier-free part has a single positive inequality. This paper decides the whole first level of the quantifier hierarchy.

▶ **Theorem 1.1** (Main theorem). *Let f be a constructor of arity $n \geq 0$, each of whose arguments is covariant or contravariant. It is decidable whether a Boolean combination of Σ_1 and Π_1 sentences over subtype constraints with the signature $\{\bot, f, \top\}$ is true, over arbitrary trees, over regular trees, and over finite trees. A sentence is true over the regular trees exactly when it is true over all trees. The same holds for the signature $\{\bot, f\}$ when f is covariant. The decision procedures are uniform in the arity and the variance.*

The function type constructor is the case $n = 2$ with a contravariant first argument. Since a Π_1 sentence is the negation of a Σ_1 sentence, the sentences of the theorem are the Boolean combinations of Σ_1 sentences; we call the set of the true ones the Σ_1 *theory*. Truth over finite trees can differ from truth over all trees: for a unary constructor, $\exists x.\ x \leq f(x) \wedge f(x) \leq x$ is true over regular trees, with the infinite tree $f(f(\cdots))$ as witness, and false over finite trees.

Disjunctive entailment. A Σ_1 sentence is true exactly when some clause of a disjunctive normal form of its quantifier-free part has a solution. After every compound subterm is named by a variable, such a clause has the form

$$\Gamma \ \wedge\ \neg(x_1 \leq y_1) \ \wedge\ \ldots \ \wedge\ \neg(x_k \leq y_k), \tag{1}$$

where Γ is a constraint, a conjunction of inequalities, and $x_1, y_1, \ldots, x_k, y_k$ are variables. It has a solution exactly when Γ does *not* entail the disjunction $x_1 \leq y_1 \vee \cdots \vee x_k \leq y_k$. So the theorem is equivalent to the decidability of *disjunctive entailment*

$$\Gamma \ \models\ x_1 \leq y_1 \vee \cdots \vee x_k \leq y_k,$$

uniformly in the number k of *queries*. Plain entailment is the case $k = 1$ and satisfiability the case $k = 0$.

Disjunctive entailment does not reduce to plain entailment in any direct way, because a disjunction can be entailed although none of its disjuncts is. For the function type constructor every tree is $\bot$ or lies above the least function type $f(\top, \bot)$, so $\forall x.\ x \leq \bot \vee f(\top, \bot) \leq x$ is true, but neither disjunct holds for all x. A countermodel of a disjunction must refute all disjuncts with a single solution (Section 3.3).

From countermodels to holes. A failed inequality $x \leq y$ has a *witness*: a path at which the labels of the values of x and y are in the wrong order. For k queries we obtain k witness paths $w_1, \ldots, w_k$, which are words over the argument positions $\{1, \ldots, n\}$. We show that a tuple of words is the tuple of witnesses of some countermodel exactly when it is a *hole* of a finite automaton compiled from Γ and the queries, a *structural reader* (Section 4). The reader has one start state for each query. It reads w_i from the ith start state, and the tuple

is a hole if every run stays alive and ends in a *rejected* state, and no *admitted* pair of states occurs at a *comparison*: a pair of cuts, one in a word w_i and one in a word w_j, such that the rest of w_j after its cut is a prefix of the rest of w_i after its cut. The words w_i and w_j may be the same word or different ones. The pairs of states at the comparisons between w_i and w_j form the *channel* between the two words. A comparison is a place where a subtree that one witness forces into the solution meets a subtree that another witness must keep out of it. Here a state is rejected when the reader has rejected every way in which the constraint could refute the witness, so rejected states are the good ones.

Whether a reader has a hole is not a regular property of the tuple of words, since comparisons relate unboundedly long suffixes at unboundedly distant cuts. Decidability follows from a bound on the length of a shortest hole. We write $\exp_0(x) = x$ and $\exp_{h+1}(x) = 2^{\exp_h(x)}$ for iterated exponentials.

▶ **Theorem 1.2** (Small holes). *There is a computable function* H *such that every structural reader with* N *states and* k *start states that has a hole has a hole of total length at most* $\mathsf{H}(k, N)$*, whatever the size of its alphabet, and*

$$\mathsf{H}(k, N) \;\leq\; \exp_4\big((k + N + 20)^{2k+6}\big).$$

For a fixed number k of queries the bound is four iterated exponentials of a polynomial in N. A reader compiled from a constraint with v variables has at most $2^{4v} + 1$ states, so the witnesses of a countermodel can be chosen of length at most five iterated exponentials of a polynomial in the size of the sentence, and the Σ_1 theory is decidable in elementary space (Section 9). The problem is PSPACE-hard, because plain entailment is [15]. The procedures establish decidability and an elementary upper bound; they are not practical algorithms.

Why several queries are harder than one. The decision procedure for plain entailment [46] also reduces a countermodel to a hole and bounds a shortest hole, for readers with one start state. Its argument does not extend to several words, for three reasons, which are summarized in Table 1.

Several witnesses in one solution. With k queries a hole is a tuple of words, and there are comparisons between different words. Each query can fail in two ways, at a path where the smaller side is too large or at a path where the larger side is too small, so there is one reader for each of the 2^k vectors of *modes*.

Rejected tails. A run enters the rejected states at some cut and then stays in them. With one word, no admitted pair can involve a cut of the *rejected tail*, the part of the word after that cut, so a hole can be cut off shortly after it. With several words, a cut in the rejected tail of one word can still be the host of an admitted comparison with another word. If the run can be continued forever, the tail may have to be long in order to avoid the other words, and its length must be bounded independently of their lengths. This is the largest part of the proof (Section 7). We replace such tails by ultimately periodic words, *rays*; we show that rays can be chosen so that the constraints between their lengths form a solvable system of difference constraints; and we compress the periodic stretches of the resulting hole without creating an admitted pair on any channel.

Shortening all words together. Deleting a block from one word of a tuple can create a comparison with another word. The shortening of Section 6 therefore follows one hierarchy of *markers* for the whole tuple and deletes the same repeated blocks from all words at the same time.

Table 1 Plain entailment and disjunctive entailment.

	Plain entailment [46]	Disjunctive entailment (this paper)
witness of a countermodel	one path	one path per query, all realized by one solution
readers	one per failure side, with one start state	one per mode vector, 2^k in all, with k start states
comparisons	within one word	within and between the words of a tuple
after the first rejection	no admitted pair; short tails	tails can still be hosts; bounded by rays and compression
shortening	hierarchy of one word	one marker hierarchy, the same blocks deleted in all words

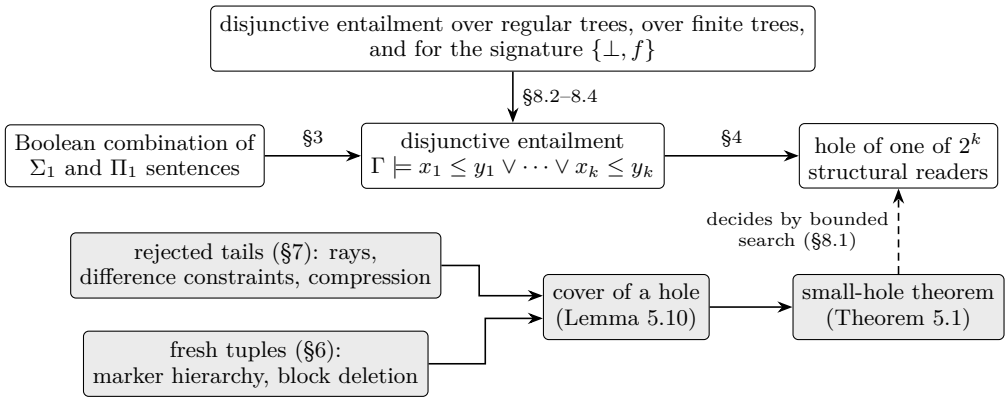


Figure 1 The proof chain. Solid arrows between white boxes are reductions: a sentence is reduced to disjunctive entailments (naming of subterms), and a failed entailment over arbitrary trees with the signature $\{\bot, f, \top\}$ to a hole of a finite reader (witnesses and modes); the other classes of trees and the signature $\{\bot, f\}$ are reduced to this case. The gray boxes are the small-hole theorem and the results it follows from. It bounds the length of a shortest hole, so a bounded search decides the entailment (dashed arrow). Its proof bounds the rejected tails of a hole and then the hole itself, read as a tuple of fresh words in a larger automaton, its cover.

Contributions. Figure 1 shows the structure of the proof.

- We prove that the Σ_1 theory of non-structural subtyping is decidable for one constructor of any arity and variance with $\bot$ and $\top$, over arbitrary, regular and finite trees, and for a covariant constructor with $\bot$ only (Theorem 1.1, proved in Section 8).
- We give an exact reduction of the countermodels of a disjunctive entailment to the holes of structural readers with k start states (Theorem 4.14).
- We prove the small-hole theorem for structural readers (Theorem 5.1) from two results of independent interest in combinatorics on words with automata: a bound on rejected tails (Theorem 5.7) and a bound on minimal tuples of fresh words (Theorem 5.8).
- We show that the bound is elementary (Theorem 9.1), discuss the gap to the PSPACE lower bound, and give families of sentences whose shortest witnesses are long (Section 9).
- The decision procedures, their correctness, the small-hole theorem and its elementary bound are mechanized in Lean 4 with Mathlib (Section 10).

Section 2 fixes trees, the order and sentences. Section 3 reduces sentences to disjunctive entailment, and Section 4 reduces disjunctive entailment to holes. Section 5 states the two bounds and composes them. Sections 6 and 7 explain their proofs, with the longer arguments in the appendices. Section 8 gives the decision procedures, including regular and finite trees and the signature $\{\bot, f\}$, and Section 9 the complexity bounds. Section 10 describes the mechanization, Section 11 discusses related work, and Sections 12 and 13 discuss future work and conclude. The notation is collected in Appendix H.

2 Preliminaries

2.1 Words

Words are finite sequences of letters over a finite alphabet; ε is the empty word and $|w|$ the length of w. A *cut* of w is a number a with $0 \le a \le |w|$; it separates the prefix $w[:a]$ of length a from the suffix $w[a:]$, the *rest* of w after a. For cuts $a \le b$, $w[a:b]$ is the factor between them, and the *positions* of w are $0, \dots, |w| - 1$, with the letter $w[p]$ at position p. We write $u \preceq v$ if u is a prefix of v. Two words are *comparable* if one is a prefix of the other. A word u *ends at the cut* e of w if u is a suffix of $w[:e]$, and *occurs at* a if $u \preceq w[a:]$. For example, the cut 2 of $w = 0110$ separates $w[:2] = 01$ from the rest $w[2:] = 10$; the word 1 occurs at the cuts 1 and 2 and ends at the cuts 2 and 3; and the rest 10 is comparable with 1 and with 100, but not with 11.

2.2 Trees and the Subtype Order

We fix a constructor f of arity $n \ge 0$ and write $[n] = \{1, \dots, n\}$ for its argument positions. A *path* is a word over $[n]$.

▶ **Definition 2.1** (Trees). *A* tree *over* $\{\bot, f, \top\}$ *is a partial function* t *from paths to* $\{\bot, f, \top\}$ *whose domain* $\operatorname{dom} t$ *contains* ε *and is closed under prefixes, such that for every* $\pi \in \operatorname{dom} t$*: if* $t(\pi) = f$ *then* $\pi i \in \operatorname{dom} t$ *for every* $i \in [n]$*, and otherwise* $\pi i \notin \operatorname{dom} t$ *for every* i*. The elements of* $\operatorname{dom} t$ *are the* positions *of* t*, and the positions labeled* $\bot$ *or* $\top$ *are its* leaves*. The* subtree *of* t *at a position* π *is the tree* $\sigma \mapsto t(\pi\sigma)$*.*

We write $\bot$ and $\top$ for the trees with the single position ε, and $f(t_1, \dots, t_n)$ for the tree with label f at ε and subtree t_i at the position i. Trees may be infinite; for $n \ge 1$ the tree f^ω with f at every path is an example. A tree is *finite* if it has finitely many positions and *regular* if it has finitely many distinct subtrees. We write $\mathcal{T}$, $\mathcal{T}_{\text{reg}}$ and $\mathcal{T}_{\text{fin}}$ for the classes of all, regular and finite trees; $\mathcal{T}_{\text{fin}} \subseteq \mathcal{T}_{\text{reg}} \subseteq \mathcal{T}$.

Each argument position is *covariant* or *contravariant*; the *variance* is a map $c\colon [n] \to \{0, 1\}$ with $c(i) = 1$ for the contravariant positions. The *polarity* of a path $\pi = i_1 \cdots i_m$ is $\operatorname{pol}(\pi) = c(i_1) + \cdots + c(i_m) \bmod 2$, the parity of the number of contravariant steps. The labels are ordered $\bot < f < \top$.

▶ **Definition 2.2** (Subtype order). *For trees* s *and* t*,* $s \le t$ *holds if for every path* $\pi \in \operatorname{dom} s \cap \operatorname{dom} t$*,*

$$s(\pi) \le t(\pi) \quad \text{if } \operatorname{pol}(\pi) = 0, \qquad t(\pi) \le s(\pi) \quad \text{if } \operatorname{pol}(\pi) = 1.$$

The order depends on the variance, which we keep fixed and do not write. It is a partial order on trees (antisymmetry follows from Theorem 4.2), and it is characterized on

constructor trees by the usual rules:

$$\bot \le t \le \top \quad \text{for every tree } t,$$

$$f(s_1,\dots,s_n) \le f(t_1,\dots,t_n) \iff \begin{cases} s_i \le t_i & \text{for every covariant } i, \\ t_i \le s_i & \text{for every contravariant } i. \end{cases} \tag{2}$$

Indeed $\bot$ and $\top$ have the single position ε, of polarity 0, and the common paths of two constructor trees are ε and the paths $i\pi$ with π a common path of s_i and t_i, where $\mathrm{pol}(i\pi) = c(i) + \mathrm{pol}(\pi)$. The order is called *non-structural* because trees with different domains can be comparable; in the *structural* order, comparable trees have the same domain.

▶ **Example 2.3** (Function types). For function types, $n = 2$, $f(a,b)$ is written $a \to b$, and $c(1) = 1$, $c(2) = 0$. By (2), $\top \to \bot \le a \to b \le \bot \to \top$ for all trees a and b: the least function type accepts every argument and returns the least type. So every tree other than $\bot$ lies above $\top \to \bot$. The types $\bot \to \bot$ and $\top \to \top$ are incomparable: at the path 1, of polarity 1, the labels $\bot$ and $\top$ would have to be in decreasing order for $\bot \to \bot \le \top \to \top$, and at the path 2, of polarity 0, the labels $\top$ and $\bot$ would have to be in increasing order for $\top \to \top \le \bot \to \bot$.

2.3 Sentences

Terms and quantifier-free formulas are built from variables $x, y, \dots$:

$$s,t ::= x \mid \bot \mid \top \mid f(t_1,\dots,t_n) \qquad\qquad \varphi,\psi ::= s \le t \mid \neg\varphi \mid \varphi \wedge \psi \mid \varphi \vee \psi,$$

and $s = t$ abbreviates $s \le t \wedge t \le s$. Given an assignment ρ of trees to the variables, a term denotes a tree and a formula is true or false, with $\le$ interpreted by the subtype order. A Σ_1 *sentence* is $\exists x_1 \dots x_m.\, \varphi$ and a Π_1 *sentence* is $\forall x_1 \dots x_m.\, \varphi$, where φ is a quantifier-free formula with variables among $x_1,\dots,x_m$.

A sentence is evaluated over a class $\mathcal{C}$ of trees: the quantifiers range over the trees of $\mathcal{C}$. We consider the three classes $\mathcal{T}$, $\mathcal{T}_{\mathrm{reg}}$ and $\mathcal{T}_{\mathrm{fin}}$. Each of them contains $\bot$ and $\top$ and is closed under the constructor, so every term denotes a tree of the class when its variables do. A *Boolean combination* of Σ_1 and Π_1 sentences is built from such sentences with $\neg$, $\wedge$ and $\vee$; its truth over $\mathcal{C}$ is defined from the truth of its component sentences.

2.4 Constraints and Disjunctive Entailment

The proof works with constraints in a flat form.

▶ **Definition 2.4** (Flat constraints). *Let V be a finite set of variables. A* flat literal *over V is one of*

$$x \le y, \qquad f(x_1,\dots,x_n) \le y, \qquad y \le f(x_1,\dots,x_n), \qquad x = \bot, \qquad x = \top,$$

with $x, y, x_1, \dots, x_n \in V$. A constraint Γ *is a finite set of flat literals, read as their conjunction. A* solution *of Γ in a class $\mathcal{C}$ is an assignment $\rho\colon V \to \mathcal{C}$ that satisfies every literal of Γ, and Γ is* satisfiable in $\mathcal{C}$ *if it has a solution in $\mathcal{C}$.*

▶ **Definition 2.5** (Disjunctive entailment). *Let Γ be a constraint over V, and let (x_1, y_1), …, (x_k, y_k) be pairs of variables of V, the* queries*; $k = 0$ is allowed and queries may repeat. Then Γ* entails *$x_1 \le y_1 \vee \dots \vee x_k \le y_k$ over $\mathcal{C}$, written*

$$\Gamma \models_{\mathcal{C}} x_1 \le y_1 \vee \dots \vee x_k \le y_k,$$

if every solution ρ of Γ in $\mathcal{C}$ satisfies $\rho(x_i) \le \rho(y_i)$ for some i. A countermodel *in $\mathcal{C}$ is a solution of Γ in $\mathcal{C}$ with $\rho(x_i) \not\le \rho(y_i)$ for every i.*

The entailment holds exactly when there is no countermodel. For $k = 0$ the disjunction is empty, and the entailment holds exactly when Γ is unsatisfiable in $\mathcal{C}$. For $k = 1$ this is *plain* entailment $\Gamma \models_{\mathcal{C}} x \leq y$, non-structural subtype entailment. With $\bar{x}$ the variables of Γ, the classical problems are sentences of our fragment:

Problem	Sentence
Γ is satisfiable	$\exists \bar{x}.\ \bigwedge \Gamma$
Γ entails $x \leq y$	$\forall \bar{x}.\ \neg \bigwedge \Gamma \vee x \leq y$
Γ entails $x_1 \leq y_1 \vee \cdots \vee x_k \leq y_k$	$\forall \bar{x}.\ \neg \bigwedge \Gamma \vee x_1 \leq y_1 \vee \cdots \vee x_k \leq y_k$

Here a literal $x = \bot$ is the inequality $x \leq \bot$, and $x = \top$ is $\top \leq x$. *Existential entailment* $\Gamma \models \exists \bar{z}.\, \Gamma'$, with further variables in the consequence, is a Π_2 sentence and lies outside the Σ_1 theory; it is decided in [44].

▶ **Example 2.6** (A disjunction that no disjunct explains)**.** Let $n = 1$. Every tree is $f^m(\bot)$, $f^m(\top)$ or f^ω, and any two trees are comparable, whatever the variance. So the empty constraint entails $x \leq y \vee y \leq x$ over each class, but it entails neither $x \leq y$ nor $y \leq x$. For a binary covariant constructor the disjunction is not entailed: $x = f(f(\bot, \bot), \bot)$ and $y = f(\bot, f(\bot, \bot))$ are incomparable. We return to this example in Example 4.16.

The signature $\{\bot, f\}$. A tree over $\{\bot, f\}$ is a tree in which $\top$ does not occur. For this signature we assume f covariant, so the order is the inclusion of the sets of positions labeled f. Terms have no $\top$, flat constraints have no literal $x = \top$, and the variables range over the trees without $\top$ of the class. Section 8.4 reduces this signature to $\{\bot, f, \top\}$.

3 From Sentences to Disjunctive Entailment

This section reduces the truth of a sentence to finitely many disjunctive entailments between flat constraints. The reduction is the same for every class of trees that contains $\bot$ and $\top$ and is closed under the constructor; we fix such a class $\mathcal{C}$.

3.1 Normal Forms

The truth value of a Boolean combination is computed from the truth values of its component sentences, and a Π_1 sentence $\forall \bar{x}.\, \varphi$ is true exactly when the Σ_1 sentence $\exists \bar{x}.\, \neg\varphi$ is false. It therefore suffices to decide Σ_1 sentences.

A *clause* is a conjunction of inequalities $s \leq t$ and negated inequalities $\neg(s \leq t)$ between terms. Pushing negations inward and distributing conjunctions over disjunctions turns a quantifier-free formula φ into an equivalent disjunction of clauses, a disjunctive normal form of φ. Hence $\exists \bar{x}.\, \varphi$ is true over $\mathcal{C}$ exactly when some clause of the normal form is *realizable* in $\mathcal{C}$: some assignment of trees of $\mathcal{C}$ to $\bar{x}$ makes all its conjuncts true.

3.2 Naming Subterms

Let K be a clause. We introduce a variable v_t for every term t that occurs in K, as a side of an inequality or as a subterm of one, with $v_x = x$ for a variable x. The *flat constraint* Γ_K consists of

- $v_t = \bot$ for $t = \bot$, and $v_t = \top$ for $t = \top$, if these terms occur;
- $f(v_{t_1}, \ldots, v_{t_n}) \leq v_t$ and $v_t \leq f(v_{t_1}, \ldots, v_{t_n})$ for every occurring term $t = f(t_1, \ldots, t_n)$;
- $v_s \leq v_t$ for every inequality $s \leq t$ of K.

The *queries* of K are the pairs (v_s, v_t) for the negated inequalities $\neg(s \leq t)$ of K.

▶ **Lemma 3.1** (Clauses). *A clause K with queries $(x_1, y_1), \ldots, (x_k, y_k)$ is realizable in $\mathcal{C}$ if and only if $\Gamma_K \not\models_{\mathcal{C}} x_1 \leq y_1 \vee \cdots \vee x_k \leq y_k$.*

Proof. Let ρ be an assignment in $\mathcal{C}$ of the variables of K that realizes K. Extend it by giving v_t the value of the term t under ρ; this value lies in $\mathcal{C}$ because $\mathcal{C}$ contains $\bot$ and $\top$ and is closed under the constructor. The first two groups of literals of Γ_K hold because the order is reflexive, the third because ρ satisfies the inequalities of K, and every query fails because ρ satisfies the negated inequalities. So the extension is a countermodel. Conversely, let ρ be a countermodel in $\mathcal{C}$. Since the order is antisymmetric, the two literals for $t = f(t_1, \ldots, t_n)$ give $\rho(v_t) = f(\rho(v_{t_1}), \ldots, \rho(v_{t_n}))$, so by induction on terms $\rho(v_t)$ is the value of t under the restriction of ρ to the variables of K. This restriction then satisfies every inequality of K, by the third group, and no inequality $s \leq t$ that is negated in K, because ρ refutes the query (v_s, v_t). ◀

▶ **Example 3.2.** The sentence $\forall x.\ x \leq \bot \vee f(\top, \bot) \leq x$ is the negation of $\exists x.\ \neg(x \leq \bot) \wedge \neg(f(\top, \bot) \leq x)$, whose body is a single clause. Its flat constraint over the variables x, b, t, a is

$$\Gamma = \{\, b = \bot, \quad t = \top, \quad f(t, b) \leq a, \quad a \leq f(t, b) \,\},$$

with the queries (x, b) and (a, x). By Lemma 3.1 the sentence is true over $\mathcal{C}$ if and only if $\Gamma \models_{\mathcal{C}} x \leq b \vee a \leq x$.

▶ **Proposition 3.3** (Reduction to disjunctive entailment). *Suppose disjunctive entailment between flat constraints over $\mathcal{C}$ is decidable, uniformly in the number of queries. Then it is decidable whether a Boolean combination of Σ_1 and Π_1 sentences is true over $\mathcal{C}$. Moreover, if two classes give every disjunctive entailment the same truth value, they give every such sentence the same truth value.*

Proof. By Section 3.1 it suffices to treat a Σ_1 sentence, which is true exactly when one of the finitely many clauses of a normal form of its body is realizable. By Lemma 3.1 this is the failure of a disjunctive entailment that is computed from the clause. The second statement follows because the reduction does not depend on the class. ◀

The number of queries is the number of negated inequalities of a clause, so the reduction needs disjunctive entailment for every k, and a single decision procedure that takes k as part of its input.

3.3 Disjunctive Entailment Does Not Reduce to Plain Entailment

Consider function types (Example 2.3) and the entailment of Example 3.2,

$$\Gamma \ \models\ x \leq b \ \vee\ a \leq x \qquad \text{with } b = \bot \text{ and } a = \top \to \bot \text{ in every solution.} \tag{3}$$

It holds, since every tree is $\bot$ or lies above the least function type. But Γ entails neither $x \leq b$, refuted by $x = \top \to \bot$, nor $a \leq x$, refuted by $x = \bot$. For a covariant constructor, $f(\bot, \ldots, \bot)$ plays the role of $\top \to \bot$. So deciding the disjuncts one at a time is not enough, and countermodels of the single disjuncts do not help: a countermodel of the disjunction must refute every disjunct with one solution, and the two refuting values of x above do not combine into one. Moving negated disjuncts into the constraint takes us outside plain

entailment as well: $\Gamma \models \varphi_1 \vee \varphi_2$ holds if and only if $\Gamma \wedge \neg\varphi_1 \models \varphi_2$, but $\neg\varphi_1$ is not a constraint, and plain entailment allows only constraints on the left. The k failures therefore have to be decided jointly. The next section describes what it means for k failures to be realized by one solution.

4 Witnesses, Modes and Readers

This section characterizes the countermodels of a disjunctive entailment over arbitrary trees by the holes of finite automata. We first represent trees by two sets of paths, in which the order becomes inclusion and a constraint becomes a set of implications between atoms (Sections 4.1 and 4.2). A failed query then has a witness path, in one of two modes (Section 4.3), and a family of witnesses belongs to one solution exactly under three conditions on derivations (Section 4.4). Finite automata follow these derivations along the witness paths (Section 4.5), which gives the readers and the reduction (Sections 4.6 and 4.7). The representation by leaf cones is that of [44]; proofs omitted here are in Appendix A. Throughout, trees are trees over $\{\bot, f, \top\}$ and assignments range over all trees.

4.1 Leaf Cones

▶ **Definition 4.1** (Signs and cones). *The* sign *of a leaf v of a tree t is $+$ if $t(v) = \top$ and* $\mathrm{pol}(v) = 0$*, or $t(v) = \bot$ and* $\mathrm{pol}(v) = 1$*; otherwise it is $-$. For a sign $s \in \{+, -\}$, the* cone *of sign s of t is*

$$C^s(t) = \{\, w \in [n]^* \mid \textit{some prefix of } w \textit{ is a leaf of } t \textit{ of sign } s \,\}.$$

We write $\overline{s}$ for the opposite sign.

For a covariant constructor all polarities are 0, so $C^+(t)$ is the set of paths along which t reaches $\top$ and $C^-(t)$ the set of paths along which it reaches $\bot$; Figure 2 shows an example. With contravariant arguments the two labels exchange their roles at paths of polarity 1. A path has at most one prefix that is a leaf, and it has none exactly when it is a position labeled f. A set L of paths is *extension closed* if $w \in L$ implies $wi \in L$ for every $i \in [n]$.

▶ **Theorem 4.2** (Leaf representation). **1.** *For every tree t, the cones $C^+(t)$ and $C^-(t)$ are disjoint and extension closed. Conversely, every pair of disjoint extension-closed sets of paths is the pair of cones of exactly one tree.*
2. *For all trees t and u: $t \le u \iff C^+(t) \subseteq C^+(u)$ and $C^-(u) \subseteq C^-(t)$.*
3. *$\varepsilon \in C^+(t)$ iff $t = \top$, and $\varepsilon \in C^-(t)$ iff $t = \bot$. If $t = f(t_1, \ldots, t_n)$, then for every $i \in [n]$, path w and sign s: $iw \in C^s(t)$ iff $w \in C^s(t_i)$ when i is covariant, and iff $w \in C^{\overline{s}}(t_i)$ when i is contravariant.*

So going up in the order adds leaves of sign $+$ and removes leaves of sign $-$. The variance enters only through the signs of the leaves; there are two cones for each tree, whatever the arity.

4.2 Solutions as Closed Sets of Atoms

Let Γ be a constraint over V. A *signed variable* is a pair x^s of a variable and a sign; $V^{\pm}$ is the set of signed variables and $\overline{x^s} = x^{\overline{s}}$. An *atom* is a pair (p, w) of a signed variable and a path, and $\overline{(p, w)} = (\overline{p}, w)$ is its *opposite*. An assignment ρ determines the set of atoms

$$S_\rho = \{\, (x^s, w) \mid w \in C^s(\rho(x)) \,\}.$$

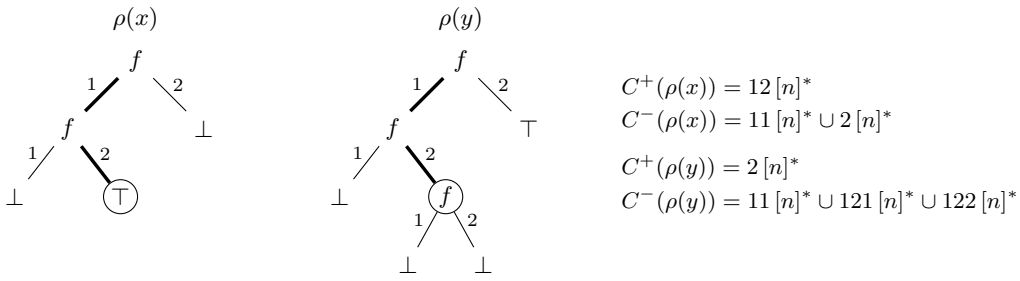


Figure 2 A witness of $\rho(x) \not\leq \rho(y)$ for a binary covariant constructor. Along the path $w = 12$ (thick) the tree $\rho(x)$ reaches $\top$ and $\rho(y)$ does not: $w \in C^+(\rho(x)) \setminus C^+(\rho(y))$. In the terms of Section 4.3 the query (x, y) fails in mode $\top$ at w, with the seed (x^+, w) and the exclusion (y^+, w).

Table 2 The rule form of the flat literals, for all paths w and all $i \in [n]$.

Literal	Moves		Facts	Forbidden atoms
$x \leq y$	$(x^+, w) \to (y^+, w)$,	$(y^-, w) \to (x^-, w)$		
$f(x_1, \ldots, x_n) \leq y$	$(x_i^{[i]}, w) \to (y^+, iw)$,	$(y^-, iw) \to (\overline{x_i^{[i]}}, w)$		(y^-, ε)
$y \leq f(x_1, \ldots, x_n)$	$(y^+, iw) \to (x_i^{[i]}, w)$,	$(\overline{x_i^{[i]}}, w) \to (y^-, iw)$		(y^+, ε)
$x = \bot$			(x^-, ε)	
$x = \top$			(x^+, ε)	

For an argument position i we write $x^{[i]}$ for x^+ if i is covariant and for x^- if i is contravariant. Table 2 associates with every literal of Γ a set of *moves*, implications between two atoms, together with *facts*, atoms that must hold, and *forbidden atoms*, which must not. The *rules* of Γ are its moves and the *extensions* $(p, w) \to (p, wi)$, for all atoms (p, w) and all $i \in [n]$.

A set S of atoms is *closed* if it is closed under the rules, *consistent* if it contains no atom together with its opposite, and a *solution set* of Γ if it is closed and consistent, contains the facts and contains no forbidden atom.

▶ **Lemma 4.3** (Rule form). *The map $\rho \mapsto S_\rho$ is a bijection from the solutions of Γ onto the solution sets of Γ.*

The proof translates each literal with Theorem 4.2: for instance, $f(x_1, \ldots, x_n) \leq y$ says that the cone of sign $+$ of the constructor tree, which consists of the paths iw with w in a cone of $\rho(x_i)$, is contained in that of $\rho(y)$, and that the cone of sign $-$ of $\rho(y)$ is contained in that of the constructor tree, which does not contain ε.

We write $a \vdash b$ if the atom b is *derivable* from the atom a by finitely many rules, and $a \Rightarrow b$ if it is derivable by moves only. The *least closure* LC of Γ is the set of atoms derivable from facts. It is closed and contained in every solution set, so Γ is satisfiable if and only if LC is consistent and contains no forbidden atom, and then LC is the least solution set.

The moves read the path of an atom as a stack whose top is its first letter: a move replaces the signed variable and pops or pushes one letter at the top, or leaves the path unchanged. Extensions add letters at the bottom. Two properties of the moves are used throughout.

▶ **Lemma 4.4** (Duality and normal form). **1.** *If $a \to b$ is a move, so is $\overline{b} \to \overline{a}$. Hence $a \Rightarrow b$ implies $\overline{b} \Rightarrow \overline{a}$.*

2. *$(p, u) \vdash (q, v)$ if and only if $(p, u\sigma) \Rightarrow (q, v)$ for some path σ.*

Two finite sets of signed variables record where the forbidden atoms and the facts enter. Both kinds of atoms have the empty path, and we let

$$\begin{aligned} \mathsf{Neg} &= \{\, r \in V^{\pm} \mid (r,\varepsilon) \text{ is a forbidden atom of } \Gamma \,\}, \\ \mathsf{Fct} &= \{\, r \in V^{\pm} \mid (p,\varepsilon) \Rightarrow (r,\varepsilon) \text{ for some fact } (p,\varepsilon) \text{ of } \Gamma \,\}. \end{aligned} \tag{4}$$

An atom whose derivations reach a signed variable of Neg with the empty path cannot belong to a solution set, and an atom that is reached from a signed variable of Fct with the empty path belongs to every solution set.

▶ **Example 4.5** (Rules of a small constraint)**.** Let f be binary and covariant, and let

$$\Gamma = \{\, u \le f(u,u), \quad x \le u, \quad y = \bot \,\}$$

over the variables x, y, u. The inequality $u \le f(u,u)$ ensures that $\top$ does not occur in u (this is the guard of Section 8.4); since $x \le u$, $\top$ does not occur in x either, and y is $\bot$. By Table 2 the moves are, for all paths w and $i \in \{1,2\}$,

$$\begin{aligned} &(u^+, iw) \to (u^+, w), && (u^-, w) \to (u^-, iw) && \text{from } u \le f(u,u), \\ &(x^+, w) \to (u^+, w), && (u^-, w) \to (x^-, w) && \text{from } x \le u. \end{aligned}$$

The first pops a letter, the second pushes one, and the last two keep the path; in each row the second move is the dual of the first. The only forbidden atom is (u^+, ε) and the only fact is (y^-, ε), so $\mathsf{Neg} = \{u^+\}$ and $\mathsf{Fct} = \{y^-\}$. The least closure consists of the atoms (y^-, w), which are derived from the fact by extensions: it describes the solution with the least set of atoms, $y = \bot$ and $x = u = f^\omega$. The derivation

$$(x^+, 12) \to (u^+, 12) \to (u^+, 2) \to (u^+, \varepsilon)$$

pops the path 12 and ends in a forbidden atom. So no solution set contains $(x^+, 12)$: the tree of x cannot reach $\top$ along the path 12, nor along any other path, because every path can be popped in this way.

4.3 Witnesses and Modes

By Theorem 4.2(2), an inequality fails in one of two ways.

▶ **Lemma 4.6** (Witnesses)**.** *Let ρ be an assignment and (x,y) a query. Then $\rho(x) \not\le \rho(y)$ if and only if there are a path w and a* mode *$\mu \in \{\top, \bot\}$ such that*

- $\mu = \top$, $(x^+, w) \in S_\rho$ *and* $(y^+, w) \notin S_\rho$; *or*
- $\mu = \bot$, $(y^-, w) \in S_\rho$ *and* $(x^-, w) \notin S_\rho$.

In mode $\top$ the smaller side is too large along w: it has a leaf of sign $+$ that the larger side lacks. In mode $\bot$ the larger side is too small. We call the atom that must belong to S_ρ the *seed* of the witness and the atom that must not belong to it the *exclusion*; both have the path w. For queries $(x_1, y_1), \dots, (x_k, y_k)$, a vector of modes $\bar\mu = (\mu_1, \dots, \mu_k)$ and a tuple of paths $W = (w_1, \dots, w_k)$, we write $A_i = (p_i, w_i)$ for the seed and $B_i = (p_i', w_i)$ for the exclusion of query i, so that

$$(p_i, p_i') = (x_i^+, y_i^+) \text{ if } \mu_i = \top, \qquad (p_i, p_i') = (y_i^-, x_i^-) \text{ if } \mu_i = \bot. \tag{5}$$

A solution ρ of Γ is a countermodel if and only if, for some $\bar\mu$ and W, the set S_ρ contains every A_i and no B_i. The signed variables p_i, p_i' depend only on the queries and the modes; the tuple W is what the decision procedure has to find. In Figure 2, for instance, the query (x,y) fails in mode $\top$ with the witness 12.

4.4 Realization

▶ **Theorem 4.7** (Realization). *Let Γ be satisfiable over $\mathcal{T}$, and let $A_1, \dots, A_k$ and $B_1, \dots, B_k$ be atoms. Some solution set of Γ contains every A_i and no B_i if and only if*

(R1) *no A_i derives a forbidden atom;*
(R2) *$B_i \notin \mathrm{LC}$ and $\overline{A_i} \notin \mathrm{LC}$, for every i;*
(R3) *$A_j \not\vdash B_i$ and $A_j \not\vdash \overline{A_i}$, for all i and j, including $i = j$.*

In this case the hull $\mathrm{LC} \cup \{b \mid A_j \vdash b \text{ for some } j\}$ *is the least such solution set.*

The conditions are clearly necessary: a solution set is closed, so it contains the hull, and it is consistent and contains no forbidden atom. For the converse one has to show that the hull is consistent. If it contained an atom b derived from A_j together with $\overline{b}$ derived from A_i, the duality of Lemma 4.4 would turn the second derivation around and compose it with the first into a derivation from A_j toward $\overline{A_i}$. The extensions, which have no dual, are handled by (R1).

Condition (R3) is where the k witnesses interact. It involves pairs of queries, and also a single query with itself: a seed must not derive its own exclusion or its own opposite.

4.5 Runs along a Path

The conditions of Theorem 4.7 are conditions on derivations between atoms whose paths are the unknown words w_i. Two families of finite sets summarize the derivations by moves along a path.

▶ **Definition 4.8** (Upper and lower runs). *For a signed variable p and a path w,*

$$\mathsf{U}(p, w) = \{\, r \in V^{\pm} \mid (p, w) \Rightarrow (r, \varepsilon) \,\}, \qquad \mathsf{D}(p, w) = \{\, r \in V^{\pm} \mid (r, \varepsilon) \Rightarrow (p, w) \,\}.$$

The *upper run* $\mathsf{U}(p, w)$ collects the signed variables reached from (p, w) when the whole path has been popped, and the *lower run* $\mathsf{D}(p, w)$ those from which (p, w) is built by pushing the whole path. Since moves only see the top of the stack, $r \in \mathsf{U}(p, w)$ also means that $(p, w\sigma) \Rightarrow (r, \sigma)$ for every σ by moves that never pop a letter of σ, and similarly for D. By duality, $\mathsf{D}(p, w) = \{\overline{r} \mid r \in \mathsf{U}(\overline{p}, w)\}$.

▶ **Lemma 4.9** (Runs are deterministic). *There are computable functions $\delta^{\uparrow}, \delta^{\downarrow} \colon 2^{V^{\pm}} \times [n] \to 2^{V^{\pm}}$ such that $\mathsf{U}(p, wi) = \delta^{\uparrow}(\mathsf{U}(p, w), i)$ and $\mathsf{D}(p, wi) = \delta^{\downarrow}(\mathsf{D}(p, w), i)$ for all p, w and i. Moreover $\delta^{\uparrow}(P, i) = \varnothing$ whenever $P \cap \mathsf{Neg} = \varnothing$.*

So both runs are computed by reading w from the root, with two deterministic automata whose states are sets of signed variables. The last statement holds because a letter can be popped only from an atom (y^{+}, iw) with a literal $y \le f(\dots)$, or from (y^{-}, iw) with a literal $f(\dots) \le y$, and in both cases the signed variable lies in Neg.

▶ **Lemma 4.10** (Derivations through cuts). *Let (p, u) and (q, v) be atoms with $\mathsf{U}(p, u) \cap \mathsf{Neg} = \varnothing$. Then $(p, u) \vdash (q, v)$ if and only if there are cuts a of u and b of v such that*

$$u[a{:}] \preceq v[b{:}] \qquad \text{and} \qquad \mathsf{U}(p, u[{:}a]) \cap \mathsf{D}(q, v[{:}b]) \neq \varnothing.$$

Moreover, (p, u) derives a forbidden atom if and only if $\mathsf{U}(p, u) \cap \mathsf{Neg} \neq \varnothing$.

A derivation from (p, u) first extends u, then pops a prefix of the path down to a lowest point and pushes a prefix of v. The hypothesis says that the pops do not go beyond u, so the lowest point is a cut a of u, and the rest $u[a{:}]$ is never touched: it survives as a prefix of the rest of v. This is the origin of the comparisons "the rest of one word is a prefix of the rest of another word" in the definition of holes. In Example 4.5, the atom $(u^-, 2)$ derives $(x^-, 12)$ by pushing the letter 1 and moving to x^-. The cuts of the lemma are 0 in the path 2 and 1 in the path 12: the rest 2 is not touched, and $\mathsf{U}(u^-, \varepsilon) = \{u^-, x^-\}$ meets $\mathsf{D}(x^-, 1) = \{u^-\}$. The same analysis, applied to derivations from facts, shows that $(q, v) \in \mathrm{LC}$ if and only if $\mathsf{D}(q, v[{:}b])$ meets the set Fct of (4) for some cut b of v; this uses that Γ is satisfiable.

4.6 Structural Readers and Holes

We now define the automata and their holes abstractly. The alphabet is an arbitrary finite set A; in the reduction it is $[n]$.

▶ **Definition 4.11** (Structural reader)**.** *A* structural reader *with k* starts *over A consists of a finite set S of states, a transition function $\delta\colon S \times A \to S$, k start states $s_1, \dots, s_k \in S$, which need not be distinct, a* dead *state $\dagger \in S$ with $\delta(\dagger, c) = \dagger$ for every letter c, a set $\mathsf{Rej} \subseteq S \setminus \{\dagger\}$ of* rejected *states and a relation $\mathsf{Adm} \subseteq S \times S$ of* admitted *pairs, such that for every $q \in \mathsf{Rej}$ and every letter c with $\delta(q, c) \neq \dagger$:*

(S1) $\delta(q, c) \in \mathsf{Rej}$*;*
(S2) $(q', \delta(q, c)) \notin \mathsf{Adm}$ *for every state q'.*

Its size *is $N = |S|$.*

The names anticipate the reduction of Section 4.7, where a reader follows the derivations from the seed of a witness. A run is *dead* when the constraint alone already contradicts the witness. A state is *rejected* when the seed can no longer derive a forbidden atom: the reader has rejected every way of refuting the witness by a forbidden atom. So a rejected state is what a witness has to reach, although the word may suggest the opposite. Condition (S1) says that rejection is permanent, and (S2) that a state reached after a rejected state is never the second component of an admitted pair. A pair of states is *admitted* when it shows a derivation from the seed of one witness that contradicts a witness, the same one or another; such pairs matter only at the places defined next.

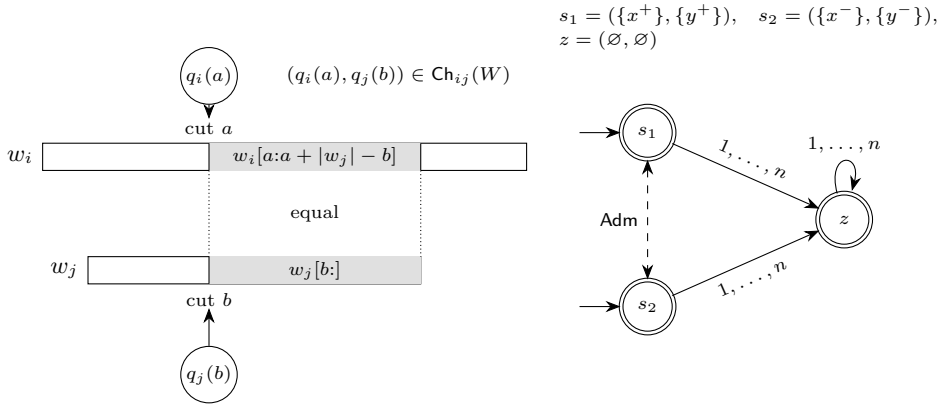
Let $W = (w_1, \dots, w_k)$ be a tuple of words. The word w_i is read from s_i; we write $q_i(a) = \delta^*(s_i, w_i[{:}a])$ for the state of coordinate i at the cut a, and $q_i(|w_i|)$ is its *endpoint*. The *total length* of W is $\|W\| = |w_1| + \cdots + |w_k|$.

▶ **Definition 4.12** (Comparisons, channels and holes)**.** *Let i and j be coordinates, possibly equal. A* comparison *from the* host *i to the* target *j is a pair (a, b) of a cut a of w_i and a cut b of w_j such that $w_j[b{:}] \preceq w_i[a{:}]$: the rest of the target is a prefix of the rest of the host. The pair of states $(q_i(a), q_j(b))$ is the* channel pair *at this comparison, and the* channel *from i to j is the set of all channel pairs,*

$$\mathsf{Ch}_{ij}(W) = \{\, (q_i(a), q_j(b)) \mid a \text{ a cut of } w_i,\ b \text{ a cut of } w_j,\ w_j[b{:}] \preceq w_i[a{:}] \,\}.$$

Coordinate i is eligible *if $q_i(a) \neq \dagger$ for every cut a and its endpoint is rejected. The tuple W is a* hole *if every coordinate is eligible and $\mathsf{Ch}_{ij}(W) \cap \mathsf{Adm} = \varnothing$ for all i and j.*

All cuts are included, also the two ends: at $b = |w_j|$ the rest of the target is empty, so (a, b) is a comparison for every cut a of the host. For example, in the tuple (0110, 10) the



(a) a comparison (a, b): host i, target j (b) the reader of Example 4.16

Figure 3 (a) A comparison of a tuple W and its channel pair: the rest of the target word w_j after the cut b is a prefix of the rest of the host word w_i after the cut a, and the pair of the states at the two cuts belongs to the channel from i to j. The tuple is a hole only if no such pair is admitted, for all i and j, including $i = j$. (b) The reader of the empty constraint with the queries (x, y) and (y, x) in the mode vector $(\top, \bot)$. All three states are rejected (double circles), the dead state is not shown, and the admitted pairs are (s_1, s_2) and (s_2, s_1).

comparisons from the host 1 to the target 2 are $(2, 0)$, since 10 is the rest of 0110 after the cut 2, then $(0, 1)$ and $(3, 1)$, since the rest 0 of the target is a prefix of the rests 0110 and 0, and $(a, 2)$ for the five cuts a of the host. In a pair $(q, q') \in \mathsf{Adm}$ the first component is the state of a host and the second the state of a target; the relation need not be symmetric. Figure 3(a) shows a comparison and its channel pair.

For an eligible coordinate, (S1) gives a *first rejected cut* J_i: the states $q_i(a)$ are rejected exactly for $a \geq J_i$. The prefix $w_i[{:}J_i]$ is the *head* of w_i and the suffix $w_i[J_i{:}]$ its *rejected tail*. By (S2), the target cut of an admitted channel pair is at most J_j. So the tail of a word is never involved as a target, but it can be involved as a host, and for $i = j$ a channel pair has $a \leq b$.

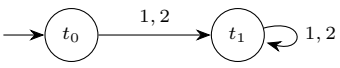
4.7 The Reader of a Mode and the Reduction

Fix a satisfiable constraint Γ, queries $(x_1, y_1), \ldots, (x_k, y_k)$ and a vector of modes $\bar{\mu}$, with the signed variables p_i, p_i' of (5) and the sets Neg and Fct of (4).

▶ **Definition 4.13** (The reader $M_{\bar{\mu}}$). *The states of $M_{\bar{\mu}}$ are the pairs (P, Q) of sets of signed variables with $(Q \cup \overline{P}) \cap \mathsf{Fct} = \varnothing$, where $\overline{P} = \{\overline{r} \mid r \in P\}$, and a dead state $\dagger$. The transition on a letter i takes (P, Q) to $(\delta^{\uparrow}(P, i), \delta^{\downarrow}(Q, i))$ if this pair is a state, and to $\dagger$ otherwise. The ith start state is $(\mathsf{U}(p_i, \varepsilon), \mathsf{D}(p_i', \varepsilon))$ if this pair is a state, and $\dagger$ otherwise. Moreover*

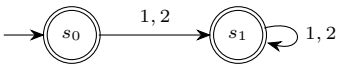
$$(P, Q) \in \mathsf{Rej} \iff P \cap \mathsf{Neg} = \varnothing,$$
$$((P_h, Q_h), (P_t, Q_t)) \in \mathsf{Adm} \iff P_t \cap (Q_h \cup \overline{P_h}) \neq \varnothing.$$

Coordinate i follows the upper run of its seed, in the *upper component* P, and the lower run of its exclusion, in the *lower component* Q. It dies when a prefix of the path shows that the exclusion or the opposite of the seed lies in the least closure, so that every solution contradicts the witness. It is rejected when its seed derives no forbidden atom. A pair is admitted when the seed of the target derives the exclusion of the host or the opposite of the



$t_0 = (\{x^+, u^+\}, \{y^+\})$
$t_1 = (\{u^+\}, \varnothing)$

(a) mode $\top$: no state is rejected



$s_0 = (\{y^-\}, \{x^-, u^-\})$
$s_1 = (\varnothing, \{u^-\})$

(b) mode $\bot$: both states are rejected

Figure 4 The readers of $\Gamma = \{u \le f(u,u),\ x \le u,\ y = \bot\}$ and the query (x, y) (Example 4.15). A state is the pair of the upper run of the seed and the lower run of the exclusion; rejected states are drawn with a double circle, and the dead state, which is not reachable here, is not shown. (a) In mode $\top$ the upper component always contains $u^+ \in \mathsf{Neg}$, so there is no hole. (b) In mode $\bot$ every word is a hole, since no pair of states is admitted.

seed of the host, at a comparison of the two paths as in Lemma 4.10. With these readings the three conditions of Theorem 4.7 are exactly the conditions of a hole: (R2) says that no run dies, (R1) that every endpoint is rejected, and (R3) that no channel contains an admitted pair.

▶ **Theorem 4.14** (Reduction). *$M_{\bar\mu}$ is a structural reader over $[n]$ with k starts and at most $2^{4|V|}+1$ states, computable from Γ, the queries and $\bar\mu$. A tuple W of paths is a hole of $M_{\bar\mu}$ if and only if some solution of Γ contains the seeds and none of the exclusions of $(\bar\mu, W)$. Consequently, for a constraint Γ that is satisfiable over $\mathcal{T}$,*

$$\Gamma \not\models_{\mathcal{T}} x_1 \le y_1 \vee \cdots \vee x_k \le y_k \iff M_{\bar\mu} \text{ has a hole for some } \bar\mu \in \{\top, \bot\}^k.$$

Proof sketch. By Lemma 4.9, the state of an eligible coordinate i at a cut a is the pair $(\mathsf{U}(p_i, w_i[:a]), \mathsf{D}(p_i', w_i[:a]))$. Once $P \cap \mathsf{Neg}$ is empty, the next upper set is empty, which gives (S1) and (S2). A run stays alive exactly when no prefix puts B_i or $\overline{A_i}$ into the least closure, which is (R2), and its endpoint is rejected exactly when (R1) holds. Under (R1), Lemma 4.10 states that $A_j \vdash B_i$ or $A_j \vdash \overline{A_i}$ if and only if there are cuts b of w_j and a of w_i with $w_j[b:] \preceq w_i[a:]$ and $\mathsf{U}(p_j, w_j[:b]) \cap (\mathsf{D}(p_i', w_i[:a]) \cup \mathsf{D}(\overline{p_i}, w_i[:a])) \neq \varnothing$, that is, an admitted pair on the channel from i to j. The last statement follows with Lemma 4.6. ◀

▶ **Example 4.15** (One query: the reader of a small constraint). Consider the constraint $\Gamma = \{u \le f(u,u),\ x \le u,\ y = \bot\}$ of Example 4.5, with $\mathsf{Neg} = \{u^+\}$ and $\mathsf{Fct} = \{y^-\}$, and the query (x, y). Figure 4 shows the two readers.

Mode $\top$. The seed is (x^+, w) and the exclusion (y^+, w): the tree of x should reach $\top$ along w. The upper run of the seed is $\mathsf{U}(x^+, \varepsilon) = \{x^+, u^+\}$ and $\mathsf{U}(x^+, w) = \{u^+\}$ for $w \neq \varepsilon$, because the only way to pop the path of the seed is to move to u^+ and to pop there. No move ends in (y^+, w), so the lower run of the exclusion is $\{y^+\}$ at ε and empty afterwards. The upper component always contains $u^+ \in \mathsf{Neg}$, so no state of the run is rejected, and the reader has no hole: the seed always derives the forbidden atom (u^+, ε), as in Example 4.5.

Mode $\bot$. The seed is (y^-, w) and the exclusion (x^-, w): the tree of y should reach $\bot$ along w and the tree of x should not. No move starts from y^-, so the upper run is $\{y^-\}$ at ε and empty afterwards. The lower run of the exclusion is $\mathsf{D}(x^-, \varepsilon) = \{x^-, u^-\}$ and $\mathsf{D}(x^-, w) = \{u^-\}$ for $w \neq \varepsilon$, since (x^-, w) is built from (u^-, ε) by pushing w at u^- and moving to x^-. Both states are rejected, neither is dead, because $y^- \in \mathsf{Fct}$ belongs neither to the lower components nor to the flipped upper components, and no pair is admitted: for the only nonempty upper component, $\{y^-\} \cap (\{x^-, u^-\} \cup \{y^+\}) = \varnothing$. So every word is a hole, in particular $w = \varepsilon$. The hull of its seed (y^-, ε) is the least closure, the solution $y = \bot$, $x = u = f^\omega$, and indeed $f^\omega \not\le \bot$: the entailment $\Gamma \models x \le y$ fails.

A dead run. For the reversed query (y, x), the start state of mode $\top$ would be $(\{y^+\}, \{x^+\})$, whose flipped upper component contains the fact y^-; and the start state of mode $\bot$ would be $(\{x^-\}, \{y^-\})$, whose lower component contains it. So both readers start in † and have no hole: Γ entails $y \leq x$, as it must, since $y = \bot$ in every solution.

▶ **Example 4.16** (Two queries, unary and binary)**.** Take the empty constraint and the queries (x, y) and (y, x) of Example 2.6. There are no moves, no facts and no forbidden atoms, so $\mathsf{U}(p, \varepsilon) = \mathsf{D}(p, \varepsilon) = \{p\}$, both sets are empty for nonempty paths, nothing is dead and every state is rejected. In the mode vector $(\top, \bot)$ the seeds are (x^+, w_1) and (x^-, w_2) and the exclusions (y^+, w_1) and (y^-, w_2): the tree of x should have a leaf of sign $+$ on w_1 and a leaf of sign $-$ on w_2. The reader is shown in Figure 3(b). The pair (s_1, s_2) is admitted because the upper component $\{x^-\}$ of the target s_2 meets the flipped upper component $\overline{\{x^+\}} = \{x^-\}$ of the host s_1, and symmetrically for (s_2, s_1).

A channel pair (s_1, s_2) arises from the comparison $(0, 0)$, which exists when $w_2 \preceq w_1$, and (s_2, s_1) when $w_1 \preceq w_2$. So (w_1, w_2) is a hole if and only if w_1 and w_2 are incomparable: two leaves of opposite signs cannot lie on one path. For $n = 1$ any two words are comparable and there is no hole; the other three mode vectors give the same reader up to renaming, so the entailment holds. For $n = 2$ the tuple $(1, 2)$ is a hole. The hull of its seeds is the solution $x = f(\top, \bot)$, $y = f^\omega$ for a covariant constructor, a countermodel.

5 Small Holes: Overview

By Theorem 4.14, deciding a disjunctive entailment amounts to deciding whether a structural reader has a hole. The rest of the proof is about readers and words; it uses no property of trees or constraints, and the alphabet A is arbitrary. We fix a structural reader $M = (S, \delta, (s_i)_i, \dagger, \mathsf{Rej}, \mathsf{Adm})$ with k starts and N states.

▶ **Theorem 5.1** (Small holes)**.** *There is a computable function* H *such that every structural reader with* k *starts and* N *states that has a hole has a hole of total length at most* $\mathsf{H}(k, N)$*.*

The definition of H is given in Equation (6) below, and its growth is estimated in Section 9. This section introduces the notions that organize the proof and reduces the theorem to two bounds.

5.1 Domination, Fresh Words and Immortal States

A *table* with k starts is the part $(S, \delta, (s_i)_i)$ of a reader: a complete deterministic automaton with k start states. Runs, endpoints and channels of a tuple are defined for tables as in Section 4.6. We write $\mathsf{Vis}_i(W) = \{q_i(a) \mid 0 \leq a \leq |w_i|\}$ for the states that coordinate i visits.

▶ **Definition 5.2** (Domination)**.** *A tuple* U dominates *a tuple* W *in a table, written* $U \sqsubseteq W$*, if for all coordinates* i *and* j*: the endpoints of coordinate* i *in* U *and in* W *are equal,* $\mathsf{Vis}_i(U) \subseteq \mathsf{Vis}_i(W)$*, and* $\mathsf{Ch}_{ij}(U) \subseteq \mathsf{Ch}_{ij}(W)$*.*

Domination is reflexive and transitive. A dominating tuple does nothing that the dominated tuple does not do: it ends in the same states, passes only through states that were already visited, and compares only pairs of states that were already compared. Since the definition of a hole only forbids things, we obtain:

▶ **Lemma 5.3** (Holes under domination)**.** *If* W *is a hole of a structural reader and* $U \sqsubseteq W$ *in its table, then* U *is a hole with the same endpoints.*

Proof. The run of u_i visits only states of $\mathsf{Vis}_i(W)$, none of which is dead, and its endpoint is that of w_i, which is rejected. No pair of $\mathsf{Ch}_{ij}(U) \subseteq \mathsf{Ch}_{ij}(W)$ is admitted. ◀

So a hole may be replaced by any tuple that dominates it, and the task is to find short dominating tuples. The block deletions of Section 6 and the compression of periodic stretches in Section 7.6 delete factors from the words of a tuple in such a way that the result dominates it. The replacement of rejected tails by rays in Section 7.5 is of a different kind: it can lengthen words and change endpoints, and it only preserves the existence of a hole. Deleting a factor between two cuts with the same state preserves endpoints and visited states; the difficulty is always the channels, where a deletion can bring two cuts into a comparison that did not exist before.

▶ **Definition 5.4** (Fresh words). *Coordinate i of a tuple W is* fresh *in a table if its endpoint is not visited earlier: $q_i(a) \neq q_i(|w_i|)$ for every $a < |w_i|$. A tuple is* fresh *if all its coordinates are. A tuple U* freshly dominates *W if U is fresh and $U \sqsubseteq W$. A fresh tuple is* minimal *if no tuple that freshly dominates it has a smaller total length.*

The empty word is fresh. Since domination is transitive, every fresh tuple is freshly dominated by a minimal fresh tuple: a fresh dominator of least total length.

▶ **Definition 5.5** (Immortal states). *A state q of a structural reader is* immortal *if for every m some word of length m can be read from q without reaching †, and* mortal *otherwise.*

A state is immortal exactly when a cycle of states other than † is reachable from it without passing through †. Freshness comes for free when the endpoint is mortal, and it yields the basic counting principle of the whole proof: *cuts at which comparable rests begin carry different states.*

▶ **Lemma 5.6** (Distinct states). *Let W be a tuple and $a < b$ cuts of w_i such that $w_i[b{:}] \preceq w_i[a{:}]$.*

1. *If coordinate i is fresh, then $q_i(a) \neq q_i(b)$.*
2. *If coordinate i is eligible with a mortal endpoint, then it is fresh.*
3. *If coordinate i is eligible and $b \leq J_i$, then $q_i(a) \neq q_i(b)$.*

Proof. (1) If $q_i(a) = q_i(b)$, reading $w_i[b{:}]$ from the cut a leads to the endpoint, at the cut $a + |w_i| - b < |w_i|$. (2) If the endpoint occurs at a cut $a < |w_i|$, the run from a to the end is a cycle of states other than †, so the endpoint is immortal. (3) If $q_i(a) = q_i(b)$, reading $w_i[b{:}J_i]$ from the cut a leads to the rejected state $q_i(J_i)$ at the cut $a + J_i - b < J_i$, before the first rejected cut. ◀

In particular, a fixed text has fewer than N nonempty rests of one fresh word as prefixes, since these rests are comparable.

5.2 The Two Bounds

The first bound concerns the coordinates with immortal endpoints. Recall that the rejected tail of an eligible coordinate is the part of its word after the first rejected cut.

▶ **Theorem 5.7** (Rejected tails). *There is a computable function Λ such that every structural reader with k starts and N states that has a hole has a hole in which the rejected tail of every coordinate with an immortal endpoint has length at most $\Lambda(k, N)$.*

The second bound is a statement about tables, which have no rejected states, no dead state and no admitted pairs.

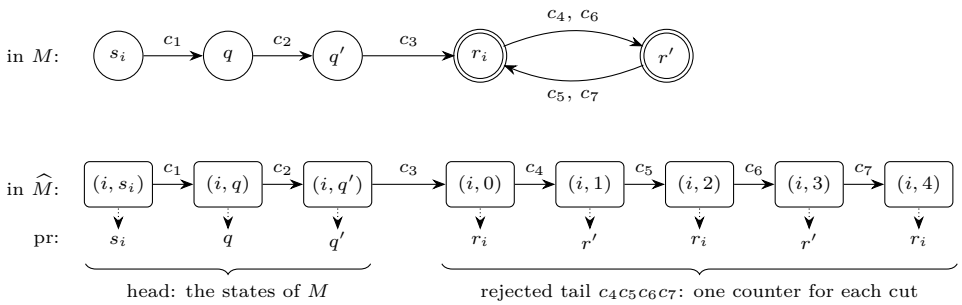


Figure 5 The cover on a tracked coordinate i with $w_i = c_1 \cdots c_7$, $J_i = 3$ and $\ell_i = 4$. In M the rejected tail runs twice through the cycle of the rejected states r_i and r', so the endpoint r_i is visited three times and the word is not fresh. In the cover the tail is spelled out by the counters $(i, 0), \ldots, (i, 4)$, the endpoint $(i, 4)$ is visited once. From each displayed counter, every transition other than the displayed next-letter transition leads to the absorbing state o, and all transitions from $(i, 4)$ lead to o; other transitions from copies of head states follow Definition 5.9. On the displayed run, the projection pr (last row) maps each state to the state of M at the same cut.

▶ **Theorem 5.8** (Fresh tuples). *There is a computable function* J *such that every minimal fresh tuple of a table with* k *starts and* N *states has total length at most* $\mathsf{J}(k, N)$*. Hence every fresh tuple is freshly dominated by a tuple of total length at most* $\mathsf{J}(k, N)$*.*

Neither bound depends on the size of the alphabet. Theorem 5.8 is proved in Section 6 and Theorem 5.7 in Section 7.

Why two bounds? A hole need not be fresh: a coordinate with an immortal endpoint may run through its endpoint many times. Deleting such a loop is tempting, but it changes the rests of the word after all earlier cuts, and with them the channels in which this coordinate is the host. For a coordinate with a mortal endpoint the situation is rigid, by Lemma 5.6. Theorem 5.7 is the tool that brings immortal coordinates under control: once the rejected tail is short, the coordinate can be made fresh in a larger table that counts the letters of the tail.

5.3 The Cover of a Hole

A hole whose immortal tails are short is not yet fresh, but it becomes fresh in a larger table that counts the letters of these tails (Figure 5). Let W be a hole of M and L a number such that every coordinate with an immortal endpoint has a rejected tail of length at most L. Call a coordinate i *tracked* if its rejected tail has length $\ell_i = |w_i| - J_i \leq L$; the other coordinates have mortal endpoints.

▶ **Definition 5.9** (Cover of a hole). *The* cover $\widehat{M}$ *of* W *is the following table with* k *starts. Its states are an absorbing state* o *and, for each coordinate* i*, the pairs* (i, s) *with* $s \in S$ *and the* counters (i, m) *with* $0 \leq m \leq L$*, all of them distinct.*

- *For an untracked coordinate* i*, the start is* (i, s_i) *and a letter* c *takes* (i, s) *to* $(i, \delta(s, c))$*.*
- *For a tracked coordinate* i *with first rejected state* $r_i = q_i(J_i)$*, a letter* c *takes* (i, s) *with* $s \notin \mathsf{Rej}$ *to the counter* $(i, 0)$ *if* $\delta(s, c) = r_i$*, to* $(i, \delta(s, c))$ *if* $\delta(s, c) \notin \mathsf{Rej}$*, and to* o *otherwise. It takes a counter* (i, m) *with* $m < \ell_i$ *to* $(i, m + 1)$ *if* c *is the letter of* w_i *at the position* $J_i + m$*, and to* o *otherwise; since* $\ell_i \leq L$*, the counter* $(i, m + 1)$ *exists. The start is* $(i, 0)$ *if* $J_i = 0$ *and* (i, s_i) *otherwise.*

- *All remaining transitions lead to o. In particular, so do all transitions from the counter (i, ℓ_i) of a tracked coordinate and from the counters of an untracked one.*

The projection pr *maps (i, s) to s, the counter (i, m) of a tracked coordinate with $m \le \ell_i$ to $q_i(J_i + m)$, and all other counters and o to $\dagger$.*

So the cover follows M on the head of a tracked word and then spells out its rejected tail, letter by letter. It has $k(N + L + 1) + 1$ states.

▶ **Lemma 5.10** (Cover). *The tuple W is fresh in its cover $\widehat{M}$, and every tuple that dominates W in $\widehat{M}$ is a hole of M with the same endpoints as W.*

Proof. On a tracked coordinate i, the run of w_i in the cover is $(i, q_i(a))$ at the cuts $a < J_i$ and the counter $(i, a - J_i)$ at the cuts $a \ge J_i$; on an untracked coordinate it is $(i, q_i(a))$. In the first case the endpoint is the counter (i, ℓ_i), which occurs only at the last cut. In the second case the endpoint of w_i in M is mortal, so coordinate i is fresh in M by Lemma 5.6, hence in the cover. So W is fresh in the cover, and its runs avoid o.

Whenever a transition of the cover from a state z on a letter c does not lead to o, it leads to a state with projection $\delta(\mathrm{pr}(z), c)$; this is checked case by case, and for a counter it holds because the letter read is the letter of w_i. The starts project to the starts of M. Now let U dominate W in the cover. The runs of U in the cover visit only states visited by W, so they avoid o, and therefore the run of U in M is the projection of its run in the cover. Applying pr to the three conditions of domination in the cover gives $U \sqsubseteq W$ in the table of M, because the run of W in M is also the projection of its run in the cover. By Lemma 5.3, U is a hole of M with the endpoints of W. ◀

Proof of Theorem 5.1 from Theorems 5.7 and 5.8. Let M have a hole. By Theorem 5.7 it has a hole W whose coordinates with immortal endpoints have rejected tails of length at most $L = \Lambda(k, N)$. By Lemma 5.10, W is fresh in its cover, a table with k starts and $k(N + L + 1) + 1$ states, so by Theorem 5.8 it is freshly dominated there by a tuple U of total length at most

$$\mathsf{H}(k, N) = \mathsf{J}\big(k,\ k\,(N + \Lambda(k, N) + 1) + 1\big). \tag{6}$$

By Lemma 5.10 again, U is a hole of M. ◀

The composition explains the shape of the final bound: the fresh-tuple bound is applied to a table whose size is the rejected-tail bound. For each fixed number k of starts, both are doubly exponential in N (Theorem 9.1), which gives four iterated exponentials in N.

6 Fresh Tuples

This section explains the proof of Theorem 5.8: a minimal fresh tuple of a table with k starts and N states has bounded total length. We fix such a table and a fresh tuple $W = (w_1, \ldots, w_k)$. Full proofs are in Appendix B.

6.1 The Idea

To shorten W we delete a factor between two cuts of one word that carry the same state. This preserves endpoints and visited states, but in general not the channels. A channel compares the rest of a target word with a factor of a host word. If we delete a factor from the host in the middle of such a comparison, the two sides no longer match; and after the

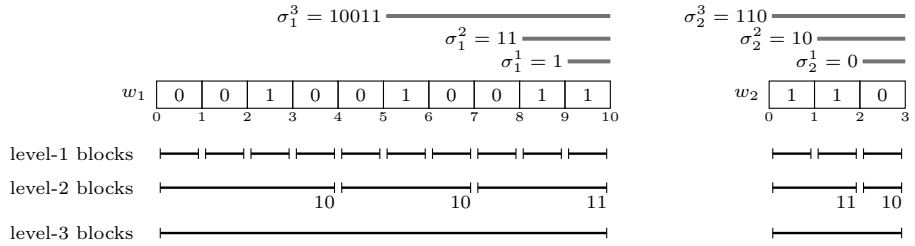


Figure 6 The marker hierarchy of Example 6.2. Above each word: its markers of the levels 1, 2, 3, nested suffixes of the word. Below: the blocks of each level; a block ends where a marker of that level ends, in any of the words. Under each level-2 block the level-2 marker that ends there is shown: 10 ends at the cuts 4 and 7 of w_1 and at the end of w_2, and 11 ends at the end of w_1 and at the cut 2 of w_2.

deletion the rest of some word may match a factor of the edited host that it did not match before, at cuts whose states were never compared. Both effects disappear if the deletion is *invisible to every comparison*: whenever a comparison runs over a deleted factor on one side, the same factor is deleted at the corresponding place on the other side.

This suggests deleting a repeated factor at *all* places that look alike, in all words at once. The question is what "look alike" should mean. Comparing bounded windows of letters is not enough, because a comparison can run over arbitrarily long equal texts. We therefore cut all words into blocks by a rule that depends only on the text, so that equal texts are cut in the same way, up to a bounded number of blocks at their left ends. The cutting rule is given by *markers*, suffixes of the words w_i themselves, and it is iterated: the blocks of one level are grouped into the blocks of the next level, until every word is a single block. Deleting equal stretches of blocks at equal distances from the right ends of equal block sequences is then invisible to comparisons, and a pigeonhole argument finds such stretches whenever some block consists of too many blocks of the level below.

6.2 The Marker Hierarchy

▶ **Definition 6.1** (Markers and anchors). *For each* level $t \geq 1$ *we define a set* Act_t *of* active *coordinates and, for* $l \in \mathrm{Act}_t$*, a* marker σ_l^t*, a nonempty suffix of* w_l*.*

- Act_1 *consists of the coordinates with a nonempty word, and* σ_l^1 *is the last letter of* w_l*.*
- *A* level-t anchor *of a text* u *is a cut* $e \geq 1$ *of* u *at which* σ_m^t *ends, for some* $m \in \mathrm{Act}_t$*. The level-*t *anchors of the word* w_l *are its anchors as a text together with its end* $|w_l|$*.*
- *A coordinate* $l \in \mathrm{Act}_t$ *belongs to* Act_{t+1} *if* w_l *has a level-*t *anchor* $e < |w_l|$*. Let* e *be the last such anchor and* σ_m^t *the longest level-*t *marker that ends at* e*. Then* $\sigma_l^{t+1} = w_l[e - |\sigma_m^t| :]$*.*

The level-0 anchors *of a text are all its cuts* $e \geq 1$*. The level-*t *anchors cut each nonempty word into its* level-t blocks*, the factors between consecutive anchors, the first block beginning at the cut* 0*. The* top level *is the largest* t *with* $\mathrm{Act}_t \neq \varnothing$*, and* 0 *if all words are empty.*

So the blocks of level 0 are single letters, the blocks of level 1 end with the last letter of some word, and the marker of the next level consists of the last block of the word together with the marker in front of it. A coordinate stops being active when its word is a single block.

▶ **Example 6.2.** Let $W = (w_1, w_2)$ with $w_1 = 0010010011$ and $w_2 = 110$ (Figure 6). The level-1 markers are the letters 1 and 0, so every cut is a level-1 anchor and the level-1 blocks

are single letters. In w_1 the last anchor before the end is the cut 9, where the marker 1 ends, so $\sigma_1^2 = 11$; in w_2 it is the cut 2, and $\sigma_2^2 = 10$. The level-2 anchors are the cuts at which 11 or 10 ends: the cuts 4, 7 and 10 of w_1, and the cuts 2 and 3 of w_2. They cut the words into the level-2 blocks

$$w_1 = 0010 \cdot 010 \cdot 011, \qquad w_2 = 11 \cdot 0.$$

At level 3 the markers are $\sigma_1^3 = 10\,011$ (the level-2 marker 10 that ends at the cut 7, followed by the last block) and $\sigma_2^3 = 11\,0 = w_2$. Neither occurs at another place, so the level-3 blocks are the whole words, no coordinate is active at level 4, and the top level is 3.

The hierarchy has three structural properties. The first is the reason why blocks behave well under comparisons.

▶ **Lemma 6.3** (Markers). *For every level $t \geq 1$:*

1. Factor antichain. *If $l, m \in \mathrm{Act}_t$ and σ_l^t is a factor of σ_m^t, then $\sigma_l^t = \sigma_m^t$. Consequently, if an occurrence of an active level-t marker in a text lies inside an occurrence of another one, the two occurrences coincide.*
2. *For $l \in \mathrm{Act}_{t+1}$, the marker σ_l^t is a proper suffix of σ_l^{t+1}, and the text σ_l^{t+1} has exactly two level-t anchors: the end of the level-t marker it begins with, and its own end.*
3. *Every level-$(t+1)$ anchor of a text is a level-t anchor of it.*

By (3) the blocks of level $t+1$ are concatenations of blocks of level t. The freshness of W enters through Lemma 5.6: a fixed text has at most $N-1$ nonempty rests of one word w_l as prefixes.

▶ **Lemma 6.4** (Depth and span). 1. *The top level is at most $k(N-1)$.*
2. *In every text, an occurrence $u[a{:}b]$ of a level-$(t+1)$ marker contains at most $k(N-1)+2$ level-t anchors e of u with $a < e \leq b$.*

For (1), a marker of level t has at least t prefixes that are suffixes of words, namely the markers nested at its left end, and these give rests of words that are prefixes of one text. For (2), a level-t marker that ends inside the occurrence lies inside it, which happens at two anchors by Lemma 6.3, or it begins to the left of it, and then the part inside is a rest of a word and a prefix of the marker.

6.3 Parses and the Reverse Trie

Fix a level t between 0 and the top level. We call the blocks of level $t+1$ *upper blocks* and those of level t *lower blocks*; at the top level the upper blocks are the nonempty words. An *upper occurrence* is an upper block together with its place, a word and two cuts. Its *parse* is the sequence of its lower blocks, a word over the alphabet of lower blocks, and for $d \geq 0$ its *cut at depth d* is the cut that lies d lower blocks to the left of its end.

Equal texts are parsed alike only up to their left ends: a marker that begins before an upper block can end inside it and create an additional anchor there. Different occurrences of one text can therefore have different parses, but by Lemma 5.6 only at boundedly many places, all to the left of the first anchor that the text has on its own (Lemma B.2). What matters is how parses end. For two parses Z and Y let $\mathrm{lcs}(Z, Y)$ be the number of lower blocks in their longest common suffix.

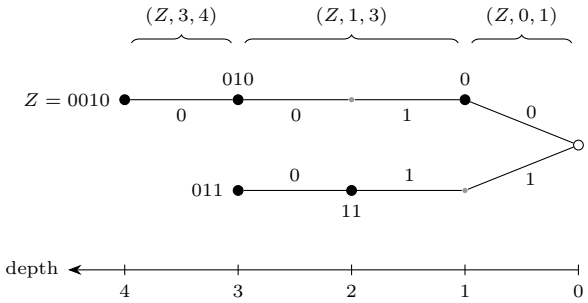


Figure 7 The reverse trie of Example 6.6: the parses of the five upper blocks, read from the right; the edges of the drawing are labeled with lower blocks, here letters. A black node is the end of the parse written next to it. The path of $Z = 0010$ has the nodes $0, 1, 3, 4$; nothing branches off at depth 2, so $(Z, 1, 3)$ is one edge, which is traversed by the upper blocks 0010 and 010.

▶ **Definition 6.5** (Reverse trie). *The* nodes *on the path of a parse Z are* 0*, the length of Z, and the numbers* $\mathrm{lcs}(Z, Y)$ *for the parses Y of all upper occurrences. An* edge *is a parse Z with two consecutive nodes $a < b$ on its path. An upper occurrence* traverses *the edge if its parse Y satisfies* $\mathrm{lcs}(Y, Z) \geq b$.

This is the trie of all parses read from the right; an edge is a stretch of a parse along which no other parse branches off. All upper occurrences that traverse an edge end with the same b lower blocks, and whether an occurrence traverses the edge is decided by a bounded part of the text at its right end (Lemma B.3).

▶ **Example 6.6.** In Example 6.2 take $t = 1$. The lower blocks are the letters, and the upper blocks are the level-2 blocks 0010, 010, 011 of w_1 and 11, 0 of w_2; the parse of an upper block is the sequence of its letters. For $Z = 0010$ we have $\mathrm{lcs}(Z, 010) = 3$, $\mathrm{lcs}(Z, 0) = 1$ and $\mathrm{lcs}(Z, 011) = \mathrm{lcs}(Z, 11) = 0$. So the nodes on the path of Z are $0, 1, 3, 4$, and its edges are $(Z, 0, 1)$, $(Z, 1, 3)$ and $(Z, 3, 4)$ (Figure 7). The edge $(Z, 1, 3)$ is traversed by the upper blocks 0010 and 010 of w_1, which end with the same three lower blocks $0, 1, 0$; the edge $(Z, 0, 1)$ is traversed by these two and by the upper block 0 of w_2. The upper block 010 lies between the cuts 4 and 7 of w_1, and its cuts at the depths $0, 1, 2, 3$ are $7, 6, 5, 4$.

6.4 Block Deletions

For a cut c of a word w_l, let $\tau_r(l, c)$ be the text of the level-r marker that ends at c, if there is one; it is unique by Lemma 6.3. Let $\omega(l, c)$ be the largest length of these texts for $1 \leq r \leq t$, and 0 if there are none. The *label* of the cut is

$$\mathrm{lab}(l, c) = \big(\tau_1(l, c), \ldots, \tau_t(l, c),\ q_l(c - \omega(l, c))\big) :$$

the markers of all levels up to t that end at the cut, and the state before the longest of them. Two cuts of one word with equal labels have the same ω letters to their left and the same states at all distances up to ω, in particular the same state. In Example 6.2 with $t = 2$, the markers 0 of level 1 and 10 of level 2 end at the cut 7 of w_1, so $\omega(1, 7) = 2$ and $\mathrm{lab}(1, 7) = (0,\ 10,\ q_1(5))$; likewise $\mathrm{lab}(1, 4) = (0,\ 10,\ q_1(2))$. The two labels are equal exactly when $q_1(2) = q_1(5)$, and then also $q_1(3) = q_1(6)$ and $q_1(4) = q_1(7)$, because the same letters $1, 0$ are read from equal states.

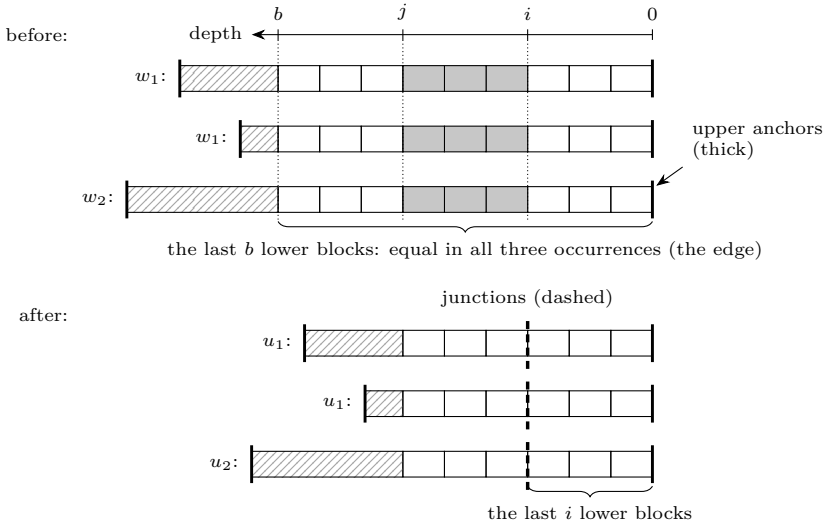


Figure 8 A block deletion at level t, before and after. Three upper blocks (blocks of level $t+1$), two in w_1 and one in w_2, are cut into lower blocks (blocks of level t) by the anchors of level t. Their parses end with the same b lower blocks, so all three traverse the same edge of the reverse trie; they differ above it (hatched). If the cuts at the depths j and i carry equal labels in every traversing upper block, the lower blocks between these two depths (gray) are deleted in all of them at once. In the result (u_1, u_2) the edited upper blocks still end with equal texts: every junction, the place where a factor was removed, is followed by the same i lower blocks.

Deletions will be restricted to the interior of an edge. The *margin*

$$\mathsf{m} = 3kN - k + 5$$

is the number of lower blocks that are kept free at both ends of the edge. At the end b, where the parses branch, the margin ensures that the markers ending at the cuts in question lie inside the text that all traversing occurrences share, so that their labels can be compared (Lemma B.6). At the end a it leaves enough common lower blocks to the right of a deleted factor to recognize, from the text alone, every upper block that traverses the edge (Lemma B.3).

▶ **Definition 6.7** (Block deletion). *A* block deletion *at level t consists of an edge (Z, a, b) and two depths $i < j$ with $a + \mathsf{m} \le i$ and $j \le b - \mathsf{m}$ such that, in every upper occurrence o that traverses the edge, the cuts of o at the depths j and i have equal labels. It deletes, from every traversing upper occurrence, the factor between its cuts at the depths j and i.*

Figure 8 illustrates the definition. The deleted factors are equal words, since traversing occurrences end with the same b lower blocks, and at least one factor is deleted. The words of the example are far too short for a block deletion, since an edge must have more than $2\mathsf{m}$ lower blocks; in general, deletions exist as soon as some edge is long (Section 6.5).

▶ **Theorem 6.8** (Block deletions). *If W is a fresh tuple, the result of a block deletion is a strictly shorter tuple that freshly dominates W. Hence a minimal fresh tuple admits no block deletion, at any level.*

The proof has three steps (Appendix B.4). First, the equal labels make every deleted factor a loop of the run, so endpoints are preserved, no new state is visited and the result is

fresh. Second, the deletion *preserves the anchors* of all levels up to $t+1$: a marker occurrence of the edited words that runs over a deletion site has only a short part to the left of the site, because of the antichain property and of Lemma 6.3(2), and this part lies in the window that the equal labels make identical on both sides of the deleted factor. So the edited words are cut into the same blocks. Third, the channels: consider a comparison of the edited tuple that runs over a deletion site of the host. The i lower blocks between the site and the end of its upper block survive in the host, and they are copied into the target as long as the comparison continues. If it continues to the end of the upper block, the target has an upper block ending with the same text, which therefore traverses the same edge, and so the target has a deletion site at the corresponding cut; the deleted factor can be put back on both sides. If the comparison ends earlier, the target is a short rest of its word, and the equal labels provide a comparison of the original words with the same states. The place of a word of the result where a factor was removed is called a *junction*.

6.5 The Bound

On an edge, the label of the cut at a depth d is the same function of the state at the top of the edge in every traversing occurrence, when d is at least m below the top: by Lemma 6.4 the markers that end at this cut lie in the common text of the edge. There are at most $(k+1)^t N^N$ such functions. So an edge of more than

$$E = (k+1)^{kN+1} N^N + 2\mathsf{m} + 1$$

lower blocks has two depths with the same function between its margins, and admits a block deletion. In a minimal fresh tuple every edge is therefore shorter. This bounds the number of lower blocks of an upper block by the number of nodes on its path times E, hence by the number of parses. The number of parses is at most $2^{k(N-1)}$ times the number of distinct texts of upper blocks, and each text of a lower block is the text of an upper block one level below. Starting from the top level, where there are at most k upper blocks, this gives a bound for every level.

▶ **Theorem 6.9** (Fresh tuples, with the bound). *Let $F = 2^{k(N-1)}$, $\nu_0 = k+1$ and $\nu_{d+1} = \nu_d + \nu_d F(\nu_d F + 2)E$, and let $X = (\nu_{kN} F + 2)E$. Every minimal fresh tuple of a table with k starts and N states has total length at most*

$$\mathsf{J}(k,N) = k\, X^{kN+1} \;\leq\; \exp_2\big((k+N+16)^4\big).$$

Here ν_d bounds the number of distinct texts of upper blocks d levels below the top level. In a minimal fresh tuple every upper block has at most X lower blocks at every level, there are at most $kN+1$ levels, and the blocks of level 0 are letters; this gives $k\, X^{kN+1}$.

7 Rejected Tails

This section explains the proof of Theorem 5.7: a reader with a hole has a hole whose coordinates with immortal endpoints have short rejected tails. If $k = 0$, the empty tuple is a hole without tails, and Theorem 5.7 holds with $\Lambda(0,N) = 0$. So let $k \geq 1$. We fix a structural reader M with k starts and N states and a hole $W = (w_1, \ldots, w_k)$, with first rejected cuts J_i. The endpoint of w_1 is a rejected state, which differs from $\dagger$, so $N \geq 2$. Full proofs are in Appendices C and D.

7.1 The Problem and the Plan

Let i be a coordinate with an immortal endpoint. After the cut J_i its run stays in rejected states, and no cut of the tail is the target of an admitted pair. With one word this makes the tail useless, and it can be cut off near J_i. With several words the tail still matters in the channels where i is the *host*: the rest of another word w_j after a cut $b \le J_j$ may be a prefix of the rest of w_i after a cut a with an admitted pair of states, unless w_i ends too early or its text deviates. The tail of w_i may be exactly what makes the texts deviate. Cutting it off, or deleting a loop from it, can therefore create an admitted pair, and how long the tail must be can depend on the other words, whose lengths are not bounded at this stage.

The plan is to make the tails *periodic* first, and to shorten them afterwards.

1. *Rays.* We replace the tail of w_i, from a cut $c_i \ge J_i$ on, by a periodic continuation τ_i^ω, where τ_i is a factor of w_i that is a loop of the run. The infinite word $w_i[:c_i]\tau_i^\omega$ is a *ray* (Section 7.2).
2. *No new comparisons with the other words.* The factor τ_i is chosen so that repeating it creates no comparison with the words that are not replaced (Section 7.3). Which words are replaced is decided by measuring how far each tail is from being periodic, at a suitable scale (Section 7.4).
3. *Compatible rays.* Two rays can agree forever from some cuts on. Whether the corresponding pair of states is compared then depends only on the lengths at which the rays are cut off, through a difference constraint. We choose the rays so that these constraints have a solution (Section 7.5). This yields a hole whose immortal tails consist of a bounded number of periodic stretches and a bounded number of other letters.
4. *Compression.* Long periodic stretches are shortened by deleting blocks of periods, in all stretches with the same period pattern at once (Section 7.6).

7.2 Rays and Difference Constraints

▶ **Definition 7.1** (Rays). *A* ray *is an ultimately periodic infinite word $\Omega = u\tau^\omega$, with τ nonempty. Its* period $\mathrm{per}(\Omega)$ *is the least $p \ge 1$ such that $\Omega[x+p] = \Omega[x]$ for all sufficiently large positions x, and its* onset $\mathrm{on}(\Omega)$ *is the least position from which this holds. We write $\Omega[:L]$ for its prefix of length L and $\Omega[a:]$ for its infinite rest after the cut a.*

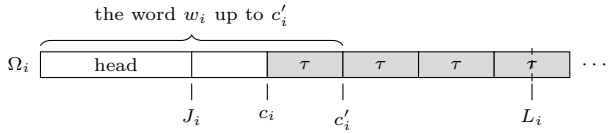
For example, $00(10)^\omega = 0\,(01)^\omega$ is a ray with period 2 and onset 1. The presentation $u\tau^\omega$ of a ray is not unique, but its period and its onset are.

A ray is read by the reader like a word, from the start state of its coordinate; we consider rays whose runs never reach †. Two rays that agree on a long stretch agree forever:

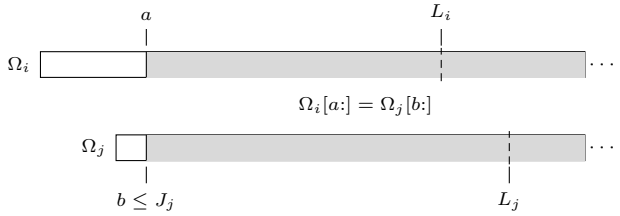
▶ **Lemma 7.2** (Comparisons of rays). *Let Ω and Ω' be rays and a, b cuts.*

1. *If $\Omega[a:]$ and $\Omega'[b:]$ agree on their first $\max(\mathrm{on}(\Omega), \mathrm{on}(\Omega')) + \mathrm{per}(\Omega) + \mathrm{per}(\Omega')$ letters, then $\Omega[a:] = \Omega'[b:]$.*
2. *If $\Omega[a:] = \Omega'[b:]$ and $b < \mathrm{on}(\Omega')$, then $\mathrm{on}(\Omega) - a = \mathrm{on}(\Omega') - b$ and $\mathrm{per}(\Omega) = \mathrm{per}(\Omega')$.*

Now let rays Ω_i be given for the coordinates i of a set I, each with the first rejected cut J_i, and consider the prefixes $\Omega_i[:L_i]$ for lengths L_i that exceed J_i by more than the bound of Lemma 7.2(1). An admitted pair on the channel from i to j has a target cut $b \le J_j$, so the rest of the target is long and the two rays agree forever from the two cuts. Conversely, if they agree forever, the prefixes are compared at these cuts exactly when the rest of the target is not longer than the rest of the host (Figure 9).



(a) the ray $\Omega_i = w_i[:c_i]\,\tau^\omega$ of a factor $\tau = w_i[c_i{:}c_i']$ and its prefix of length L_i



(b) an admitted pair at (a, b) with equal infinite rests is harmless if $L_j - b > L_i - a$

Figure 9 (a) A ray continues a word periodically after its first rejected cut. (b) If the states at the cuts a and b are admitted and the rays agree forever from these cuts, the prefixes of lengths L_i and L_j carry this pair on the channel from i to j unless the rest of the target is longer than the rest of the host. This gives the difference constraint $L_j \ge L_i + (b - a + 1)$.

▶ **Definition 7.3** (Ray system). *The* ray system *of* $(\Omega_i)_{i\in I}$ *has a constraint*

$$L_j \;\ge\; L_i + (b - a + 1)$$

for all $i \neq j$ *in* I *and all cuts* a *of* Ω_i *and* $b \le J_j$ *of* Ω_j *such that the pair of states at* a *and* b *is admitted and* $\Omega_i[a:] = \Omega_j[b:]$. *A* self event *of* Ω_i *is a pair of cuts* $a \le b \le J_i$ *with an admitted pair of states and* $\Omega_i[a:] = \Omega_i[b:]$.

The states along a ray are ultimately periodic, so the constraints whose host cut a is smaller than $\mathrm{on}(\Omega_i) + N\,\mathrm{per}(\Omega_i)$ imply all others, and the target cuts are at most J_j (Appendix C.1). So the ray system is equivalent to a finite system of difference constraints, whose graph has an edge $i \to j$ of weight $b - a + 1$ for each constraint. For example, let $\Omega_1 = (01)^\omega$ and $\Omega_2 = 1(01)^\omega$, and suppose that the states at the cut $a = 1$ of Ω_1 and the cut $b = 0$ of Ω_2 form an admitted pair. The rests after these cuts are both $(10)^\omega$. In the prefixes of lengths L_1 and L_2, the rest of the target has L_2 letters and the rest of the host $L_1 - 1$, so $(1, 0)$ is a comparison exactly when $L_2 \le L_1 - 1$. It is avoided by the constraint $L_2 \ge L_1$, which is $L_2 \ge L_1 + (b - a + 1)$: the target must outlast the host.

▶ **Lemma 7.4** (Prefixes of rays). **1.** *If no ray has a self event and the lengths* L_i *satisfy the ray system and exceed* $J_i + \max_j \mathrm{on}(\Omega_j) + 2\max_j \mathrm{per}(\Omega_j)$, *with the maxima over* $j \in I$ *(a maximum over the empty set is zero, and for* $I = \emptyset$ *the assertion is vacuous), then no channel between two coordinates of* I, *equal or not, of the tuple* $(\Omega_i[:L_i])_{i\in I}$ *contains an admitted pair.*

2. *A system of difference constraints has a solution in natural numbers above any given lower bounds if and only if its graph has no cycle of positive weight.*

So rays turn the unbounded problem of avoiding each other into a finite graph condition. The remaining difficulty is to choose the rays so that there is no positive cycle, and so that they create no comparisons with the words that stay as they are.

7.3 Closed Factors

Let x be the word of an eligible coordinate, with the first rejected cut J, and let $t_1, \dots, t_m$ be words, the *target words*: they stand for the words of the other coordinates, whose rests may be compared with rests of x. We want cuts $J \le c < c'$ of x such that replacing the tail of x after c by repetitions of $\tau = x[c{:}c']$ creates no comparison in which this coordinate is the host and the target is one of the target words. A comparison ends where the target word ends, so what matters at a cut e of the host is which rests of target words end there, and with which state before them. For rests of bounded length this is finite information.

▶ **Definition 7.5** (Suffix nodes and closed factors)**.** *Let $B \ge 0$ be a* window*. For a target word t, $v_B(t)$ denotes its suffix of length B if $|t| \ge B$; if $|t| < B$, there is no such word, and we say that $v_B(t)$ has no occurrences. The* suffix dictionary *$\mathcal{D}_B$ consists of the empty word and the factors of the words $v_B(t_j)$, or of t_j itself if $|t_j| < B$. The* suffix node *of a cut e of x is the pair*

$$\mathrm{node}_B(e) = \big(\sigma,\ q(e - |\sigma|)\big), \qquad \sigma \text{ the longest suffix of } x[{:}e] \text{ in } \mathcal{D}_B,$$

where $q(\cdot)$ is the state of the run of x. A factor $x[c{:}c']$ with $c < c'$ is closed at B *if* $\mathrm{node}_B(c) = \mathrm{node}_B(c')$*, and it is* free at B *if no occurrence of a word $v_B(t_j)$ ends at a cut of $(c, c']$.*

▶ **Example 7.6.** Let $t = 0110$ be the only target word and $B = 2$. Then $v_2(t) = 10$ and $\mathcal{D}_2 = \{\varepsilon, 1, 0, 10\}$. In the host word $x = 101100$, the longest suffix of $x[{:}2] = 10$ in the dictionary is 10, so $\mathrm{node}_2(2) = (10, q(0))$; at the cut 3 it is 1, and $\mathrm{node}_2(3) = (1, q(2))$; at the cut 5 it is again 10, and $\mathrm{node}_2(5) = (10, q(3))$. The factor $x[2{:}5] = 110$ is closed at 2 if $q(0) = q(3)$. It is not free at 2, because an occurrence of $v_2(t) = 10$ ends at the cut 5.

The dictionary has at most $mB(B+1)/2 + 1$ words, whatever the alphabet, so there are at most $N(mB(B+1)/2 + 1)$ suffix nodes. Since the dictionary is closed under factors, the suffix node after one more letter is a function of the suffix node before and of the letter, as in the automaton of Aho and Corasick [1]. Hence along a repetition of a closed factor the suffix nodes repeat (Figure 10).

▶ **Lemma 7.7** (Pumping a closed factor)**.** *Let $x[c{:}c']$ be closed and free at B, with $c \ge J$, and let $x' = x[{:}c]\,(\tau^\omega)[{:}L]$ with $\tau = x[c{:}c']$ and $L \ge c' - c$. Then x' is eligible with the first rejected cut J and an immortal endpoint, and for every target word t_j and all cuts b of t_j and a of x': if $t_j[b{:}]$ is a prefix of $x'[a{:}]$, then it is a prefix of $x[a^\circ{:}]$ for a cut a° of x that carries the same state as the cut a of x'.*

A rest of a target word of length at most B that ends at a cut of x' lies in the dictionary, so it is a suffix of the word of the suffix node, and it ends at a cut of x with the same node. A longer rest ends with $v_B(t_j)$, which by freedom ends only at cuts up to c, where x' and x coincide.

Closed factors exist by the pigeonhole principle; the point is to find one that is free. The occurrences of $v_B(t)$ can be avoided if they are sparse, and they are sparse unless t has a periodic suffix at the scale of B. This is measured by *suffix chains*: for a window b with $1 \le b \le |t|$ let $\mathrm{gap}_t(b)$ be the distance from the end of t to the end of the previous occurrence of $v_b(t)$ in t, if there is one; a chain is a sequence $b_0 < b_1 < \cdots$ of windows with $b_{r+1} = b_r + \mathrm{gap}_t(b_r)$, and its number of *steps* is the number of its members after the first. Along a chain the suffixes $v_{b_r}(t)$ are prefixes of one another.

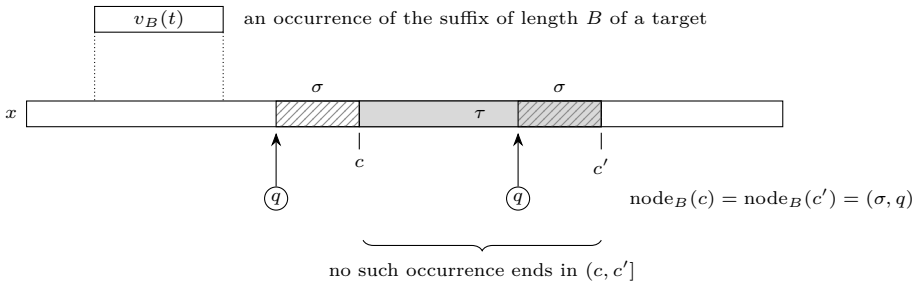


Figure 10 A closed factor. The cuts $c < c'$ of the host word x have the same suffix node: the same longest suffix σ in the suffix dictionary of the targets at the window B, and the same state q before it. If moreover no occurrence of the suffix of length B of a target ends in $(c, c']$, the factor $\tau = x[c{:}c']$ can be repeated any number of times without creating a comparison with a target (Lemma 7.7).

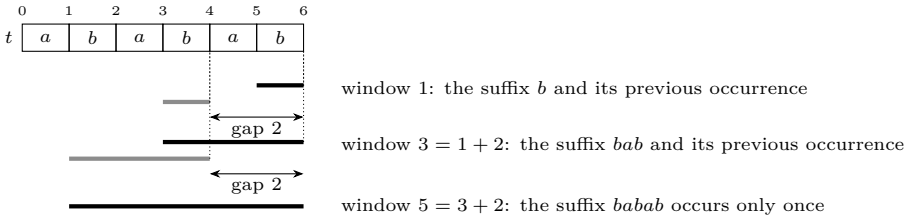


Figure 11 The suffix chain $1 < 3 < 5$ of $t = ababab$ (Example 7.8). At each window the suffix (black) is compared with its previous occurrence (gray); the distance between their ends is the gap, which is added to the window. The suffixes b, bab, $babab$ are prefixes of one another.

▶ **Example 7.8.** Let $t = ababab$. The suffix $v_1(t) = b$ also ends two letters before the end of t, so $\mathrm{gap}_t(1) = 2$ and the chain from the window 1 continues with 3. The suffix $v_3(t) = bab$ again ends two letters before the end, so $\mathrm{gap}_t(3) = 2$, and the next window is 5. The suffix $v_5(t) = babab$ has no other occurrence, and the chain $1 < 3 < 5$ stops after two steps (Figure 11). A periodic suffix thus gives a chain whose gap is the period, with about as many steps as the suffix has periods. In contrast, for $t = aabbab$ the chain from the window 1 is $1 < 3$: the gap of b is again 2, but bab occurs only once. A chain stops at a window whose suffix occurs only at the end of t, and a long chain with small gaps indicates a long periodic suffix.

▶ **Lemma 7.9** (Closed factors free of the target words). *For every number s there are computable functions L_m and B_m, which depend on s and are nondecreasing in the window, with the following property. Let $1 \le b_0 \le b$, and suppose that every suffix chain of every target word that starts at a window of at least b_0 has at most s steps. Then every sequence of $\mathsf{L}_m(b)$ consecutive cuts of x contains cuts $c < c'$ and there is a window B with $b \le B \le \mathsf{B}_m(b)$ such that $x[c{:}c']$ is closed and free at B and $B < c' - c \le \Phi(B)$, where $\Phi(B) = (B+1)\,N\,(mB(B+1)/2+1)$.*

The proof treats the target words one after the other. For one target word it follows a suffix chain as long as the gaps stay below the length needed for the remaining ones; the chain stops within s steps, at a window where the occurrences of the suffix are far apart. Either a long stretch of the given cuts is free at this window, or the occurrences come so regularly that a slightly larger window has no occurrence at all after the first one. The base case is a pigeonhole on the suffix nodes of cuts that are $B+1$ apart, which makes the closed

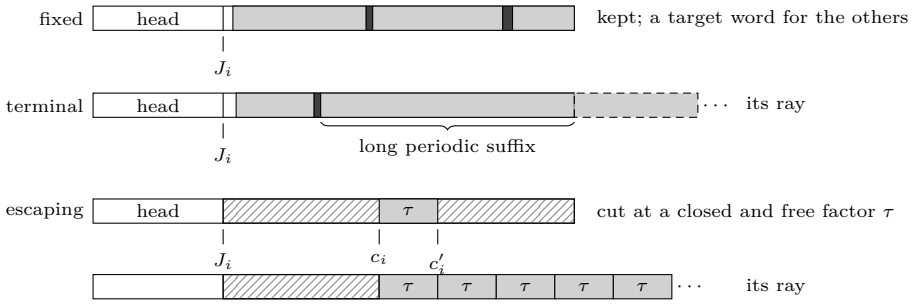


Figure 12 The three types of coordinates at a stable scale. Periodic factors with short periods are gray, exceptional positions black, and text without such a structure is hatched. A fixed coordinate with an immortal endpoint has a tail of few periodic factors and no long periodic suffix; its word is kept. A terminal coordinate ends with a long periodic suffix and is continued by the ray of this suffix. An escaping coordinate is far from periodic at this scale; its tail is cut at one of many closed and free factors τ and continued by τ^ω.

factor longer than its window. The functions L_m are polynomials in the window whose degree depends on s and m; this matters for the complexity bound (Section 9).

7.4 Scales of Periodicity

Which coordinates should be replaced by rays, and at which scale? A tail that already consists of few periodic stretches should be kept: its word is a harmless target word, with short suffix chains. A tail that is far from periodic offers many closed factors with different onsets to choose from. The two cases must be separated by a gap in scale, because the closed factors of one word have to avoid the suffixes of the others. We therefore measure periodicity at a sequence of *scales* $l = 0, 1, 2, \ldots$, each much coarser than the one before. Scale l comes with four numbers, defined by a recursion on l in Appendix C.3 and nondecreasing in l:

- e_l bounds the periods that count as short at scale l;
- T_l bounds the number of positions of a tail that may lie outside its periodic factors;
- Y_l is the length from which a periodic suffix counts as long;
- b_l is the *initial window* of scale l, the smallest window at which closed factors are searched.

The recursion makes each scale dominate the previous ones: for instance $b_{l+1} > (N+1)\, e_l$, and Y_{l+1} exceeds every length that can be assembled from T_l exceptional positions and boundedly many periods e_{l+1} (inequalities (L1)–(L5) of Appendix C.3). The number of periodic factors that we allow is governed by the constant

$$\mathsf{Q} = 4 + 2(k-1)(N-1).$$

Its value is chosen for the count in Section 7.5: a coordinate that is not almost periodic offers at least $\mathsf{Q} - 1$ suitable rays, and each of the other $k - 1$ coordinates can spoil at most $2(N-1)$ of them, so one remains because $\mathsf{Q} - 1 > 2(k-1)(N-1)$.

▶ **Definition 7.10** (Almost periodic tails). *An eligible coordinate i has property F_l if all positions of its rejected tail, except at most T_l, lie in at most $\mathsf{Q}+1$ factors of w_i with periods at most e_l. It has property K_l if it has F_l and the longest suffix of w_i with a period of at most e_l is shorter than Y_l.*

So F_l says that the tail consists of few periodic stretches, up to few exceptional positions, and K_l says in addition that the word does not end with a long periodic stretch. For instance, for three different letters a, b, c, a tail $(ab)^r c\,(ba)^{r'}$ has F_l for every l, with two factors of period 2 and one exceptional position, whatever r and r' are, and it has K_l exactly when $2r' < Y_l$. In contrast, a tail fails F_l when its periodic stretches with periods up to e_l are so many and so short that any $\mathsf{Q}+1$ of them leave more than T_l positions uncovered.

Both properties are inherited from a scale to the next (Lemma C.3); for K_l this uses the periodicity lemma of Fine and Wilf [10]. There are $2k$ properties F_l, K_l of the k coordinates, each of which changes at most once as l grows, so among the scales up to $3(2k+1)$ there is a *stable* scale $l \geq 2$: no coordinate changes its status with respect to F between $l-1$ and l, or with respect to K between $l-2$ and $l-1$. At a stable scale every coordinate is of exactly one of three types (Figure 12).

- *Fixed*: the endpoint is mortal, or the coordinate has K_{l-2}. Its word is kept. Its suffix chains from the initial window b_l of scale l on have at most $\mathsf{Q}+N$ steps (Lemma C.4); for a mortal endpoint this follows from Lemma 5.6.
- *Terminal*: the endpoint is immortal and the coordinate has F_{l-1} but not K_{l-1}. Its word ends with a long suffix of period at most e_{l-1}, and it is continued by the ray of this suffix. The ray contains the whole word w_i.
- *Escaping*: the endpoint is immortal and the coordinate does not have F_l. Its tail is cut at a suitable closed factor and continued periodically.

The number of steps of a suffix chain of a fixed word does not grow with the scale; this is what keeps the recursion of the numbers e_l, T_l, Y_l, b_l, and with it the final bound, elementary.

7.5 Choosing Compatible Rays

We fix a stable scale l. The target words are the words of the fixed coordinates. For a terminal coordinate, the last $pN!$ letters of its periodic suffix, p the period, form a factor that is closed and free at a suitable window (Lemma C.5). For an escaping coordinate v, Lemma 7.9 provides a closed and free factor in every stretch of $\mathsf{L}_k(b_l)$ cuts of the tail, a *candidate*. If the rays of the candidates had at most $\mathsf{Q}+1$ different onsets, the periodic stretches from these onsets would cover the tail up to T_l positions, and v would have F_l. So there are candidates whose rays have at least $\mathsf{Q}+2$ different onsets, of which at most three lie at or before J_v (Lemma C.6); this leaves $\mathsf{Q}-1$ candidate rays with different onsets after J_v.

It remains to exclude positive cycles in the ray system. A constraint $i \to j$ requires the rays to agree forever from cuts a and $b \leq J_j$. If the onset of Ω_j lies after J_j, Lemma 7.2 aligns the onsets, and the stretch $X_j = \Omega_j[J_j{:}\,\mathrm{on}(\Omega_j)]$ of the target between its first rejected cut and its onset is copied into the host just before the onset of the host. Following a cycle backwards from an escaping coordinate v, these stretches grow strictly, so the cycle cannot return to v, unless one of three *forbidden overlaps* of v with the ray of another coordinate u occurs (Definition C.7). In the first, X_v is a suffix of the text of the ray of u before its onset, the onset of u lies after J_u, and X_u is not longer than X_v. In the second, X_v is such a suffix, the onset of u lies at or before J_u, and the two rays have the same periodic part. In the third, the roles are exchanged: X_u is a nonempty suffix of the text of the ray of v before its onset, and X_v is not longer than X_u.

▶ **Lemma 7.11** (Compatible candidates). **1.** *For a terminal ray or an escaping ray chosen earlier, of a coordinate $u \neq v$, at most $2(N-1)$ candidates of v with pairwise different onsets after J_v have a forbidden overlap with it.*

2. *A ray with its onset after its first rejected cut that has no forbidden overlap with any other ray lies on no cycle of the ray system.*

The count in (1) is again an application of Lemma 5.6: different candidates with the same kind of overlap give cuts at or before a first rejected cut that carry different states. Since $\mathsf{Q} - 1 > 2(k-1)(N-1)$, the escaping coordinates can be treated one after the other, each taking one of its $\mathsf{Q} - 1$ candidates that has no forbidden overlap with the at most $k-1$ rays chosen before. Cycles then pass through terminal rays only, and for these the lengths of the original words satisfy all constraints, because W is a hole. So there is no positive cycle. Solving the ray system above sufficiently large lower bounds and applying Lemmas 7.4 and 7.7 gives:

▶ **Proposition 7.12** (Holes with periodic tails)**.** *There are computable numbers D and E_0, depending on k and N, such that every structural reader with a hole has a hole in which the rejected tail of every coordinate with an immortal endpoint is covered, up to E_0 positions in total, by at most $\mathsf{Q} + 3$ maximal periodic factors with periods at most D and lengths at least $2D$.*

7.6 Compressing the Pieces

A *piece* of period d in a tuple is a maximal factor of one word that has period d, length at least $2d$ and a primitive *root*, its first d letters; a word is primitive if it is not a power of a shorter word. Pieces are the runs, or maximal repetitions, of combinatorics on words [16]; we say piece because run already means the run of an automaton. The maximal periodic factors of Proposition 7.12 are pieces. The pieces whose roots are conjugates (rotations) of a primitive word ζ form the *class* of ζ. The *follower* of a piece is the rest of its word after it. For example, the word *abababb* has the piece *ababab* of period 2, with the root *ab* and the follower *b*, and the piece *bb* of period 1, with the root *b* and the empty follower; in *bababa* the whole word is a piece of the class of *ab*, with the root *ba*.

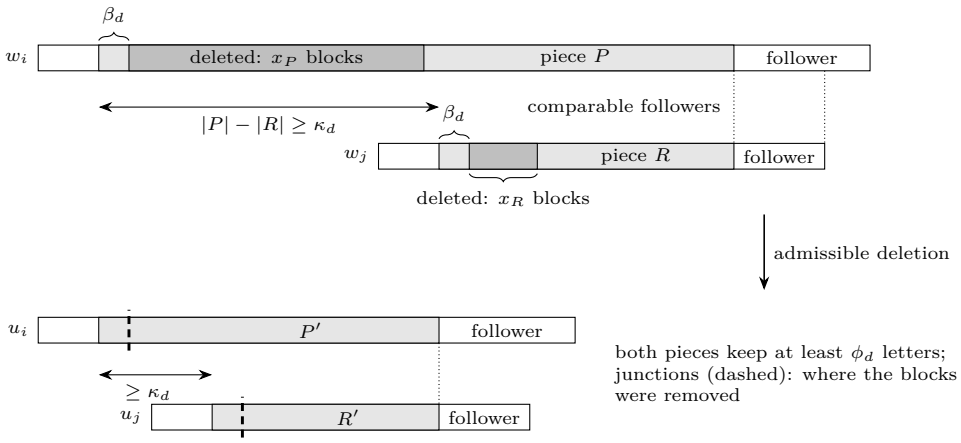


Figure 13 An admissible deletion (schematic). Two pieces P and R of the same class whose followers are comparable can be aligned by a comparison, so they must be shortened consistently. Each loses a number of blocks of m_d letters (darker), starting β_d letters after its start, where the states of the run have become periodic with period m_d. Here P is longer than R by at least the threshold κ_d; it loses more blocks, but remains longer by at least κ_d. Pieces whose lengths differ by less than κ_d lose the same number of blocks.

Inside a piece of period d the states of the run are periodic with period $m_d = d \cdot N!$ after $\alpha_d = Nd$ letters. So a block of m_d letters can be deleted from the interior of a piece without changing the later states. Deleting blocks from one piece only would be visible to comparisons that align this piece with a piece of the same class in another word, or in the same word. Two pieces can be aligned by a comparison that continues beyond them only if their followers are comparable, which leads to the following conditions (Figure 13). They use three lengths:

$$\beta_d = \alpha_d + m_d, \qquad \kappa_d = 2\beta_d + m_d, \qquad \phi_d = 6\beta_d + 2m_d.$$

The deleted blocks begin β_d letters after the start of a piece: this skips the transient α_d of the states and one more block. The number κ_d is the threshold for the difference of the lengths of two pieces with comparable followers. Pieces that differ by less are shortened alike. A difference of at least κ_d is kept, and it ensures that where the shorter piece begins, the states of the longer one are already periodic, so that it does not matter how many blocks each of them has lost. Finally, ϕ_d is the length that a shortened piece keeps; it leaves room for the β_d letters before the deleted blocks and for $5\beta_d + 2m_d$ letters after them, so that both ends of the piece and the letters next to them are not affected. For instance, for $N = 3$ and $d = 2$ we get $\alpha_d = 6$, $m_d = 12$, $\beta_d = 18$, $\kappa_d = 48$ and $\phi_d = 132$.

▶ **Definition 7.13** (Admissible deletion). *An* admissible deletion *of the class of ζ, with $d = |\zeta|$, deletes x_P blocks of m_d letters from every piece P of the class, starting β_d letters after the start of P, such that*

- *a piece that is shortened keeps at least ϕ_d letters;*
- *two pieces with comparable followers whose lengths differ by less than κ_d lose the same number of blocks;*
- *if P and R have comparable followers and R is longer than P by at least κ_d, it remains longer by at least κ_d.*

▶ **Lemma 7.14** (Admissible deletions). *The result of an admissible deletion dominates the original tuple. If the tuple is a hole, the result is a hole with the same endpoints, its first rejected cuts are the images of the original ones, and pieces of other classes with periods at most d and lengths at least $2d$ are not affected.*

The numbers x_P are the solution of another system of difference constraints, and the least solution is small unless long chains of constraints force a piece to stay long. Such a chain is a path in the graph that joins pieces with comparable followers. Comparability of followers is comparability in the forest of the prefix order, and a simple path in the comparability graph of a forest of height h has at most $2^h - 1$ vertices. For the pieces that end at or before the first rejected cut of their word, or lie in words with mortal endpoints, comparable followers begin at cuts with different states (Lemma 5.6), so their height is at most $k(N-1)$. The other pieces end in the rejected tails of immortal coordinates, and their number is bounded by Proposition 7.12. Splitting a path at these pieces gives a bound that is *linear* in their number.

▶ **Lemma 7.15** (The cap). *Suppose that every set of pieces of the class that have pairwise comparable followers and are not among r exceptional pieces has at most h elements. Then there is an admissible deletion after which every piece of the class has length at most*

$$\phi_d + m_d + \big((r+1)\,2^h - 1\big)(\kappa_d + m_d).$$

Starting from the hole of Proposition 7.12, we call the at most $k(\mathsf{Q}+3)$ pieces that cover its immortal tails the *designated* pieces. We compress the class of a designated piece of the largest period, count the positions of the shortened designated pieces of this class among the uncovered positions, and repeat with the remaining designated pieces. After at most $k(\mathsf{Q}+3)$ passes all immortal tails are short.

▶ **Theorem 7.16** (Rejected tails, with the bound)**.** *For $k \geq 1$ and $N \geq 1$, let E_0 and D be as in Proposition 7.12, let $g = k(\mathsf{Q}+3)$, and define*

$$E_{r+1} = E_r + (g-r)\Big(\phi_D + m_D + \big((2(g-r)+E_r+1)\,2^{k(N-1)} - 1\big)(\kappa_D + m_D)\Big).$$

Then Theorem 5.7 holds with $\Lambda(k,N) = E_g \leq \exp_2\big((k+N+20)^{2k+5}\big)$ for $k \geq 1$ and $N \geq 1$, and with $\Lambda(0,N) = 0$.

The closed form is proved in Appendix F.1.

8 The Decision Procedures

We now assemble the decision procedures for disjunctive entailment: over arbitrary trees, where the previous sections apply directly; over regular trees, where the answers are the same; over finite trees, where they differ only through satisfiability; and for the signature $\{\bot, f\}$, which reduces to $\{\bot, f, \top\}$. Theorem 1.1 then follows with Proposition 3.3. Proofs omitted here are in Appendix E.

8.1 Arbitrary Trees

▶ **Theorem 8.1** (Arbitrary trees)**.** *Disjunctive entailment between flat constraints over $\{\bot, f, \top\}$ is decidable over $\mathcal{T}$, uniformly in the number of queries, the arity and the variance.*

Proof. Let Γ be a constraint and $(x_1, y_1), \ldots, (x_k, y_k)$ queries.

1. Decide whether Γ is satisfiable over $\mathcal{T}$ (Lemma 8.2). If it is not, the entailment holds.
2. Otherwise, for each of the 2^k mode vectors $\bar{\mu}$, compute the reader $M_{\bar{\mu}}$ of Definition 4.13 and test all tuples of paths of total length at most $\mathsf{H}(k,N)$, where N is the number of its states, for being holes. The entailment fails if and only if a hole is found. For $k = 0$ there is one mode vector, the empty one, and $\mathsf{H}(0,N) = 0$; the empty tuple is a hole, and the entailment fails, as it must for a satisfiable constraint.

The second step is correct by Theorems 4.14 and 5.1. Whether a tuple is a hole is checked directly from Definition 4.12. ◀

▶ **Lemma 8.2** (Satisfiability)**.** *It is decidable whether a flat constraint over $\{\bot, f, \top\}$ is satisfiable over $\mathcal{T}$.*

Satisfiability is in fact decidable in polynomial time [17, 28]. In our setting it is the question whether the least closure is consistent and contains no forbidden atom (Section 4.2), and for each signed variable the paths of its atoms in the least closure form a regular language accepted by the automaton of lower runs (Appendix A.3).

8.2 Regular Trees

▶ **Theorem 8.3** (Regular trees). *For every flat constraint* Γ *over* $\{\bot, f, \top\}$ *and all queries,*

$$\Gamma \models_{\mathcal{T}_{\text{reg}}} x_1 \le y_1 \vee \cdots \vee x_k \le y_k \iff \Gamma \models_{\mathcal{T}} x_1 \le y_1 \vee \cdots \vee x_k \le y_k.$$

Proof sketch. A regular countermodel is a countermodel, which gives the implication from right to left. Conversely, let ρ be a countermodel over $\mathcal{T}$, with seeds A_i and exclusions B_i as in Section 4.3. By Theorem 4.7 the hull of the seeds is a solution set that contains the seeds and none of the exclusions, so the corresponding assignment is a countermodel. For each signed variable x^s, the paths w with (x^s, w) in the hull form a regular language: those from the least closure do, and by Lemma 4.10 those derived from the seed (p_i, w_i) are the words $v\, w_i[a{:}]\, \sigma$, for a cut a of w_i, a word v with $\mathsf{D}(x^s, v) \cap \mathsf{U}(p_i, w_i[{:}a]) \neq \varnothing$ and an arbitrary word σ. A tree whose two cones are regular languages has finitely many subtrees, because the subtree at a position is determined by the residuals of the two cones and the polarity of the position (Lemma E.1). ◀

So the hull of the seeds, the least countermodel with the given witnesses, is a regular tree in every component, and it is computable from a hole.

8.3 Finite Trees

Over finite trees a constraint can be unsatisfiable although it is satisfiable over $\mathcal{T}$; for a unary constructor, $x \le f(x)$, $f(x) \le x$ is an example. This is the only difference.

▶ **Theorem 8.4** (Finite trees). *For every flat constraint* Γ *over* $\{\bot, f, \top\}$ *and all queries,* $\Gamma \models_{\mathcal{T}_{\text{fin}}} x_1 \le y_1 \vee \cdots \vee x_k \le y_k$ *holds if and only if* Γ *is unsatisfiable over* $\mathcal{T}_{\text{fin}}$ *or* $\Gamma \models_{\mathcal{T}} x_1 \le y_1 \vee \cdots \vee x_k \le y_k$*. Satisfiability over* $\mathcal{T}_{\text{fin}}$ *is decidable, and hence so is disjunctive entailment over* $\mathcal{T}_{\text{fin}}$*.*

The nontrivial direction is to turn a countermodel over $\mathcal{T}$ into a finite one when Γ has some finite solution. A countermodel is described by finitely many *observations*: the k seeds belong to its set of atoms and the k exclusions do not. Each observation is equivalent to a bound of a variable by a finite tree. For instance, for a covariant constructor, $(x^+, w) \in S_\rho$ says that $\rho(x)$ lies above the finite tree that has f along the path w, $\top$ at its end and $\bot$ beside it, and $(x^+, w) \notin S_\rho$ says that $\rho(x)$ lies below the finite tree that has f along w and at its end, and $\top$ everywhere else (Lemma E.2). The theorem then follows from a pinning property of finite solutions:

▶ **Lemma 8.5** (Pinning). *Suppose that* Γ *has a solution over* $\mathcal{T}_{\text{fin}}$ *and that some solution of* Γ *over* $\mathcal{T}$ *satisfies finitely many bounds* $g \le x$ *and* $x \le g'$ *of variables by finite trees. Then some solution of* Γ *over* $\mathcal{T}_{\text{fin}}$ *satisfies these bounds.*

Its proof (Appendix E.2) first treats a covariant constructor. It compiles the bounds into flat literals on fresh variables, which only hang finite trees below or above existing variables, and shows that they create no *cycle clash*, the obstruction to finite satisfiability of [46]. With variance, it translates the constraint into a covariant constraint on the signed variables and makes the finite solution obtained there symmetric under the exchange of the two signs, by a median of trees.

8.4 The Signature $\{\bot, f\}$

Let f be covariant. The trees over $\{\bot, f\}$ are the trees over $\{\bot, f, \top\}$ without $\top$, with the same order, so a constraint over $\{\bot, f\}$ can be read over $\{\bot, f, \top\}$; but it may then have additional solutions, in which $\top$ occurs. A single *guard* variable excludes them.

▶ **Lemma 8.6** (Guard). *Let f be covariant. If $u \leq f(u, \dots, u)$, then $\top$ occurs neither in u nor in any tree $t \leq u$.*

Proof. Suppose $\top$ occurs in u, and let π be a position of minimal length with $u(\pi) = \top$. Since $u \leq f(u, \dots, u)$ and $u \neq \top$, either $u = \bot$, which is impossible, or $u = f(u_1, \dots, u_n)$ with $u_i \leq u$ for every i. Then $\pi = i\pi'$ and $u_i(\pi') = \top$. Let v be the longest prefix of π' that is a position of u. If $v = \pi'$, then $u_i \leq u$ gives $u(\pi') = \top$; otherwise $u(v)$ is a leaf above the label f of $u_i(v)$, so $u(v) = \top$. In both cases $\top$ occurs in u at a position shorter than π. The same argument, applied to a position of $\top$ in a tree $t \leq u$, shows that $\top$ occurs in u. ◀

For a constraint Γ over $\{\bot, f\}$ with variables V, let Γ^g be Γ together with the literals $u \leq f(u, \dots, u)$ and $x \leq u$ for every $x \in V$, where u is a fresh variable.

▶ **Theorem 8.7** (Reduction of $\{\bot, f\}$). *Let f be covariant and $\mathcal{C}$ one of $\mathcal{T}$, $\mathcal{T}_{\mathrm{reg}}$, $\mathcal{T}_{\mathrm{fin}}$. A constraint Γ over $\{\bot, f\}$ entails $x_1 \leq y_1 \vee \dots \vee x_k \leq y_k$ over the trees of $\mathcal{C}$ without $\top$ if and only if Γ^g entails it over $\mathcal{C}$ as a constraint over $\{\bot, f, \top\}$.*

Proof. By Lemma 8.6, the restriction of a solution of Γ^g to V is a solution of Γ without $\top$. Conversely, a solution of Γ in trees of $\mathcal{C}$ without $\top$ extends to a solution of Γ^g in $\mathcal{C}$: over $\mathcal{T}$ and $\mathcal{T}_{\mathrm{reg}}$ let u be the tree f^ω with f at every path (for $n = 0$, the tree f), which lies above every tree without $\top$ and satisfies $u = f(u, \dots, u)$; over $\mathcal{T}_{\mathrm{fin}}$ let u be the complete tree of a depth d that exceeds the lengths of all positions of the finitely many values, with f at all paths shorter than d and $\bot$ at the paths of length d. The queries do not involve u, so the two entailments have the same countermodels up to the value of u. ◀

The guard is needed: at arity 0, the inequality $x \leq f$ holds for both trees $\bot$ and f over $\{\bot, f\}$, but it fails for $x = \top$.

8.5 Proof of the Main Theorem

Proof of Theorem 1.1. Disjunctive entailment between flat constraints over $\{\bot, f, \top\}$ is decidable over $\mathcal{T}$ by Theorem 8.1, over $\mathcal{T}_{\mathrm{reg}}$ by Theorem 8.3, and over $\mathcal{T}_{\mathrm{fin}}$ by Theorem 8.4. Each of the three classes contains $\bot$ and $\top$ and is closed under the constructor, so Proposition 3.3 decides the Boolean combinations of Σ_1 and Π_1 sentences over each class, and its second statement together with Theorem 8.3 shows that truth over $\mathcal{T}_{\mathrm{reg}}$ and over $\mathcal{T}$ agree. For $\{\bot, f\}$ with a covariant constructor, the trees without $\top$ of each class contain $\bot$ and are closed under the constructor, Lemma 3.1 holds for terms without $\top$, and Theorem 8.7 decides the resulting disjunctive entailments; regular and arbitrary trees agree again, because they agree for the guarded constraints. All steps are uniform in the arity and the variance. ◀

9 Complexity

The decision procedures search for holes up to the bound of Theorem 5.1. This section estimates the bound, in the size of the reader and in the size of the sentence, and compares it with the known lower bounds. Recall that $\exp_0(x) = x$ and $\exp_{h+1}(x) = 2^{\exp_h(x)}$.

9.1 An Elementary Upper Bound

▶ **Theorem 9.1** (Elementary bounds). *For all $k \geq 0$ and $N \geq 1$,*

$$\mathsf{J}(k,N) \leq \exp_2\big((k+N+16)^4\big), \qquad \Lambda(k,N) \leq \exp_2\big((k+N+20)^{2k+5}\big),$$
$$\mathsf{H}(k,N) \leq \exp_4\big((k+N+20)^{2k+6}\big).$$

The bounds do not depend on the size of the alphabet.

Proof. The inequalities are proved in Appendix F.1, by estimating the recurrences of Theorems 6.9 and 7.16 and the composition (6). ◀

In outline: for J, the recurrence $\nu_{d+1} \leq 4EF^2\nu_d^2$ of Theorem 6.9 squares its value kN times, so $\log_2 \nu_{kN}$ is at most 2^{kN} times a polynomial in k and N, and $\mathsf{J} = kX^{kN+1}$ is doubly exponential. For Λ, one shows by induction on the scale that the logarithms of all scale functions of scale l are bounded by a common value H_l with $H_{l+1} \leq 2^\ell H_l$, where $\ell = \ell(k,N) \leq (k+N+10)(\mathsf{Q}+N+3)^k + 5$: by Lemma 7.9 the search length L_k is bounded by a polynomial in the window of degree $2^{(k+N+10)(\mathsf{Q}+N+3)^k}$, because the chain budget is the constant $\mathsf{Q}+N$, and the other scale functions are polynomials of bounded degree in L_k, $N!$ and the functions of the previous two scales. After the $3(2k+1)+1$ scales, the logarithms of the period bound D and of E_0 are at most $2^{O(k\ell)}$. Each compression pass replaces E by a value that is linear in E, with a coefficient of the form $2^{k(N-1)} \cdot \mathrm{poly}(D, N!)$ (Theorem 7.16), so the $k(\mathsf{Q}+3)$ passes multiply the logarithm by a factor polynomial in k and N. Finally, $\mathsf{H}(k,N) = \mathsf{J}(k, k(N+\Lambda(k,N)+1)+1)$ applies the bound on fresh tuples to a table whose size is about $k\Lambda(k,N)$.

The innermost argument $(k+N+20)^{2k+5}$ is a polynomial in N for each fixed k, but not a polynomial in k and N jointly. So, for each fixed number k of starts, the bounds on J and Λ are doubly exponential in N and the bound on H is fourfold exponential in N. Uniformly in k and N, since $(k+N+20)^{2k+6} = 2^{(2k+6)\log_2(k+N+20)}$, the estimates give $\Lambda(k,N) \leq \exp_3(O(k\log(k+N+20)))$ and $\mathsf{H}(k,N) \leq \exp_5(O(k\log(k+N+20)))$, while $\mathsf{J}(k,N) \leq \exp_2((k+N+16)^4)$ is doubly exponential in a polynomial in k and N. These are upper bounds; we do not know how far they are from the truth.

Three counts in the proof of Theorem 5.7 keep the bound elementary; each replaces a cruder count that would give a tower of exponentials whose height grows with k or N.

- *Suffix nodes instead of windows.* A closed factor is found by a pigeonhole on suffix nodes (Definition 7.5), of which there are polynomially many in the window B. A pigeonhole on the last B letters before a cut would cost $|A|^B$ at each of the nested uses in Lemma 7.9, and would make the bound depend on the alphabet.
- *A constant chain budget.* The suffix chains of a fixed word have at most $\mathsf{Q}+N$ steps from the initial window of the scale on (Lemma C.4). The obvious bound on the number of steps, by the length of the non-periodic part of the tail, grows from scale to scale, and the number of iterations in Lemma 7.9 with it.
- *A linear cap.* In a compression pass, the simple-path bound 2^h is applied only to the pieces whose number of pairwise comparable members is bounded by states; the pieces in the rejected tails are counted linearly (Lemma 7.15). Applying the bound 2^h to all pieces would add one exponential in each of the $k(\mathsf{Q}+3)$ passes.

▶ **Corollary 9.2** (Sentences). *Let φ be a Boolean combination of Σ_1 and Π_1 sentences of size m. Every clause of its normal forms that is realizable over $\mathcal{T}$ is realized by a solution whose witness paths have total length at most $\exp_5(p(m))$ for a fixed polynomial p, and the truth of φ over each of the three classes is decidable in space $\exp_5(p(m))$.*

Proof. A clause has at most m subterms and at most m negated inequalities, so its flat constraint has $v \leq m$ variables, $v \leq m+1$ with the guard of Section 8.4, and $k \leq m$ queries. The readers have at most $2^{4v}+1$ states (Theorem 4.14), and $\exp_4((m+2^{4m+4}+21)^{2m+6}) \leq \exp_5((4m+6)(2m+6))$, because $m+2^{4m+4}+21 \leq 2^{4m+6}$. The search of Theorem 8.1 enumerates the clauses, the mode vectors and the tuples one after the other and stores one tuple at a time; the tests of satisfiability, of finite satisfiability and of the hole conditions need space polynomial in the size of the reader and the tuple. ◀

The bound is far from practical. It is a bound on a brute-force search, obtained from shortening arguments that were not designed for efficiency.

9.2 Lower Bounds

▶ **Proposition 9.3** (Hardness). *For a covariant constructor of every fixed arity $n \geq 2$, the Σ_1 theory over $\{\bot, f, \top\}$ is PSPACE-hard over each of the three classes of trees, already for plain entailment.*

Proof. Plain entailment is a Π_1 sentence (Section 2.4), and it is PSPACE-hard: for $n = 2$ over infinite trees by Henglein and Rehof [15], and for every $n \geq 2$ and each of the three classes by [46], by a reduction from the universality of nondeterministic finite automata. ◀

For the arities 0 and 1, to which Proposition 9.3 does not apply, Boolean combinations of sentences are still NP-hard, since propositional satisfiability reduces to them as soon as two distinct trees, such as $\bot$ and $\top$, are definable; for $n \geq 2$ this is subsumed by the proposition.

Long witnesses. Shortest holes can be long. For one query and a covariant constructor of arity $n \geq 2$, there are constraints with $O(\beta)$ variables that have finite solutions and whose shortest witness paths have length exponential in β, over each of the three classes [46]: the constraint lets an automaton accept every word except the extensions of the record of a binary counter. With two queries there is a second, different mechanism, which uses the channels between the two words.

▶ **Example 9.4** (Witnesses forced by the least common multiple). For $r \geq 1$ moduli $p_1, \ldots, p_r \geq 1$ with least common multiple L, there is a structural reader with two starts and $O(p_1+\cdots+p_r)$ states, over the alphabet $\{1, 2\}$, whose holes are exactly the tuples

$$W_t = \big((1^{tL}2)^r,\ 1^{tL}2\big), \qquad t \geq 1,$$

so its shortest hole has total length $(r+1)(L+1)$. The first word passes r stages; stage j counts a block of letters 1 modulo p_j and lets the word proceed only if the block length is divisible by p_j. The admitted pairs are the pairs of a state at the beginning of a block and a state inside a block, in either order and in both words, which forces all blocks to have the same length. For a binary covariant constructor, there is a constraint with $O(p_1+\cdots+p_r)$ variables and literals and two queries that has finite solutions and whose countermodels, over each of the three classes, have exactly these tuples as their tuples of witness paths; so the same lower bound holds for the witnesses of countermodels. With the first r primes as moduli, $L = 2^{\Theta(r \log r)}$ while the constraint has $O(r^2 \log r)$ variables. Appendix F.2 gives the construction and the proofs.

9.3 The Gap

Between PSPACE-hardness and the upper bound of Corollary 9.2 there are five exponentials. Already for plain entailment the exact complexity is open, between PSPACE and 2-EXPSPACE [46]. For disjunctive entailment we know no witnesses longer than exponential and no hardness result beyond PSPACE. The upper bound is unlikely to be tight: the two shortening arguments are composed through the cover of Section 5, which, for a fixed number of queries, applies a doubly exponential bound to a table of doubly exponential size, and each of the two bounds is itself obtained by pigeonhole arguments on finite summaries rather than by a search over summaries. For plain and for existential entailment, sharper analyses that do not enumerate words give procedures in 2-EXPSPACE [46, 44]; whether the shortening of tuples admits a similar analysis is open (Section 12).

10 Mechanization

The decision procedures and their correctness are mechanized in Lean 4 [5] with Mathlib [41]. The development defines trees over $\{\bot, f, \top\}$ and over $\{\bot, f\}$, the order with variance, the three classes of trees, terms, quantifier-free formulas, Σ_1 sentences and their Boolean combinations, and it contains

- a decision procedure for Boolean combinations of Σ_1 and Π_1 sentences over each class, for every arity and variance over $\{\bot, f, \top\}$ and for a covariant constructor over $\{\bot, f\}$, with the equivalence of truth over regular and over arbitrary trees (Theorem 1.1);
- decision procedures for disjunctive entailment between flat constraints over the three classes and both signatures, and the statements that satisfiability and (disjunctive) entailment are the sentences of Section 2.4;
- the small-hole theorem for structural readers over an arbitrary finite alphabet (Theorem 5.1), with the computable bound H, and its closed form (Theorem 9.1).

The results of Sections 3–8 and the inequalities of Theorem 9.1 have counterparts in the development; Appendix G gives the correspondence and the points where the formal statements or proofs are arranged differently. The complexity statements of Section 9 other than the numerical bound, that is, Corollary 9.2, Proposition 9.3 and Example 9.4, are not mechanized.

Each decision procedure is a computable definition together with a theorem that it returns *true* exactly when its proposition holds. The procedures have no hypotheses beyond a finite type of variables with decidable equality: satisfiability is tested inside them, the number of queries may be zero, and the arity may be zero. Decidability comes from bounded searches. Classical reasoning is used in the proofs of correctness and of the bounds, where it is harmless, and not to choose answers. The procedures are executable in principle and useless in practice, because of the size of the bound; they establish decidability.

The statements one has to trust are the definitions of trees and of the order, of the syntax and truth of sentences, of the classes of trees, of disjunctive entailment, and the statements of the main theorems, which are collected in one file. The kernel checks everything else. The development uses no axioms beyond the three standard ones of Lean and Mathlib (propositional extensionality, quotient soundness and choice), no unfinished proofs, and no native evaluation; a file of the development checks the axioms of every public declaration and that every decision procedure compiles to executable code. Examples show that the definitions are not vacuous: they include the entailments of Example 2.6 and (3) together

with the failure of their single disjuncts, sentences that separate finite from regular trees, a reader compiled from an entailment with two queries, with a hole, and a minimal fresh tuple.

The chain of shortening arguments is long, and several of its steps are of a kind in which informal proofs are fragile: the transport of comparisons through deletions has many boundary cases, with cuts at the ends of deleted factors and empty rests. Both deletion arguments are therefore built on one small library for deleting a set of positions from a word and following cuts and runs through the deletion (Appendices B.4 and D).

The development has about 27,500 lines. It is available at

`https://github.com/noti0na1/deci-sigma1`.

11 Related Work

Satisfiability and entailment. Subtype constraints over trees were introduced for type inference with subtyping [21, 12]. For non-structural subtyping, satisfiability is decidable in polynomial time over finite and infinite trees [17, 28]; over finite structural trees it depends on the poset of base types [42, 34, 11, 27]. Entailment was studied for constraint simplification [31, 43, 36, 33]. For structural subtyping it is coNP-complete over finite trees and PSPACE-complete over infinite trees [14, 15]. For non-structural subtyping, Henglein and Rehof proved PSPACE-hardness [15]; Niehren and Priesnitz characterized entailment by automata with comparisons between suffixes of paths, proved PSPACE-completeness for a fragment and decided the unary case [25, 26]; decidability, TLCA Problem 16 [35], was proved in [46]. Existential entailment, which underlies the subsumption of constrained types [43], is decided in [44]. The present paper uses the leaf cones of [44] for the reduction to automata and follows [46] in reducing decidability to a bound on shortest holes; the readers with several starts, the treatment of rejected tails and the simultaneous shortening of tuples are new.

The first-order theory. Su, Aiken, Niehren, Priesnitz and Treinen proved that the first-order theory of non-structural subtyping is undecidable, for finite and for infinite trees, and decidable when all constructors are unary or nullary [39, 40]. Their sentences have several quantifier alternations; undecidability already holds for the prefix $\exists\forall\exists$, for a binary covariant constructor with or without $\top$, over finite, regular and arbitrary trees [45]. For structural subtyping of non-recursive types the full first-order theory is decidable [18]. For non-structural subtyping with one binary covariant constructor, our result places the boundary: the Σ_1 level is decidable, the Σ_3 level is not, and the level in between is open except for the existential entailments of [44]. For unary and nullary constructors the whole theory is decidable. Su [38] reduces first-order questions to the emptiness of constrained tree automata, which is undecidable in general; our readers are a decidable special case in which the tests compare rests of words.

Ordering constraints over feature trees. For feature trees ordered by information, entailment and the existential fragment of the first-order theory are well understood: entailment with existential quantifiers is PSPACE-complete, and the full first-order theory is undecidable [23, 24, 22]. Feature trees have an open alphabet of features, so a countermodel can always use a fresh feature. A ranked signature offers no fresh direction, and this is the source of the comparisons between different witness paths in our holes.

Word equations and combinatorics on words. Niehren and Priesnitz observed that entailment involves word equations of a restricted kind [26]. The existential theory of words

with concatenation is decidable by Makanin's algorithm [20, 30, 37], but the conditions of a hole quantify universally over the cuts of the unknown words, and theories of words with a quantifier alternation are undecidable [8]. Our proof uses elementary combinatorics on words: the periodicity lemma of Fine and Wilf [10, 19], squares and maximal repetitions [4, 16], and an automaton of suffixes in the style of Aho and Corasick [1]. The marker hierarchy of Section 6 is a parsing of several words by their own suffixes, in the spirit of the hierarchical decompositions used for plain entailment [46].

Other approaches to subtyping. Algebraic subtyping [7, 6, 29] changes the model of types so that constraints can be solved and types simplified by automata-theoretic means, and semantic subtyping [3, 13] interprets types as sets, with decidable subtyping in the presence of Boolean connectives. These systems have union, intersection and often negation *types*; our sentences have disjunction and negation of *constraints* over the classical tree model with $\bot$, $\top$ and one constructor.

12 Future Work

The signature $\{\bot, f\}$ with contravariance. Theorem 1.1 covers $\{\bot, f\}$ only for a covariant constructor. The guard of Lemma 8.6 relies on covariance, and with a contravariant argument the readers need an additional parity condition on the witness paths, under which the rejected states are not closed under transitions. Over arbitrary and regular trees, plain entailment in this setting, and disjunctive entailment when all arguments are contravariant, can be decided by guarded variants of the present method, which we do not include here; these variants do not settle finite trees. Mixed variance with at least two queries, including function types without $\top$, is open.

More constructors. Type languages have several constructors, and base types. With several constructors a witness path records constructors together with argument positions, and labels can clash in more ways. It is open which comparisons between rests of paths replace our channels, and whether a small-hole theorem holds for the resulting automata.

The Σ_2 level. The $\exists\forall\exists$ fragment is undecidable for a binary covariant constructor [45], and the Σ_1 level is decidable. Sentences with one quantifier alternation are open in general. Existential entailment [44] is the special case $\forall \bar{x}.\, \neg \bigwedge \Gamma \vee \exists \bar{z}.\, \bigwedge \Gamma'$; allowing negated constraints in Γ', or a disjunction of existential conclusions, combines the difficulties of that paper with those of this one.

Complexity. The gap between PSPACE and five exponentials should be narrowed from both sides. On the upper side, the cover construction composes two bounds that are doubly exponential for a fixed number of queries, and a direct argument for holes, or a search over finite summaries of tuples in place of the enumeration of tuples, could remove several exponentials. On the lower side, the question is whether the channels between different words can force witnesses that are longer than exponential, or encode computations beyond polynomial space.

Algorithms. The procedures of this paper are bounded searches. A practical procedure for small numbers of queries, for instance for the simplification of constraints under case distinc-

tions, would need a characterization of entailed disjunctions that avoids the enumeration, in the way the closure rules for satisfiability avoid the enumeration of solutions.

13 Conclusion

We have shown that the Σ_1 theory of non-structural subtyping is decidable for one type constructor of any arity and variance with $\bot$ and $\top$, over arbitrary, regular and finite trees, and for a covariant constructor with $\bot$ only. Equivalently, it is decidable whether a constraint entails a disjunction of inequalities. For a binary covariant constructor, where the $\exists\forall\exists$ fragment is undecidable, this locates the boundary of decidability in the quantifier hierarchy between the first and the third level.

The proof reduces a countermodel with k failed inequalities to a tuple of k words that avoids forbidden pairs of states wherever the rest of one word is a prefix of the rest of another, and bounds the length of a shortest such tuple. The two halves of the bound treat two different phenomena. Words that can be continued forever must be prevented from running into each other, which leads to periodic continuations, difference constraints between their lengths, and the compression of their periodic stretches. After the tails are bounded, the cover turns the hole into a fresh tuple, which is shortened by deleting the same repeated blocks from all its words along a hierarchy of markers. The resulting bound is elementary, and the whole argument, from the semantics of trees to the closed form of the bound, is checked in Lean.

Use of LLM-Based Tools

Tools based on large language models (LLMs) assisted proof attempts, checks of arguments, exploratory programs, and the formalization. Kernel-checked proofs support the decision procedures and the bound on shortest holes (Section 10). Search results were treated as evidence, not as proofs of universal claims.

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A Proofs for Section 4: The Reduction to Holes

A.1 Leaf Cones

We use the following property of trees: for a tree t and a path w, at most one prefix of w is a leaf of t, and if none is, then w is a position of t labeled f. Indeed, by induction on its

prefixes, a prefix of w is a position as long as no shorter prefix is a leaf, since a position that is not a leaf has all its children, and a leaf has none.

Proof of Theorem 4.2. (1) The cones are extension closed by definition, and disjoint because a path has at most one prefix that is a leaf. Conversely, let L^+, L^- be disjoint extension-closed sets of paths. Let D be the set of paths none of whose proper prefixes lies in $L^+ \cup L^-$, and define t on D as follows: $t(w) = f$ if $w \notin L^+ \cup L^-$; if $w \in L^s$, then $t(w)$ is the label of a leaf of sign s at polarity $\mathrm{pol}(w)$, that is, $\top$ if $s = +$ and $\mathrm{pol}(w) = 0$ or $s = -$ and $\mathrm{pol}(w) = 1$, and $\bot$ otherwise. The set D contains ε and is closed under prefixes; a position labeled f has all its children in D, and a position in $L^+ \cup L^-$ has none. So t is a tree, and its leaves of sign s are the elements of L^s without a proper prefix in $L^+ \cup L^-$. A path w lies in L^s exactly when its shortest prefix in $L^+ \cup L^-$ exists and lies in L^s, because both sets are extension closed and they are disjoint; hence $C^s(t) = L^s$. A tree is determined by its cones: its leaves are the minimal elements of the union of the cones, their labels are determined by their signs and polarities, and its other positions are the paths without a prefix in the union.

(2) Let $t \leq u$ and $w \in C^+(t)$, with a prefix v of w that is a leaf of t of sign $+$. Let v' be the longest prefix of v that is a position of u; it is a position of both trees. If $v' = v$ and $\mathrm{pol}(v) = 0$, then $t(v) = \top \leq u(v)$, so $u(v) = \top$; if $\mathrm{pol}(v) = 1$, then $u(v) \leq t(v) = \bot$. In both cases v is a leaf of u of sign $+$. If v' is a proper prefix of v, then $u(v')$ is a leaf and $t(v') = f$; for $\mathrm{pol}(v') = 0$ we get $f \leq u(v')$, so $u(v') = \top$, and for $\mathrm{pol}(v') = 1$ we get $u(v') \leq f$, so $u(v') = \bot$. Again v' is a leaf of u of sign $+$, and a prefix of w. Hence $C^+(t) \subseteq C^+(u)$, and the inclusion $C^-(u) \subseteq C^-(t)$ follows in the same way with the roles of the trees and of the labels exchanged.

Conversely, assume both inclusions and let π be a position of both trees with $\mathrm{pol}(\pi) = 0$; we show $t(\pi) \leq u(\pi)$. If $t(\pi) = \top$, then $\pi \in C^+(t) \subseteq C^+(u)$, so a prefix of π is a leaf of u of sign $+$; since π is a position of u this prefix is π, and $u(\pi) = \top$. If $t(\pi) = f$ and $u(\pi) = \bot$, then $\pi \in C^-(u) \subseteq C^-(t)$, so a prefix of π is a leaf of t, contradicting $t(\pi) = f$. If $t(\pi) = \bot$ there is nothing to show. The case $\mathrm{pol}(\pi) = 1$ is symmetric.

(3) The only prefix of ε is ε, which has polarity 0. For $t = f(t_1, \ldots, t_n)$, the leaves of t are the paths iv with v a leaf of t_i, with the same label and with $\mathrm{pol}(iv) = c(i) + \mathrm{pol}(v)$; so the sign of iv in t is the sign of v in t_i if i is covariant and the opposite sign if i is contravariant. ◀

Proof of Lemma 4.3. By Theorem 4.2(1), $\rho \mapsto S_\rho$ is a bijection from the assignments onto the consistent sets of atoms that are closed under extensions. It remains to show that ρ satisfies a literal if and only if S_ρ is closed under its moves, contains its facts and contains none of its forbidden atoms. For $x \leq y$ this is Theorem 4.2(2), and for $x = \bot$ and $x = \top$ it is the first statement of (3). Consider $f(x_1, \ldots, x_n) \leq y$ and let $t = f(\rho(x_1), \ldots, \rho(x_n))$. By (3), ε lies in no cone of t, and $iw \in C^+(t)$ if and only if $(x_i^{[i]}, w) \in S_\rho$, and $iw \in C^-(t)$ if and only if $(\overline{x_i^{[i]}}, w) \in S_\rho$. So $C^+(t) \subseteq C^+(\rho(y))$ says that S_ρ is closed under the moves $(x_i^{[i]}, w) \to (y^+, iw)$, and $C^-(\rho(y)) \subseteq C^-(t)$ says that $(y^-, \varepsilon) \notin S_\rho$ and that S_ρ is closed under the moves $(y^-, iw) \to (\overline{x_i^{[i]}}, w)$. The literal $y \leq f(x_1, \ldots, x_n)$ is symmetric. ◀

A.2 Derivations

A move changes the signed variable of an atom and replaces a prefix of length at most one of its path by a prefix of length at most one; the set of moves does not depend on the rest of the path. We record this as two *stack properties*, for all atoms and all paths σ:

(M1) if $(p, w) \Rightarrow (q, w')$, then $(p, w\sigma) \Rightarrow (q, w'\sigma)$;

(M2) if $(p, w\sigma) \Rightarrow (q, w'\sigma)$ by a derivation in which every atom has a path of the form $z\sigma$, where the displayed occurrence of σ is never popped, then $(p, w) \Rightarrow (q, w')$.

In a derivation by moves, let ℓ be the path of an atom of minimal length. Then every path of the derivation has the form $z\ell$ with this occurrence of ℓ never popped, because a letter of the path can change only when all letters above it have been popped. We call ℓ a *lowest point* of the derivation.

Proof of Lemma 4.4. (1) is read off Table 2: the two moves of a literal $x \le y$ are dual to each other, and so are the two moves of each constructor literal, for every i and w. For (2), the implication from right to left holds because $(p, u) \vdash (p, u\sigma)$ by extensions. For the other implication we show, by induction on the length of a derivation of (q, v) from (p, u), that $(p, u\sigma) \Rightarrow (q, v)$ where σ consists of the letters of its extension steps in order. If the last step is a move, this follows from the induction hypothesis. If it is an extension $(q, v') \to (q, v'i)$, the induction hypothesis gives $(p, u\sigma') \Rightarrow (q, v')$, and (M1) gives $(p, u\sigma'i) \Rightarrow (q, v'i)$. ◀

Proof of Lemma 4.6. By Theorem 4.2(2), $\rho(x) \not\leq \rho(y)$ if and only if some path lies in $C^+(\rho(x)) \setminus C^+(\rho(y))$ or in $C^-(\rho(y)) \setminus C^-(\rho(x))$. ◀

Runs. For a set P of signed variables let $\mathrm{cl}^{\uparrow}(P)$ be the set of r with $(p, \varepsilon) \Rightarrow (r, \varepsilon)$ for some $p \in P$, and $\mathrm{cl}^{\downarrow}(P)$ the set of r with $(r, \varepsilon) \Rightarrow (p, \varepsilon)$ for some $p \in P$. Define

$$\delta^{\uparrow}(P, i) = \mathrm{cl}^{\uparrow}\big(\{\, r' \mid (r, i) \to (r', \varepsilon) \text{ is a move for some } r \in P \,\}\big),$$
$$\delta^{\downarrow}(P, i) = \mathrm{cl}^{\downarrow}\big(\{\, r' \mid (r', \varepsilon) \to (r, i) \text{ is a move for some } r \in P \,\}\big).$$

The relation $(p, \varepsilon) \Rightarrow (r, \varepsilon)$ between signed variables is computable: it is the least reflexive and transitive relation R that contains the moves between atoms with empty paths and satisfies: if $(p, \varepsilon) \to (p', i)$ and $(r', i) \to (r, \varepsilon)$ are moves and $p'\ R\ r'$, then $p\ R\ r$. This follows by induction on the length of a derivation from (p, ε) to (r, ε): either some intermediate atom has an empty path, and the derivation splits, or it has one step, or its first step pushes a letter i that its last step pops, and (M2) applies to the part in between.

Proof of Lemma 4.9. Let $r \in \mathsf{U}(p, w)$ and let $(r, i) \to (r', \varepsilon)$ be a move and $(r', \varepsilon) \Rightarrow (r'', \varepsilon)$. By (M1), $(p, wi) \Rightarrow (r, i) \to (r', \varepsilon) \Rightarrow (r'', \varepsilon)$, so $r'' \in \mathsf{U}(p, wi)$. Conversely, consider a derivation of (r'', ε) from (p, wi) and the first atom with an empty path in it. The atom before it is (r, j) for a letter j and a move $(r, j) \to (r', \varepsilon)$. All earlier atoms have nonempty paths, so their last letter is never popped and is i; hence $j = i$, and $(p, w) \Rightarrow (r, \varepsilon)$ by (M2). The rest of the derivation shows $r'' \in \mathrm{cl}^{\uparrow}(\{r'\})$. This proves $\mathsf{U}(p, wi) = \delta^{\uparrow}(\mathsf{U}(p, w), i)$, and $\mathsf{U}(p, \varepsilon) = \mathrm{cl}^{\uparrow}(\{p\})$. The statement for D follows in the same way from the last atom with an empty path in a derivation of (p, wi) from (r'', ε), or from the first statement by duality. Finally, by Table 2 a move $(r, i) \to (r', \varepsilon)$ exists only for $r = y^+$ with a literal $y \le f(\dots)$ or $r = y^-$ with a literal $f(\dots) \le y$, and then $r \in \mathsf{Neg}$. ◀

Note that $\delta^{\uparrow}(\varnothing, i) = \varnothing$, so $\mathsf{U}(p, u) = \varnothing$ implies $\mathsf{U}(p, uz) = \varnothing$ for every z. By Lemma 4.4(1),

$$\mathsf{D}(p, w) = \{\, \overline{r} \mid r \in \mathsf{U}(\overline{p}, w) \,\}. \tag{7}$$

Proof of Lemma 4.10. Let a, b be cuts as in the statement, $r \in \mathsf{U}(p, u[{:}a]) \cap \mathsf{D}(q, v[{:}b])$ and $v[b{:}] = u[a{:}]\,\sigma$. Then, by (M1),

$$(p, u) \vdash (p, u\sigma) = (p, u[{:}a]\,v[b{:}]) \Rightarrow (r, v[b{:}]) \Rightarrow (q, v[{:}b]\,v[b{:}]) = (q, v).$$

Conversely, let $(p,u) \vdash (q,v)$. By Lemma 4.4(2), $(p,u\sigma) \Rightarrow (q,v)$ for some σ. Let ℓ be a lowest point of this derivation, reached at an atom (r,ℓ), and write $u\sigma = u'\ell$ and $v = v'\ell$. By (M2), $r \in \mathsf{U}(p,u') \cap \mathsf{D}(q,v')$. If $|u'| > |u|$, then $u' = ujz$ for a letter j and a path z, and $\mathsf{U}(p,u') \neq \varnothing$ implies $\mathsf{U}(p,uj) \neq \varnothing$, hence $\mathsf{U}(p,u) \cap \mathsf{Neg} \neq \varnothing$ by Lemma 4.9, contrary to the hypothesis. So $|u'| \leq |u|$, and $a = |u'|$, $b = |v'|$ are cuts with $u[a{:}]\,\sigma = \ell = v[b{:}]$.

For the last statement, if $r \in \mathsf{U}(p,u) \cap \mathsf{Neg}$ then $(p,u) \Rightarrow (r,\varepsilon)$, a forbidden atom. Conversely, if $(p,u\sigma) \Rightarrow (r,\varepsilon)$ with $r \in \mathsf{Neg}$, then either $\sigma = \varepsilon$ and $r \in \mathsf{U}(p,u)$, or $\mathsf{U}(p,u\sigma) \neq \varnothing$ with $\sigma \neq \varepsilon$, which implies $\mathsf{U}(p,u) \cap \mathsf{Neg} \neq \varnothing$ as above. ◀

A.3 The Least Closure and Satisfiability

Let R_F be the set of signed variables r with $r \in \mathsf{U}(p,\sigma)$ for a fact (p,ε) and a path σ. By Lemma 4.9 it is the least set that contains $\mathsf{U}(p,\varepsilon)$ for every fact (p,ε) and is closed under the maps $\delta^{\uparrow}(\cdot,i)$ applied to its singletons; so it is computable.

▶ **Lemma A.1** (Least closure). *An atom (q,v) lies in* LC *if and only if $\mathsf{D}(q,v[{:}b]) \cap R_F \neq \varnothing$ for some cut b of v. If Γ is satisfiable, then $R_F = \mathsf{Fct}$.*

Proof. $(q,v) \in$ LC if and only if $(p,\sigma) \Rightarrow (q,v)$ for a fact (p,ε) and a path σ, by Lemma 4.4(2). A lowest point ℓ of such a derivation gives $\sigma = \sigma_1\ell$, $v = v'\ell$ and a signed variable in $\mathsf{U}(p,\sigma_1) \cap \mathsf{D}(q,v')$, as in the proof of Lemma 4.10; conversely such a variable gives a derivation with $\sigma = \sigma_1 v[b{:}]$, by (M1). If Γ is satisfiable, LC contains no forbidden atom, so $\mathsf{U}(p,\varepsilon) \cap \mathsf{Neg} = \varnothing$ for every fact, and $\mathsf{U}(p,\sigma_1) = \varnothing$ for $\sigma_1 \neq \varepsilon$ by Lemma 4.9; hence R_F is the union of the sets $\mathsf{U}(p,\varepsilon)$ over the facts, which is Fct. ◀

Proof of Lemma 8.2. By Section 4.2, Γ is satisfiable if and only if LC contains no forbidden atom and no atom together with its opposite. By Lemma A.1, a forbidden atom (r,ε) lies in LC if and only if $\mathsf{D}(r,\varepsilon)$ meets R_F. And LC contains (x^+,v) and (x^-,v) for some v if and only if the product of the two automata that compute $\mathsf{D}(x^+,\cdot)$ and $\mathsf{D}(x^-,\cdot)$ (Lemma 4.9), with one bit for each component that records whether its state has met R_F, can reach a state with both bits set. Both conditions are decidable. ◀

A.4 Realization

▶ **Lemma A.2** (Return). *Let Γ be satisfiable, let $a = (p,u)$ be an atom that derives no forbidden atom, and let $a \vdash b$.*

1. *If $\overline{b} \in$ LC, then $\overline{a} \in$ LC.*
2. *If a' is an atom that derives no forbidden atom and $a' \vdash \overline{b}$, then $a' \vdash \overline{a}$ or $a \vdash \overline{a'}$.*

Proof. Let $b = (q,v)$. By Lemma 4.10, whose hypothesis holds by its last statement, there are cuts α of u and β of v, a path σ with $v[\beta{:}] = u[\alpha{:}]\,\sigma$, and $r \in \mathsf{U}(p,u[{:}\alpha]) \cap \mathsf{D}(q,v[{:}\beta])$. By (7), $\overline{r} \in \mathsf{U}(\overline{q},v[{:}\beta]) \cap \mathsf{D}(\overline{p},u[{:}\alpha])$, so by (M1)

$$\overline{b} = (\overline{q},v[{:}\beta]\,v[\beta{:}]) \Rightarrow (\overline{r},u[\alpha{:}]\,\sigma) \Rightarrow (\overline{p},u\sigma). \tag{8}$$

We use twice that $\mathsf{D}(\overline{p},z) = \varnothing$ for every path z of which u is a proper prefix: otherwise $\mathsf{U}(p,z) \neq \varnothing$ by (7), and then $\mathsf{U}(p,u) \cap \mathsf{Neg} \neq \varnothing$ as in the proof of Lemma 4.10.

(1) Since LC is closed, (8) gives $(\overline{p},u\sigma) \in$ LC. By Lemma A.1 some prefix z of $u\sigma$ has $\mathsf{D}(\overline{p},z) \cap \mathsf{Fct} \neq \varnothing$. By the remark, z is a prefix of u, so $(\overline{p},u) \in$ LC by Lemma A.1.

(2) Let $a' = (p',u')$. By (8), $a' \vdash (\overline{p},u\sigma)$, so by Lemma 4.10 there are cuts α' of u' and γ of $u\sigma$ with $u'[\alpha'{:}] \preceq (u\sigma)[\gamma{:}]$ and $r' \in \mathsf{U}(p',u'[{:}\alpha']) \cap \mathsf{D}(\overline{p},(u\sigma)[{:}\gamma])$. By the remark,

$\gamma \le |u|$. The words $u'[\alpha':]$ and $u[\gamma:]$ are both prefixes of $u[\gamma:]\,\sigma$, so they are comparable. If $u'[\alpha':] \preceq u[\gamma:]$, then $a' \vdash (\overline{p}, u)$ by Lemma 4.10. Otherwise $u[\gamma:] \preceq u'[\alpha':]$ and, by (7), $\overline{r'} \in \mathsf{U}(p, u[:\gamma]) \cap \mathsf{D}(\overline{p'}, u'[:\alpha'])$, so $a \vdash (\overline{p'}, u')$ by Lemma 4.10. ◀

Proof of Theorem 4.7. Let S be a solution set that contains every A_i and no B_i. It is closed and contains the facts, so it contains the hull. Hence no A_i derives a forbidden atom, no B_i lies in LC or is derived from a seed, and since S is consistent the same holds for the atoms $\overline{A_i}$. This proves (R1)–(R3) and that every such solution set contains the hull.

Conversely, assume (R1)–(R3), and let H be the hull. It is closed, as a union of closed sets, and contains the facts. It contains no forbidden atom, by (R1) and because Γ is satisfiable, and no B_i, by (R2) and (R3). Suppose $b, \overline{b} \in H$. They are not both in LC, which is consistent. If $A_j \vdash b$ and $\overline{b} \in$ LC, then $\overline{A_j} \in$ LC by Lemma A.2(1), contradicting (R2). If $A_j \vdash b$ and $A_i \vdash \overline{b}$, then $A_i \vdash \overline{A_j}$ or $A_j \vdash \overline{A_i}$ by Lemma A.2(2), contradicting (R3). So H is consistent, and it is a solution set. ◀

A.5 The Reduction

Proof of Theorem 4.14. There are $2^{2|V|}$ sets of signed variables, hence at most $2^{4|V|} + 1$ states. The dead state is absorbing by definition. Let (P, Q) be rejected, that is, $P \cap \mathsf{Neg} = \varnothing$, and let its successor on a letter be a state (P', Q'). By Lemma 4.9, $P' = \varnothing$. So (P', Q') is rejected, which is (S1), and it is the target of no admitted pair, which is (S2). The functions $\delta^{\uparrow}, \delta^{\downarrow}$ and the sets Neg, Fct are computable, so the reader is.

Let W be a tuple of paths, with seeds $A_i = (p_i, w_i)$ and exclusions $B_i = (p_i', w_i)$. By Lemma 4.9 and induction on a, the state of coordinate i at a cut a is $(\mathsf{U}(p_i, w_i[:a]), \mathsf{D}(p_i', w_i[:a]))$ if the pairs at all cuts up to a are states, and † otherwise. By (7) and Lemma A.1, the run of coordinate i never reaches † if and only if $B_i \notin$ LC and $\overline{A_i} \notin$ LC. Its endpoint is rejected if and only if $\mathsf{U}(p_i, w_i) \cap \mathsf{Neg} = \varnothing$, that is, by Lemma 4.10, if and only if A_i derives no forbidden atom. So all coordinates are eligible exactly when (R1) and (R2) hold. Assume this. A pair of $\mathsf{Ch}_{ij}(W)$ at cuts a of w_i and b of w_j, with $w_j[b:] \preceq w_i[a:]$, is admitted if and only if $\mathsf{U}(p_j, w_j[:b])$ meets $\mathsf{D}(p_i', w_i[:a])$ or $\mathsf{D}(\overline{p_i}, w_i[:a])$. By Lemma 4.10, applied to the seed A_j, such cuts exist if and only if $A_j \vdash B_i$ or $A_j \vdash \overline{A_i}$. So no channel contains an admitted pair exactly when (R3) holds. By Theorem 4.7, W is a hole if and only if some solution set contains the seeds and none of the exclusions. The last statement of the theorem follows with Lemmas 4.3 and 4.6. ◀

B Proofs for Section 6: Fresh Tuples

Throughout, $W = (w_1, \ldots, w_k)$ is a fresh tuple of a table with k starts and N states, and $q_l(c)$ is the state of coordinate l at the cut c.

B.1 Width

▶ **Lemma B.1** (Width). 1. *For every text z and every coordinate l, at most $N-1$ cuts $b < |w_l|$ satisfy $w_l[b:] \preceq z$.*

2. *For every text z, the set* $\mathrm{Pre}(z)$ *of the numbers p with $1 \le p \le |z|$ such that $z[:p]$ is a suffix of some word w_l has at most $k(N-1)$ elements.*

Proof. (1) The rests $w_l[b:]$ in question are prefixes of one text, hence pairwise comparable. By Lemma 5.6, the states at their cuts are pairwise different, and they differ from the

endpoint because w_l is fresh and $b < |w_l|$. (2) If $z[{:}p]$ is a suffix of w_l, then $b = |w_l| - p$ is a cut as in (1), and different p give different b for the same l. ◀

B.2 Markers

An anchor of a text u is a cut at which a marker ends that fits into u; an occurrence of a marker is always a complete occurrence.

Proof of Lemma 6.3. By induction on t we prove (1) for level t, and (2), (3) for the passage from t to $t+1$. At level 1 the markers are single letters, and a letter is a factor of another letter only if they are equal. If an occurrence of a marker lies inside an occurrence of another marker of the same level, the first is a factor of the second, so the texts are equal and the occurrences coincide; this is the second statement of (1). In particular, two markers of one level that end at the same cut of a text have the same text, so "the longest marker ending at e" in Definition 6.1 is the only one, up to the choice of the coordinate.

Assume (1) for level t. Let $l \in \mathrm{Act}_{t+1}$, $\ell = |w_l|$, let $e < \ell$ be the last level-t anchor of w_l before its end and μ the level-t marker ending at e, so that $\sigma_l^{t+1} = w_l[e - |\mu|{:}]$. The occurrence of μ at $[e - |\mu|, e)$ does not lie inside the occurrence of σ_l^t at $[\ell - |\sigma_l^t|, \ell)$, because the two occurrences end at different cuts. Hence $e - |\mu| < \ell - |\sigma_l^t|$, and σ_l^t is a proper suffix of σ_l^{t+1}. Now consider a level-t marker μ' that ends at a cut e' of the text σ_l^{t+1} and fits into it. If $e' \le |\mu|$, its occurrence lies inside that of the prefix μ, so $e' = |\mu|$. If $|\mu| < e' < |\sigma_l^{t+1}|$, then w_l has a level-t anchor strictly between e and ℓ, contradicting the choice of e. So the anchors are $|\mu|$ and the end, which proves (2). For (3), a level-$(t+1)$ marker that ends at a cut of a text has the level-t marker of the same coordinate as a suffix, which ends at the same cut.

For (1) at level $t+1$, let σ_l^{t+1} occur in σ_m^{t+1} at the position o, and let μ_l, μ_m be the level-t markers with which they begin. The two level-t anchors of the text σ_l^{t+1} give level-t anchors $o + |\mu_l|$ and $o + |\sigma_l^{t+1}|$ of the text σ_m^{t+1}, which by (2) are $|\mu_m|$ and $|\sigma_m^{t+1}|$. So the occurrence ends at the end of σ_m^{t+1}, and μ_l and μ_m end at the same cut of σ_m^{t+1}; they are equal by (1) at level t, hence $o = 0$ and the two markers are equal. ◀

We also note that for $t \ge 1$ a coordinate $l \notin \mathrm{Act}_{t+1}$ has no level-t anchor other than its end: if $l \in \mathrm{Act}_t$ this is the definition, and otherwise l left the hierarchy at a level $r < t$, has no level-r anchor before its end, and level-t anchors are level-r anchors by Lemma 6.3(3).

Proof of Lemma 6.4. (1) We show by induction that for $l \in \mathrm{Act}_t$ the set $\mathrm{Pre}(\sigma_l^t)$ of Lemma B.1 has at least t elements; then $t \le k(N-1)$ whenever $\mathrm{Act}_t \ne \varnothing$. For $t = 1$ the marker is a suffix of w_l. For $t+1$, the marker σ_l^{t+1} begins with a level-t marker μ of some coordinate, whose t prefixes that are suffixes of words are prefixes of σ_l^{t+1} of length at most $|\mu|$, and σ_l^{t+1} itself is a longer one.

(2) For $t = 0$ the marker is a letter. Let $t \ge 1$, let σ be the level-$(t+1)$ marker with the occurrence $u[a{:}b]$, and let e be a level-t anchor of u with $a < e \le b$, with a level-t marker μ ending at e. If μ begins at or after a, then $e - a$ is one of the two anchors of the text σ. Otherwise $u[a{:}e]$ is a proper suffix of μ, hence a suffix of a word, and a prefix of σ; so $e - a \in \mathrm{Pre}(\sigma)$, a set of at most $k(N-1)$ elements. ◀

B.3 Parses

Fix a level t between 0 and the top level. Let z be a text that occurs in a text uz after the left context u. For $t \ge 1$, a position p with $1 \le p \le |z|$ is an *inner anchor* of z if a level-t

marker of length at most p ends at the cut p of z, and a *crossing anchor* in the context u if a level-t marker of length greater than p ends at the cut $|u|+p$ of uz.

▶ **Lemma B.2** (Parsing is local). *Let $t \geq 1$.*

1. *The level-t anchors of uz after $|u|$ are the cuts $|u|+p$ with p an inner anchor of z or a crossing anchor in the context u. The inner anchors do not depend on u.*
2. *Every crossing anchor is smaller than every inner anchor.*
3. *Every crossing anchor lies in* $\mathrm{Pre}(z)$.

Hence the lower anchors of two occurrences of z, in arbitrary contexts, agree from the first inner anchor of z on, and each occurrence has at most $k(N-1)$ lower anchors before it.

Proof. (1) is the definition. (2) Let a marker μ of length greater than p end at $|u|+p$ and a marker μ' of length at most p' end at $|u|+p'$, with $p' \leq p$. Then the occurrence of μ' lies inside that of μ, so by Lemma 6.3(1) they coincide, which is impossible since μ' begins inside z and μ before it. (3) The prefix $z[:p]$ is a proper suffix of the marker, which is a suffix of a word. ◀

For $t = 0$ every cut is a lower anchor, in every context.

▶ **Lemma B.3** (Texts determine traversal). *Let (Z, a, b) be an edge at level t, let i be a depth with $a + k(N-1) + 2 \leq i \leq b$, and let z be the text of the last i lower blocks of Z. If z ends at an upper anchor e of a word w_l, then the upper occurrence of w_l that ends at e traverses the edge.*

Proof. Let o_1 be an upper occurrence with parse Z; its last i lower blocks form an occurrence of z whose i lower anchors are the cuts at the depths $i-1, \dots, 0$. Let o_2 be the upper occurrence of w_l ending at e, and consider the occurrence of z that ends at e.

We claim that o_2 begins at or before the first inner anchor p_0 of z, measured in this occurrence; for $t = 0$ we let $p_0 = 1$. Suppose an upper anchor of w_l lies at a position r of z with $0 < r < |z|$, and let σ be a level-$(t+1)$ marker ending there. If σ fits into z, it also ends at the corresponding cut of o_1, which is then an upper anchor strictly inside the upper block o_1, a contradiction; for $t = 0$, where σ is a letter, this is the only case. So σ begins before z. By Lemma 6.3(2), the text σ has no level-t anchor other than the end of its first level-t marker and its own end; an inner anchor of z at a position $r' < r$ would be a third one, since its marker begins after the beginning of σ and ends before its end. So $r \leq p_0$, which proves the claim.

For $t = 0$ the two parses consist of the letters of z. For $t \geq 1$, by Lemma B.2, in the occurrence inside o_1 at most $k(N-1)$ of the i lower anchors are crossing anchors, so z has at least $i - k(N-1)$ inner anchors, and in both occurrences the lower anchors from p_0 on are exactly the inner anchors. By the claim, the lower blocks between them belong to o_2. So the parses of o_1 and o_2 end with the same $i - k(N-1) - 1 > a$ lower blocks. Since the parse of o_1 is Z, the number $\mathrm{lcs}(Y, Z)$ for the parse Y of o_2 is a node on the path of Z greater than a, and no node lies strictly between a and b; so it is at least b. ◀

B.4 Block Deletions

Fix a block deletion at level t, with the edge (Z, a, b) and the depths $i < j$. Let z_λ be the text of the last i lower blocks of Z, $\lambda = |z_\lambda|$, and let F be the text of its lower blocks at the depths $i, \dots, j-1$, $\delta = |F| \geq 1$. For a traversing upper occurrence o of a word w_l, between the cuts $s_o < e_o$, let c_o and $c'_o = c_o + \delta = e_o - \lambda$ be its cuts at the depths j and i. The *site* of

o is the interval of positions $[c_o, c'_o)$; it carries the text F. The edited tuple $U = (u_1, \ldots, u_k)$ is obtained by deleting all sites.

Deleting positions. We use the following notation for the deletion of a set of pairwise disjoint intervals of positions from a word w, with the result u. A cut of w is *retained* if it does not lie strictly inside a deleted interval. The *cut map* im sends a retained cut p of w to the number of retained positions before it, a cut of u; it is monotone, and the two ends of a deleted interval have the same image, a *junction* of u. Every cut a of u has a greatest preimage $\gamma(a)$, which is the right end of a deleted interval if a is a junction. If no junction lies strictly between the cuts $a < a'$ of u, then

$$u[a{:}a'] = w[\gamma(a){:}\gamma(a) + a' - a], \tag{9}$$

and $\gamma(a) + a' - a$ is the left end of the deleted interval of a' if a' is a junction and $\gamma(a')$ otherwise. If the two ends of every deleted interval carry the same state, then the state of u at $\mathrm{im}(p)$ is the state of w at p, for every retained cut p.

Geometry of the sites.

(G1) $s_o < c_o < c'_o < e_o$, since $1 \le i < j < b$ and the parse of o has at least b lower blocks. Sites of different traversing occurrences lie in different upper blocks. So in u_l the junction of o is followed by the λ letters z_λ up to $\mathrm{im}(e_o)$, the cuts strictly between the junction and $\mathrm{im}(e_o)$ are images of cuts strictly inside the upper block o, and two junctions of one word are more than λ apart.

(G2) A word with a traversing occurrence has a level-t anchor before its end, so it is active at level $t + 1$.

(G3) A level-$(t + 1)$ marker ends at e_o, and every such marker has length at most λ: otherwise its occurrence would contain the $i + 1$ lower anchors at the depths $i, \ldots, 0$, and $i + 1 > k(N - 1) + 2$, contradicting Lemma 6.4. For the same reason no site meets the last $|\sigma_l^{t+1}|$ letters of a word w_l with $l \in \mathrm{Act}_{t+1}$, so u_l ends with σ_l^{t+1}.

States. Let ω be the common value $\omega(l, c_o) = \omega(l, c'_o)$ of the equal labels. If $\omega > 0$, the same marker of length ω ends at both cuts, so the ω letters before c_o and before c'_o are equal, and the equal states at the distance ω give

$$w_l[c_o - x{:}c_o] = w_l[c'_o - x{:}c'_o], \qquad q_l(c_o - x) = q_l(c'_o - x) \qquad \text{for } 0 \le x \le \omega. \tag{10}$$

In particular the two ends of every site carry the same state. Hence the states of U at the images of retained cuts are the states of W: the endpoints agree, no new state is visited, and U is fresh, because a cut before the end of u_l is the image of a retained cut before the end of w_l. The tuple U is strictly shorter than W, because the occurrence with the parse Z traverses the edge.

▶ **Lemma B.4** (Anchors are preserved). *For every level r with $1 \le r \le t + 1$ and every coordinate l, the level-r anchors of the text u_l, for the markers of W, are exactly the images* $\mathrm{im}(e)$ *of the retained cuts e of w_l that are level-r anchors of the text w_l.*

Proof. We delete the sites of w_l one at a time, from right to left, and show that each deletion has the stated property. The following three facts about a site $[c, c')$ concern only the prefix $w_l[{:}c']$, which the deletions to its right do not change.

(V1) If $t \geq 1$, a level-t marker ends at c, because c is a lower anchor other than the end of the word.
(V2) For $1 \leq r \leq t$, a level-r marker ends at c if and only if it ends at c', by the equal labels.
(V3) No level-$(t+1)$ marker ends at c or at c', by (G1).

Write the current word as $x = yFv$ with $|y| = c$, and $x' = yv$. Consider an occurrence of a level-r marker μ in x'. If it lies in y or in v it is an occurrence in x ending at a retained cut with the same image, and conversely. If it ends at the junction, it is a suffix of y; by (V2) and (V3) this is the case if and only if μ ends at c and at c' in x. Now let $\mu = SR$ with S a nonempty suffix of y and R a nonempty prefix of v. We claim $|S| \leq \omega$. Otherwise $t \geq 1$, since for $t = 0$ the markers are letters, and by (V1) a level-t marker μ_t of length at most ω ends at c, so its occurrence lies inside that of μ and ends before it. If $r \leq t$, the level-r marker of the coordinate of μ_t is a suffix of μ_t, so a level-r marker occurs inside the occurrence of μ and ends before it, contradicting Lemma 6.3(1). If $r = t+1$, then by Lemma 6.3(2) μ_t is the level-t marker with which μ begins, so $S = \mu_t$ has length at most ω. This proves the claim, and by (10) S is a suffix of yF, so μ ends at the corresponding retained cut of x. Conversely, let μ end at a cut after c' in x and begin before c', and write $\mu = SR$ with S a suffix of yF and R a prefix of v. The same argument, with the level-t marker that ends at c', gives $|S| \leq \omega$, so S is a suffix of y and μ ends at the image of the cut in x'. This covers the occurrences that begin inside the site or before it. ◀

By (G1) and Lemma B.4, no cut strictly between a junction of u_l and the cut λ letters after it is a level-$(t+1)$ anchor of u_l.

▶ **Lemma B.5** (Transfer). *Let a_2 be a cut of u_l such that $u_l[a_2{:}a_2+\lambda] = z_\lambda$ and no junction of u_l lies strictly between a_2 and $a_2+\lambda$. Then a_2 is a junction of u_l.*

Proof. By (9), $w_l[c{:}c+\lambda] = z_\lambda$ for $c = \gamma(a_2)$. By (G3) a level-$(t+1)$ marker of length at most λ is a suffix of z_λ, so $c+\lambda$ is an upper anchor of w_l. By Lemma B.3, which applies since $i \geq a + \mathsf{m}$, the upper occurrence ending at $c+\lambda$ traverses the edge. Its site is $[c-\delta, c)$, so $a_2 = \mathrm{im}(c)$ is a junction. ◀

Proof of Theorem 6.8. It remains to show $\mathsf{Ch}_{hg}(U) \subseteq \mathsf{Ch}_{hg}(W)$ for a host h and a target g. Let a', b' be cuts of u_h and u_g with $T = u_g[b':] \preceq u_h[a':]$, and let $\theta = |T|$. We must find cuts of w_h and w_g with the same states whose rests are compared. A *host junction offset* is a number x with $0 < x < \theta$ such that $a' + x$ is a junction of u_h, and a *target junction offset* one such that $b' + x$ is a junction of u_g.

Step 1: junction offsets correspond. Let x be a target junction offset. By (G1), $x + \lambda \leq \theta$, the target reads z_λ from $b' + x$, and so does the host from $a' + x$. By (G3) a level-$(t+1)$ marker is a suffix of z_λ, so $a' + x + \lambda$ is a level-$(t+1)$ anchor of u_h. If a host junction offset y satisfied $x < y < x + \lambda$, this anchor would lie strictly between a junction and the cut λ letters after it, which is impossible. So Lemma B.5 applies, and x is a host junction offset. Symmetrically, a host junction offset y with $y + \lambda \leq \theta$ is a target junction offset: the target reads z_λ from $b' + y$, the cut $b' + y + \lambda$ is a level-$(t+1)$ anchor of u_g, and no target junction offset lies strictly between y and $y + \lambda$.

Step 2: the matched case. Suppose every host junction offset is a target junction offset, and let $x_1 < \cdots < x_m$ be the junction offsets. By (9), applied between consecutive junctions, $w_g[\gamma(b'){:}]$ is the word obtained from T by inserting F at the offsets $x_1, \ldots, x_m$, and the same word is a prefix of $w_h[\gamma(a'){:}]$. The states at $\gamma(a')$ and $\gamma(b')$ are those of U at a' and b'. This includes the case of an empty T.

Step 3: the unmatched case. Otherwise, by Step 1, there is a host junction offset y with $y + \lambda > \theta$, and it is the only host junction offset that is not a target junction offset, by (G1). We first show that the target has no junction at or after b'. If $g \notin \mathrm{Act}_{t+1}$, the word w_g has no site, by (G2). Let $g \in \mathrm{Act}_{t+1}$ and $\sigma = \sigma_g^{t+1}$, a suffix of u_g by (G3). If $|\sigma| \le \theta$, then σ ends at the cut $a' + \theta$ of u_h, a level-$(t+1)$ anchor strictly between the junction $a' + y$ and the cut λ letters after it, which is impossible. So $|\sigma| > \theta$, and by (G3) no junction of u_g lies at or after b'. Consequently $T = w_g[\tilde{b}{:}]$ for $\tilde{b} = |w_g| - \theta$, with the state of b', there is no target junction offset, and by Step 1 y is the only host junction offset.

Let o be the traversing occurrence of the junction $a' + y$, with the site $[c, c')$, and write $T = SR$ with $|S| = y$; R is not empty. By (9), $S = w_h[c - y{:}c]$, with the state of a' at the cut $c - y$, and $R \preceq w_h[c'{:}]$. If $y \le \omega$, then by (10) $S = w_h[c' - y{:}c']$ and the state at $c' - y$ is that at $c - y$; so $w_g[\tilde{b}{:}] = SR \preceq w_h[c' - y{:}]$ is a comparison of W with the required states.

It remains to exclude $y > \omega$. For $t = 0$, the word w_g is not empty, so $g \in \mathrm{Act}_1$ and $|\sigma_g^1| = 1 \le \theta$, which was excluded above. Let $t \ge 1$. By (V1) a level-t marker μ of length at most $\omega < y$ ends at c; it is a suffix of S, so it occurs in w_g, begins at or after $\tilde{b}$, and ends at the cut $\tilde{b} + y < |w_g|$. So w_g has a level-t anchor before its end, $g \in \mathrm{Act}_{t+1}$, and since $\theta < |\sigma|$ this occurrence of μ lies inside the final occurrence of $\sigma = \sigma_g^{t+1}$, begins after its beginning and ends before its end. This contradicts Lemma 6.3(2). ◀

B.5 The Bound

▶ **Lemma B.6** (Labels are local). *Let (Z, a, b) be an edge at level t and d a depth with $1 \le d \le b - \mathsf{m}$. There are marker texts $\tau_1, \dots, \tau_t$, some possibly undefined, and a map $g_d \colon S \to S$ such that for every traversing upper occurrence o of a word w_l, the label of its cut at depth d is $(\tau_1, \dots, \tau_t, g_d(q_l(c)))$, where c is the cut of o at depth b.*

Proof. Let y_d be the text of the lower blocks of Z at the depths $b - 1, \dots, d$; it is the text between the cuts of o at the depths b and d. A level-r marker with $1 \le r \le t$ that ends at the cut at depth d contains at most $k(N-1) + 2$ lower anchors, by Lemma 6.4 for the levels $r - 1$ and r, because lower anchors are level-$(r-1)$ anchors. The cuts at the depths $d, \dots, b$ are $b - d + 1 > k(N-1) + 2$ lower anchors, so the marker begins at or after c: it is a suffix of y_d. So τ_r is the level-r marker that is a suffix of y_d, and the state before the longest of them is obtained from $q_l(c)$ by reading a prefix of y_d. ◀

Proof of Theorem 6.9. Let W be a minimal fresh tuple. By Lemma 6.4 its top level s is at most $k(N-1)$. Fix $t \le s$. There are at most $(k+1)^t$ possibilities for the texts τ_r of Lemma B.6, since at each level at most one of at most k marker texts ends at a cut, and N^N maps g_d. If an edge (Z, a, b) had $b - a > E$, two of the $b - a - 2\mathsf{m} + 1$ depths between $a + \mathsf{m}$ and $b - \mathsf{m}$ would have the same texts and the same map, and by Lemma B.6 they would form a block deletion, contradicting Theorem 6.8. So every edge has at most E lower blocks.

Let h_t be the number of distinct texts of upper blocks at level t and P_t the number of distinct parses. By Lemma B.2, the parse of an upper block with the text z is determined by z and its set of crossing anchors, a subset of the set $\mathrm{Pre}(z)$ of at most $k(N-1)$ positions; so $P_t \le F\, h_t$. The path of a parse Z has at most $P_t + 2$ nodes, the first is 0, the last is the length of Z, and consecutive nodes are at most E apart; so Z has at most $(P_t + 2)E$ lower blocks. For $t \ge 1$, every upper block of level $t - 1$ is a lower block in the parse of an upper block of level t, hence

$$h_{t-1} \le P_t\,(P_t + 2)\,E \le F h_t\,(F h_t + 2)\,E.$$

At the top level the upper blocks are the nonempty words, so $h_s \le k$. By induction $h_{s-d} \le \nu_d$, and since the ν_d are nondecreasing and $s \le kN$, every upper block of every level has at most $X = (\nu_{kN}F + 2)E$ lower blocks. Blocks of level 0 are letters, so a block of level $t+1$ has at most X^{t+1} letters, and each word, a block of level $s+1$ if it is not empty, has at most X^{kN+1} letters. The closed form is proved in Appendix F.1.

The second statement of Theorem 5.8 follows because a fresh dominator of least total length of a fresh tuple is a minimal fresh tuple. ◀

C Proofs for Section 7: Rejected Tails

Throughout, M is a structural reader with $k \ge 1$ starts and $N \ge 2$ states, which holds for every reader with a start and a hole (Section 7), W is a hole, J_i is the first rejected cut of coordinate i and $q_i(a)$ its state at the cut a. We use the periodicity lemma of Fine and Wilf [10] in the following form: if a word of length at least $p + p'$ has the periods p and p', it has the period $\gcd(p, p')$; consequently, if a factor of length at least $p + g$ of a word with period p has period g, the whole word has period g. A word has *period* p on an interval of positions if letters of the interval at distance p are equal. We also use the following consequence of Lemma 5.6.

▶ **Lemma C.1** (Periodic stretches before the first rejected cut)**.** *Let coordinate i be eligible and let w_i have period $p \ge 1$ on an interval $[s, e)$ with $s \le J_i \le e$. Then $J_i - s < Np$.*

Proof. Otherwise the $N+1$ cuts $s, s+p, \dots, s+Np$ are at most J_i, and two of them, $a < a'$, carry the same state. The texts $w_i[a{:}a + J_i - a']$ and $w_i[a'{:}J_i]$ are equal by periodicity, so the rejected state $q_i(J_i)$ is reached at the cut $a + J_i - a' < J_i$, a contradiction. ◀

C.1 Rays and Difference Constraints

Proof of Lemma 7.2. (1) Let A be the maximum of the two onsets. From the position A on, the word $\Omega[a{:}]$ has period $\mathrm{per}(\Omega)$ and $\Omega'[b{:}]$ has period $\mathrm{per}(\Omega')$, and they agree on the $\mathrm{per}(\Omega) + \mathrm{per}(\Omega')$ positions from A on. By Fine and Wilf both have the period $g = \gcd(\mathrm{per}(\Omega), \mathrm{per}(\Omega'))$ on this interval, hence from A on, and two words with period g from A on that agree on g positions after A agree from A on. (2) The infinite word $Z = \Omega'[b{:}]$ is ultimately periodic with the least period $\mathrm{per}(\Omega')$, and the least position from which it has this period is $\mathrm{on}(\Omega') - b > 0$. Since $Z = \Omega[a{:}]$, the same holds with $\mathrm{per}(\Omega)$ and $\max(\mathrm{on}(\Omega) - a, 0)$. ◀

A system of difference constraints over a finite set I is a finite set of constraints $L_j \ge L_i + c$ with $i, j \in I$ and integers c; its graph has an edge $i \to j$ of weight c for each constraint.

Proof of Lemma 7.4. (2) Summing the constraints along a cycle shows that a solution excludes cycles of positive weight. Conversely, if there is no such cycle, let L^0_j be the largest weight of a path that ends in j, including the empty path; it is finite, attained by a path without repeated vertices, and satisfies all constraints. So does $L^0_j + c$ for every constant c, which exceeds any given lower bounds for large c.

(1) Consider an admitted pair on the channel from i to j of the tuple of prefixes, at cuts $a \le L_i$ and $b \le L_j$ with $\Omega_j[b{:}L_j] \preceq \Omega_i[a{:}L_i]$. By (S2), $b \le J_j$, so the two rests agree on $L_j - J_j$ letters, and $\Omega_i[a{:}] = \Omega_j[b{:}]$ by Lemma 7.2(1). If $i = j$, then $a \le b$ because $L_j - b \le L_i - a$, and the pair is a self event. If $i \ne j$, the ray system has the constraint $L_j \ge L_i + (b - a + 1)$, which contradicts $L_j - b \le L_i - a$. ◀

The ray system is equivalent to a finite system. Let Ω be one of the rays, with onset A and period p, and let φ be the map on states that reads the word $\Omega[A{:}A+p]$; it is also the word between the cuts $A+jp$ and $A+(j+1)p$ for every j. Among the $N+1$ states $\varphi^0(q),\dots,\varphi^N(q)$, where q is the state at the cut A, two are equal, $\varphi^h(q)=\varphi^{h+c}(q)$ with $c\geq 1$ and $h+c\leq N$, so $\varphi^j(q)=\varphi^{j-c}(q)$ for all $j\geq N$. Hence every cut $a\geq A+Np$ of Ω has a cut a^- with $A\leq a^-<A+Np$ and $a^-\equiv a$ modulo p, which carries the same state and has the same infinite rest. If the cuts a and b give a constraint $L_j\geq L_i+(b-a+1)$ of the ray system, the cuts a^- and b give the constraint $L_j\geq L_i+(b-a^-+1)$, which implies it. So the constraints with host cuts $a<\mathrm{on}(\Omega_i)+N\,\mathrm{per}(\Omega_i)$ imply all others; their target cuts are at most J_j, so there are finitely many of them, and Lemma 7.4(2) applies to them.

C.2 Closed Factors

Let x be a word read from a state q_0, with states $q(e)$, let $t_1,\dots,t_m$ be the target words and B a window, with the dictionary $\mathcal{D}_B$ of Definition 7.5. The dictionary is closed under factors. For a word y let $\sigma(y)$ be its longest suffix in $\mathcal{D}_B$.

▶ **Lemma C.2** (Suffix nodes). **1.** *$\sigma(yc)=\sigma(\sigma(y)c)$ for every word y and letter c. Hence $\mathrm{node}_B(e+1)$ is a function of $\mathrm{node}_B(e)$ and the letter at the position e.*
2. *If $\mathrm{node}_B(e)=\mathrm{node}_B(e')$ and a word $u\in\mathcal{D}_B$ ends at the cut e, then u ends at e' and $q(e-|u|)=q(e'-|u|)$. In particular $q(e)=q(e')$.*
3. *There are at most $N(mB(B+1)/2+1)$ suffix nodes.*

Proof. (1) A suffix of yc in the dictionary is empty or of the form uc with u a suffix of y and $u\in\mathcal{D}_B$, since the dictionary is closed under factors; then u is a suffix of $\sigma(y)$. The state component of the node after the letter is obtained from the node by reading a prefix of $\sigma(y)c$. (2) u is a suffix of the word σ of the common node (σ,q), and the state before u is obtained from q by reading σ without its suffix u. (3) A word of length at most B has at most $B(B+1)/2$ nonempty factors. ◀

Proof of Lemma 7.7. Let $h=c'-c$. By Lemma C.2(1) and $\mathrm{node}_B(c)=\mathrm{node}_B(c')$, the suffix nodes of x' are periodic with period h from the cut c on. So every cut e of x' has a cut $e^\circ\leq c'$ of x with the same suffix node, with $e^\circ=e$ if $e\leq c$ and $c<e^\circ\leq c'$ otherwise; in particular the states of x' are states of x at cuts in $[J,c']$ from J on, and they are periodic from c on. So x' is eligible with the same first rejected cut, by (S1), and its endpoint lies on a cycle of states other than †.

Let $u=t_j[b{:}]$ be a prefix of $x'[a{:}]$, and let $e=a+|u|$ be the cut of x' at which this occurrence of u ends. If $|u|\leq B$, then $u\in\mathcal{D}_B$, and by Lemma C.2(2) u ends at the cut e° of x, with the same state before it; so $a^\circ=e^\circ-|u|$ is as required. If $|u|>B$, then $v_B(t_j)$ is a suffix of u and ends at e in x', hence at e° in x, by the same lemma. Since the factor is free, $e^\circ\leq c$, so $e\leq c$ and the occurrence of u lies in the common prefix $x[{:}c]$ of x and x'. ◀

Suffix chains. For a word t and a window b with $1\leq b\leq|t|$, let $\mathrm{gap}_t(b)$ be the least $h\geq 1$ such that $v_b(t)$ ends at the cut $|t|-h$ of t, if there is one. A *suffix chain* of t is a sequence $b_0<b_1<\cdots<b_r$ of windows with $b_{u+1}=b_u+\mathrm{gap}_t(b_u)$; it has r *steps* and $r+1$ *members*.

(C1) If $b\leq b'$ and $\mathrm{gap}_t(b')$ is defined, then $\mathrm{gap}_t(b)\leq\mathrm{gap}_t(b')$: an occurrence of the longer suffix ends with an occurrence of the shorter one.
(C2) If $h=\mathrm{gap}_t(b)$, then $v_b(t)\preceq v_{b+h}(t)$. So along a chain the rest of t after the cut $|t|-b_{u'}$ has the rest after $|t|-b_u$ as a prefix, for $u<u'$.

(C3) Let g be a nondecreasing function, $G(b) = b + g(b)$, $b_0 \geq 1$, and suppose every chain of t that starts at a window of at least b_0 has at most s steps. Then for every $b \geq b_0$ there is a window $\hat{b}$ with $b \leq \hat{b} \leq G^s(b)$, the s-fold iterate, such that $\mathrm{gap}_t(\hat{b})$ is undefined or greater than $g(\hat{b})$: follow the chain from b as long as the gap is at most g. If $b > |t|$, then $\hat{b} = b$.

(C4) If $v_{b+g}(t)$ ends at a cut e of a word and $v_b(t)$ ends at a cut e' of it with $e - g \leq e' < e$, then $\mathrm{gap}_t(b) \leq e - e'$: both occurrences lie inside the occurrence of $v_{b+g}(t)$.

Let $\Phi(B) = (B+1)\,N\,(mB(B+1)/2 + 1)$, and for a number s define

$$\begin{aligned} \mathsf{L}_0(b) &= \Phi(b) + 1, & \mathsf{B}_0(b) &= b, \\ \mathsf{L}_{r+1}(b) &= 2\mathsf{L}_r(b^*) + \mathsf{L}_r\big(b^* + \mathsf{L}_r(b^*)\big) + 2, & \mathsf{B}_{r+1}(b) &= \mathsf{B}_r\big(b^* + \mathsf{L}_r(b^*)\big), \end{aligned}$$

where $b^* = G_r^s(b)$ is the s-fold iterate of $G_r(b) = b + \mathsf{L}_r(b)$. All these functions are nondecreasing in b, and $\mathsf{B}_r(b) \geq b$.

Proof of Lemma 7.9. We prove, by induction on $r \leq m$: if the chains of the target words $t_1, \ldots, t_r$ that start at a window of at least b_0 have at most s steps and $b \geq b_0 \geq 1$, then among any $\mathsf{L}_r(b)$ consecutive cuts of x there are cuts $c < c'$, and there is a window B with $b \leq B \leq \mathsf{B}_r(b)$, such that $\mathrm{node}_B(c) = \mathrm{node}_B(c')$, no occurrence of $v_B(t_u)$ with $u \leq r$ ends in $(c, c']$, and $B < c' - c \leq \Phi(B)$. The dictionary is always that of all m target words.

For $r = 0$ let $B = b$ and let $\nu = N(mb(b+1)/2 + 1)$ bound the number of suffix nodes. Among the $\nu + 1$ cuts $u, u + (b+1), \ldots, u + \nu(b+1)$, which lie within $\Phi(b) + 1$ consecutive cuts, two have the same node; their distance is a positive multiple of $b + 1$ and at most $\nu(b+1) = \Phi(b)$.

For the step from r to $r+1$, let $t = t_{r+1}$. By (C3) with $g = \mathsf{L}_r$ there is a window $\hat{b}$ with $b \leq \hat{b} \leq b^*$ such that $\mathrm{gap}_t(\hat{b})$ is undefined or greater than $\mathsf{L}_r(\hat{b})$. Let I be the given set of $\mathsf{L}_{r+1}(b)$ consecutive cuts and O the set of cuts of I at which $v_{\hat{b}}(t)$ ends.

If I contains $\mathsf{L}_r(\hat{b})$ consecutive cuts outside O, the induction hypothesis for the window $\hat{b}$ gives $c < c'$ among them and a window $B \geq \hat{b}$. No occurrence of $v_B(t)$ ends in $(c, c']$, because it would end with an occurrence of $v_{\hat{b}}(t)$, and $B \leq \mathsf{B}_r(\hat{b}) \leq \mathsf{B}_{r+1}(b)$.

Otherwise every $\mathsf{L}_r(\hat{b})$ consecutive cuts of I contain an element of O. Let e_0 be the first element of O; it is among the first $\mathsf{L}_r(\hat{b})$ cuts of I. Let $b^+ = \hat{b} + \mathsf{L}_r(\hat{b})$. No occurrence of $v_{b^+}(t)$ ends at a cut $e > e_0$ of I: such a cut belongs to O, the largest element e' of O below e satisfies $e - e' \leq \mathsf{L}_r(\hat{b})$, and (C4) gives $\mathrm{gap}_t(\hat{b}) \leq \mathsf{L}_r(\hat{b})$, a contradiction. The cuts of I after e_0 number at least $\mathsf{L}_{r+1}(b) - \mathsf{L}_r(b^*) \geq \mathsf{L}_r(b^+)$, since $b^+ \leq b^* + \mathsf{L}_r(b^*)$. The induction hypothesis for the window b^+, applied to these cuts, gives $c < c'$ and B with $b^+ \leq B \leq \mathsf{B}_r(b^+) \leq \mathsf{B}_{r+1}(b)$, and no occurrence of $v_B(t)$ ends in $(c, c']$. ◀

C.3 The Scale Functions

Let $\mathsf{Q} = 4 + 2(k-1)(N-1)$ and $s = \mathsf{Q} + N$, and let Φ, L_k, B_k be the functions of Appendix C.2 for $m = k$ and this s. Put $e_{-1} = 1$, $X_{-1} = X_{-2} = 0$, $Y_{-1} = T_{-1} = 0$, and for $l \geq 0$

$$\begin{aligned} b_l &= (N+1)\,e_{l-1} + (\mathsf{Q}+2)(X_{l-2} + 1), \qquad T_l' = \mathsf{L}_k(b_l), \qquad B_l = \mathsf{B}_k(b_l), \\ e_l &= \max\big(\Phi(B_l),\, e_{l-1}\big), \qquad T_l = (\mathsf{Q}+2)(3T_l' + 3), \\ Y_l &= 1 + (X_{l-1} + N + 2)\,e_l N! + Ne_l + 3(\mathsf{Q}+3)e_l + T_{l-1} + Y_{l-1}, \\ X_l &= (\mathsf{Q}+2)(1 + 2e_l) + T_l + Y_l + N. \end{aligned}$$

As in Section 7.4, b_l is the *initial window* of scale l, e_l bounds periods, T_l uncovered positions and Y_l periodic suffixes; moreover T_l' is the length of the stretches in which closed factors

are searched, B_l bounds their windows, and X_l bounds the members of suffix chains. These functions are defined for all $k \geq 1$ and $N \geq 1$. They are nondecreasing in l, and they satisfy the following inequalities:

(L1) $b_{l+1} > (N+1)e_l$ and $b_{l+2} \geq (\mathsf{Q}+2)(X_l+1)$;
(L2) $\Phi(B_l) \leq e_l < T'_l$;
(L3) $Y_{l+1} > Ne_{l+1} + T_l + 2(\mathsf{Q}+1)e_{l+1}$;
(L4) $Y_l \geq 1 + (X_{l-1}+N)\, e_l N! + 2e_l N! + Ne_l$;
(L5) $X_l \geq 2(\mathsf{Q}+1)e_l + T_l + Y_l + N$.

All but (L2) are immediate from the definitions. For (L2), induction on r gives $\Phi(\mathsf{B}_r(b)) < \mathsf{L}_r(b)$: for $r = 0$ this is $\Phi(b) < \Phi(b) + 1$, and $\mathsf{B}_{r+1}(b) = \mathsf{B}_r(y)$ with $y = b^* + \mathsf{L}_r(b^*)$, where $\Phi(\mathsf{B}_r(y)) < \mathsf{L}_r(y) \leq \mathsf{L}_{r+1}(b)$. Moreover $e_{l-1} < b_l \leq B_l < \Phi(B_l)$, so $e_l = \Phi(B_l) < \mathsf{L}_k(b_l) = T'_l$.

▶ **Lemma C.3** (Monotone scales). *For an eligible coordinate, F_l implies F_{l+1} and K_l implies K_{l+1}.*

Proof. The first statement holds because e_l and T_l are nondecreasing. Assume K_l for the word w with the first rejected cut J, and suppose w has a suffix of length at least Y_{l+1} with a period $p \leq e_{l+1}$, on the interval $[s, |w|)$. By Lemma C.1 fewer than Ne_{l+1} of its positions lie before J. Of the others at most T_l are not covered by the at most $\mathsf{Q}+1$ factors of property F_l, so by (L3) one of these factors, with a period $g \leq e_l$, meets the suffix in an interval of more than $2e_{l+1} \geq p + g$ positions. By Fine and Wilf the suffix has period $g \leq e_l$, and its length is at least $Y_{l+1} \geq Y_l$, contradicting K_l. ◀

▶ **Lemma C.4** (Chains of fixed words). *Let coordinate i be eligible.*

1. *If its endpoint is mortal, every suffix chain of w_i has at most $N-1$ members.*
2. *If it has K_m, every suffix chain of w_i has at most $X_m + N$ members.*
3. *If it has K_m, every suffix chain of w_i that starts at a window of at least $(\mathsf{Q}+2)(X_m+1)$ has at most $\mathsf{Q}+N$ steps.*

Consequently, at a stable scale l, the suffix chains of the word of a fixed coordinate that start at a window of at least b_l have at most $\mathsf{Q}+N$ steps, and all its suffix chains have at most $X_{l-2}+N$ steps.

Proof. Let $w = w_i$ and $J = J_i$. By (C2) the members b_u of a chain give cuts $|w| - b_u$ at which comparable rests begin. (1) follows from Lemma 5.6: the states at these cuts are pairwise different and differ from the endpoint.

(2) We count the members by their cuts $c = |w| - b_u$. At most $N-1$ are at most J, by Lemma 5.6(3), at most T_m are uncovered positions of the tail, and at most Y_m are among the last Y_m positions. Finally consider one of the factors of property F_m, with a period $g \leq e_m$, and extend it to a maximal interval $[s, e)$ of period g. If $e = |w|$, all its positions are among the last Y_m, by K_m. Otherwise the letter at e differs from the letter at $e - g$. If two members had cuts $c_1 < c_2 \leq e - g$ in this interval, the rest after c_2 would be a prefix of the rest after c_1, and comparing the positions $e - c_2$ and $e - c_2 - g$ of the two rests would give $w[e] = w[e-g]$. So at most one member has its cut in $[s, e-g]$, and at most g in $(e-g, e)$. With (L5) the total is at most $X_m + N$.

(3) Let b_0 be the first member of the chain, $U = v_{b_0}(w)$, and $P = \mathsf{Q}+1$. By (C2), every member b_u gives an occurrence of U in w that begins at the cut $|w| - b_u$. At most $N-1$ of them begin at or before J, by Lemma 5.6. We show that at most P occurrences of U begin in the tail; then the chain has at most $P + N - 1$ members. This is clear if

$|w| - J < b_0$. Otherwise the last b_0 positions lie in the tail. Let $R = Y_m + T_m + 2e_m P + 1$; by (L5), $b_0 \geq 1 + T_m + PR$, so one of the factors of property F_m, with a period $p \leq e_m$, meets the last b_0 positions in an interval $[a, e)$ of more than R positions. Extend this interval to the right, keeping the period p, up to the first position d with $w[d] \neq w[d-p]$, or to the end of w. It does not reach the end: otherwise w would have a suffix of period $p \leq e_m$ and length more than $R > Y_m$, contrary to K_m. So $d < |w|$, and in the coordinates of U, with $\iota = a - (|w| - b_0)$ and $\beta = d - (|w| - b_0)$, the interval $[\iota, \beta)$ has more than R positions and period p, and the letters at β and $\beta - p$ differ, a *break*. Consider an occurrence of U that begins at a cut $c \geq J$. Of the R positions before its break $c + \beta$, at most T_m are uncovered, so one of the factors of property F_m, say f, contains more than $2e_m$ of them. By Fine and Wilf f has period p, so f does not contain the break: it ends at or before $c + \beta$. We charge the occurrence to f. If a later occurrence, at $c' > c$, were charged to the same factor, then more than $2e_m \geq p$ of the R positions before $c' + \beta$ would lie before $c + \beta$, so the positions $c + \beta$ and $c + \beta - p$ would both lie in the interval of period p of the later occurrence, contradicting the break. So the occurrences that begin in the tail are charged to different factors.

The last statement follows from (1), from (2) and (3) for $m = l - 2$, and from (L1). ◀

C.4 Terminal Rays and Candidates

A ray Ω *agrees with a word w up to E* if $E \leq |w|$ and $\Omega[:E] = w[:E]$. For cuts $J \leq c < c'$ of w, the ray $w[:c]\,\tau^\omega$ of the factor $\tau = w[c{:}c']$ agrees with w up to c', its onset is at most c and its period at most $c' - c$. If a ray agrees with w_i up to E, its states at the cuts up to E are the states q_i.

▶ **Lemma C.5** (Terminal rays). *Let coordinate i be eligible, and let $w = w_i$ have a suffix of length at least Y_m with a period $p \leq e_m$. Let $h = pN!$, $c' = |w|$ and $c = c' - h$. Then $J_i \leq c$, and there is a window B such that $w[c{:}c']$ is closed at B for every dictionary of words of length at most B, w has period p on $[c - B, c')$, and no occurrence of $v_B(t)$ ends in $(c, c']$ for any word t whose suffix chains have at most $X_{m-1} + N$ steps.*

Proof. Let $[s, c')$ be the suffix of period p and $B = 1 + (X_{m-1} + N)h + h$. By (L4), $s + Np \leq c - B$, so $J_i < c - B$ by Lemma C.1. On a word of period p the states at the cuts that are at least Np after its beginning are periodic with period $pN!$: reading p letters is a map on the states whose Nth iterate lands on a cycle, of a length dividing $N!$. So $q_i(c - B) = q_i(c' - B)$, and the B letters before c and before c' are equal, since p divides h. Hence the state and the longest suffix in any dictionary of words of length at most B are the same at c and c'.

Let t be a word whose chains have at most $X_{m-1} + N$ steps. By (C3) with the constant function $g = h$ there is a window $\hat{b} \leq 1 + (X_{m-1} + N)h$ such that $\mathrm{gap}_t(\hat{b})$ is undefined or greater than h. Suppose $v_B(t)$ ends at a cut $e \in (c, c']$. The word w has period h on $[c - B, c')$, and $B \geq \hat{b} + h$, so $v_{\hat{b}}(t)$ ends at e and at $e - h$, both inside this occurrence of $v_B(t)$. By (C4), $\mathrm{gap}_t(\hat{b}) \leq h$, a contradiction. ◀

▶ **Lemma C.6** (Candidates). *Let $l \geq 1$, let v be an eligible coordinate without F_l, and let the target words be at most k words whose suffix chains from the window b_l on have at most $\mathsf{Q} + N$ steps. Then there are cuts $J_v \leq c_r < c'_r$ of w_v and windows $B_r \geq b_l$, for r in a finite set, such that*

1. *$w_v[c_r{:}c'_r]$ is closed and free at B_r, and $B_r < c'_r - c_r \leq e_l$;*
2. *the rays $\Omega^r = w_v[:c_r]\tau_r^\omega$ of these factors have at least $\mathsf{Q} - 1$ pairwise different onsets greater than J_v;*

3. *for each of them, all positions of* $[J_v, c_r)$ *except at most* $(\mathsf{Q}+3)(3T'_l+3)$ *lie in at most* $\mathsf{Q}+2$ *factors of* w_v *with periods at most* e_l.

Proof. Let $w = w_v$ and $J = J_v$, and divide the cuts from J on into segments of T'_l consecutive cuts. By Lemma 7.9, applied with $m = k$ after adding empty words to the target words if there are fewer than k, and by (L2), every complete segment r contains cuts $c_r < c'_r$ with (1). The ray Ω^r agrees with w up to c'_r and has a period of at most e_l, so w has this period on the interval $[\mathrm{on}(\Omega^r), c'_r)$, which contains the position c_r of segment r.

For a set $\mathcal{R}$ of segments, the intervals $[\mathrm{on}(\Omega^r), c'_r)$, $r \in \mathcal{R}$, with the same onset are contained in the longest of them. If the rays of the segments below some segment r_0 have at most g different onsets, we thus obtain at most g factors with periods at most e_l that contain a position of each of these segments. A maximal interval of positions of $[J, c_{r_0})$ outside these factors contains no complete segment below r_0, so it has fewer than $3T'_l$ positions, and there are at most $g+1$ such intervals. For $g = \mathsf{Q}+1$ and all segments, with the incomplete last segment, this would give F_l, by the definition of T_l. So the rays have at least $\mathsf{Q}+2$ different onsets. Let r_0 be the first segment such that the segments up to r_0 show $\mathsf{Q}+2$ different onsets, and choose one segment for each of these onsets; for these, (3) holds by the same count with $g = \mathsf{Q}+2$.

At most three of the $\mathsf{Q}+2$ onsets are at most J. Indeed, let two of the chosen rays have onsets at most J and $c'_r \geq J + 2e_l$. Both agree with w on $[J, J+2e_l)$, an interval after both onsets of at least the sum of the two periods, so they are equal by Lemma 7.2(1), and have the same onset. And $c'_r < J + 2e_l < J + 2T'_l$ holds for the first two segments only. ◀

C.5 Forbidden Overlaps

For a ray Ω_u of a coordinate u write $A_u = \mathrm{on}(\Omega_u)$, $\Pi_u = \Omega_u[A_u{:}]$ for its periodic part, and $X_u = \Omega_u[J_u{:}A_u]$ for the *stretch* between the first rejected cut and the onset, the empty word if $A_u \leq J_u$.

▶ **Definition C.7** (Forbidden overlaps). *Let* Ω_v *and* Ω_u *be rays of coordinates* $v \neq u$ *with* $J_v < A_v$. *The ray* Ω_v *has a* forbidden overlap *with* Ω_u *if one of the following holds.*

(O1) $J_u < A_u$, X_v *is a suffix of* $\Omega_u[{:}A_u]$, *and* $|X_u| \leq |X_v|$.
(O2) $A_u \leq J_u$, X_v *is a suffix of* $\Omega_u[{:}A_u]$, *and* $\Pi_v = \Pi_u$.
(O3) X_u *is not empty and is a suffix of* $\Omega_v[{:}A_v]$, *and* $|X_v| \leq |X_u|$.

If both onsets lie after the first rejected cuts, (O2) does not apply and (O1) for (v, u) is (O3) for (u, v); so the relation "no forbidden overlap" is symmetric for such rays.

Proof of Lemma 7.11. (2) Consider a constraint $h \to x$ of the ray system, from cuts a and $b \leq J_x$ with $\Omega_h[a{:}] = \Omega_x[b{:}]$, and suppose $J_x < A_x$. By Lemma 7.2(2), $A_h - a = A_x - b > 0$. Hence $\Omega_x[b{:}A_x] = \Omega_h[a{:}A_h]$, the stretch X_x is a suffix of $\Omega_h[{:}A_h]$, and $\Pi_h = \Pi_x$.

Let v be as in the statement and suppose it lies on a cycle. We follow the cycle backwards from v and show that every vertex x on the way satisfies $J_x < A_x$, $\Pi_x = \Pi_v$, X_v is a suffix of X_x, and $x = v$ or $|X_v| < |X_x|$. This holds for $x = v$. Let $h \to x$ be the edge of the cycle into such an x. By the first paragraph, X_x, and with it X_v, is a suffix of $\Omega_h[{:}A_h]$, and $\Pi_h = \Pi_v$. If $h = v$, then $x \neq v$, and X_x is a nonempty suffix of $\Omega_v[{:}A_v]$ that is longer than X_v: this is (O3). If $h \neq v$ and $A_h \leq J_h$, we have (O2). If $J_h < A_h$ and $|X_h| \leq |X_v|$, we have (O1). All three are excluded, so $h \neq v$, $J_h < A_h$ and $|X_v| < |X_h|$; and since X_v and X_h are both suffixes of $\Omega_h[{:}A_h]$, the first is a suffix of the second. So the invariant passes to h with $h \neq v$, and the cycle never returns to v.

(1) Let Ω^r, $r \in \mathcal{R}$, be candidates of v with pairwise different onsets $A^r > J_v$, each agreeing with w_v up to its cut c'_r, so that the stretches $X^r = w_v[J_v{:}A^r]$ are prefixes of one another, of different lengths. Let Ω_u agree with w_u up to some cut $E_u \geq \max(A_u, J_u)$.

For (O1), each r with this overlap gives the cut $a_r = A_u - |X^r| \leq J_u$ of w_u, at which X^r begins. If two of them, $a_{r'} < a_r$, carried the same state, the texts of w_u after them would agree up to J_u, because X^r is a prefix of $X^{r'}$ and $J_u - a_r \leq |X^r|$, and the rejected state would be reached before J_u. So at most $N-1$ candidates have the overlap (O1). For (O3), the word $Y = X_u$ ends at A^r in w_v and begins at a cut $A^r - |Y| \leq J_v$; two such cuts with the same state are followed by the same text Y, which reaches beyond J_v, with the same contradiction.

For (O2) we use in addition that each candidate agrees with w_v on at least $(N+1)\nu$ letters from its onset, where $\nu = \mathrm{per}(\Omega_u)$; this holds by Lemma C.6(1) and (L1) when $\nu \leq e_{l-1}$. Let $f = J_u - A_u$; by Lemma C.1, $f < N\nu$. The overlap gives the cut $a_r = A_u - |X^r| \leq A_u$ of w_u, followed by X^r and then by Π_u up to J_u. Let $a_{r'} < a_r$ be two such cuts; then $X^{r'} = X^r Z$ with $Z = w_v[A^r{:}A^{r'}]$. The word w_v has period ν on $[A^r, A^r + (N+1)\nu)$ and reads Π_u there. If $|Z| \leq N\nu$, the ray $\Omega^{r'}$, whose periodic part Π_u begins at $A^{r'}$ and which agrees with w_v beyond $A^{r'} + \nu$, would have period ν from A^r on, so its onset would be at most $A^r < A^{r'}$. So $|Z| > N\nu > f$, and Z begins with the first f letters of Π_u. Hence the texts of w_u after $a_{r'}$ and after a_r agree on $|X^r| + f = J_u - a_r$ letters, and equal states at the two cuts would give a rejected state before J_u. So at most $N-1$ candidates have the overlap (O2). Since (O1) and (O2) exclude each other for a fixed u, at most $2(N-1)$ candidates have a forbidden overlap with Ω_u. ◀

C.6 Holes with Periodic Tails

Proof of Proposition 7.12. Let W be a hole and l a stable scale, $2 \leq l \leq 3(2k+1)$, with the fixed, terminal and escaping coordinates of Section 7.4. The target words are the words of the fixed coordinates; by Lemma C.4 their suffix chains have at most $X_{l-2} + N$ steps, and at most $\mathsf{Q} + N$ steps from the window b_l on.

Terminal rays. A terminal coordinate i has F_{l-1} and not K_{l-1}, so w_i has a suffix of length at least Y_{l-1} with a period $p \leq e_{l-1}$. Lemma C.5 for $m = l-1$ gives the factor $\tau_i = w_i[c_i{:}|w_i|]$, closed and free at a window with respect to the target words, and the ray $\Omega_i = w_i[{:}c_i]\tau_i^\omega$, which has period p from $c_i - B$ on. It agrees with w_i up to $|w_i|$, and its period is at most e_{l-1}.

Escaping rays. We treat the escaping coordinates in some order. Let v be the next one and $\mathcal{U}$ the set of terminal coordinates and of escaping coordinates treated before, at most $k-1$ coordinates. Lemma C.6 gives $\mathsf{Q}-1$ candidates of v with different onsets after J_v. Each ray of $\mathcal{U}$ with its onset at or before its first rejected cut is terminal and has a period of at most e_{l-1}, so Lemma 7.11(1) applies: each ray of $\mathcal{U}$ excludes at most $2(N-1)$ candidates, and since $\mathsf{Q} - 1 > 2(k-1)(N-1)$ some candidate of v has no forbidden overlap with any ray of $\mathcal{U}$. We take its ray as Ω_v. By the symmetry noted after Definition C.7, in the end every escaping ray has no forbidden overlap with any other ray.

The ray system. By Lemma 7.11(2), no cycle of the ray system contains an escaping coordinate. For two terminal coordinates $i \neq j$, consider a constraint from cuts a and $b \leq J_j$. The rays contain the words w_i and w_j as prefixes. If $|w_j| - b \leq |w_i| - a$, then $w_j[b{:}] \preceq w_i[a{:}]$, and the admitted pair lies on a channel of W, with the original states; this is impossible. So the lengths of the original words satisfy all constraints between terminal rays, and there is no cycle of positive weight. No ray has a self event: for a terminal ray, a self event at $a \leq b \leq J_i$ restricts to the comparison $w_i[b{:}] \preceq w_i[a{:}]$ of W with an admitted pair; for an

escaping ray with $a < b$, the equality $\Omega_v[a{:}] = \Omega_v[b{:}]$ makes the ray periodic from $a \le J_v$ on, contradicting $J_v < \mathrm{on}(\Omega_v)$, and for $a = b$ the pair lies on the channel of W from v to v.

The new hole. By Lemma 7.4(2) there are lengths L_i, for the terminal and escaping coordinates, that satisfy the ray system, exceed the bound of Lemma 7.4(1), and satisfy $L_i \ge c'_i$ and $L_i > J_i + |w_j|$ for every fixed j. Let W^* consist of the prefixes $\Omega_i[{:}L_i]$ and of the unchanged words of the fixed coordinates. Every coordinate is eligible with the same first rejected cut, by Lemma 7.7, and the new endpoints are immortal. No channel of W^* contains an admitted pair: between two fixed coordinates nothing has changed; between two ray coordinates this is Lemma 7.4(1); if the host is a ray and the target is fixed, the pair is a pair of W by Lemma 7.7; and if the host is fixed and the target a ray, the target cut is at most J_j by (S2), and the rest of the target is longer than the host word. So W^* is a hole.

The periodic factors. In W^* a coordinate with an immortal endpoint is a ray coordinate, or it is fixed with K_{l-2}. In the second case its tail is covered by at most $\mathsf{Q}+1$ factors with periods at most e_{l-2} up to T_{l-2} positions. For a terminal coordinate the same holds at scale $l-1$ for the positions of w_i, and the rest of the new word lies in one further factor of period p. For an escaping coordinate, Lemma C.6(3) covers the tail before c_v, and the rest lies in one further factor. So every immortal tail is covered by at most $\mathsf{Q}+3$ factors with periods at most $D = e_{3(2k+1)}$ up to $T_{3(2k+1)} + (\mathsf{Q}+3)(3T'_{3(2k+1)} + 3)$ positions. A factor of period $p \le D$ and length at least $2D$ lies in a maximal periodic factor whose period is the least period of the factor, a divisor of p by Fine and Wilf; the positions of shorter factors, at most $2D$ for each, are added to the uncovered ones. This gives the proposition with

$$E_0 = k\big(T_{3(2k+1)} + (\mathsf{Q}+3)(3T'_{3(2k+1)} + 3) + 2(\mathsf{Q}+3)D\big). \qquad \blacktriangleleft$$

D Proofs for Section 7.6: Compressing the Pieces

We keep the conventions of Appendix C; in particular $k \ge 1$ and $N \ge 2$. As in Definition 4.12, a *comparison* of a tuple is a pair of cuts of a host word and a target word such that the rest of the target is a prefix of the rest of the host. A word is *primitive* if it is not a power of a shorter word, and two words are *conjugate* if one is a rotation of the other.

D.1 Pieces

Let W be a tuple. A *piece* of period d is an interval $[s, e)$ of positions of a word w_l with $e - s \ge 2d$ on which w_l has period d, whose first d letters form a primitive word, its *root*, and which is maximal: $s = 0$ or $w_l[s-1] \ne w_l[s-1+d]$, and $e = |w_l|$ or $w_l[e] \ne w_l[e-d]$. Every factor of length at least $2p$ with period p lies in a piece whose period is the least period of the factor, a divisor of p.

(P1) Two different pieces of period d in one word overlap in fewer than d positions, and their ends are more than d apart: an overlap of d positions would make the union an interval of period d.

(P2) Two pieces of one word with periods $d' \le d$ and roots that are not conjugate overlap in fewer than $d + d'$ positions: otherwise, by Fine and Wilf, the overlap has the period $\gcd(d, d')$, which is d and d' because the roots are primitive, and the roots are conjugate.

(P3) Let $\alpha_d = Nd$ and $m_d = d \cdot N!$. Inside a piece $[s, e)$ of period d, $q_l(s + o + m_d) = q_l(s + o)$ whenever $\alpha_d \le o$ and $o + m_d \le e - s$. Indeed, reading the d letters at the offsets $o', \dots, o' + d - 1$ is the same map on states for all offsets $o' \equiv o$ modulo d; its Nth iterate lands on a cycle, whose length divides $N!$.

Fix a primitive word ζ, $d = |\zeta|$, and write $\alpha, m, \beta, \kappa, \phi$ for $\alpha_d, m_d, \beta_d, \kappa_d, \phi_d$; so $\beta = \alpha + m \geq 4d$, $\kappa = 2\beta + m$ and $\phi = 6\beta + 2m$. Let $\mathcal{C}$ be the class of ζ. For $P \in \mathcal{C}$ we write s_P, e_P for its ends and $L_P = e_P - s_P$.

D.2 Admissible Deletions

Let $(x_P)_{P \in \mathcal{C}}$ be an admissible deletion (Definition 7.13). The *site* of P is the interval $[s_P + \beta,\, s_P + \beta + m\, x_P)$, and $U = (u_1, \ldots, u_k)$ is the tuple obtained by deleting all sites. We use the notation im_l, γ and the facts on the deletion of positions from Appendix B.4, writing im_l for the cut map of the word w_l; a *junction* is the image of a nonempty site. Put $L'_P = L_P - m\, x_P$.

(A1) A nonempty site lies inside its piece, at least β positions after its start and at least $5\beta + 2m$ before its end. By (P1), sites of different pieces of one word are disjoint and not adjacent, and a position of a site lies in no other piece of the class.

(A2) By (P3), the two ends of a site carry the same state. So the state of u_l at $\mathrm{im}_l(p)$ is the state of w_l at p for every retained cut p; endpoints are preserved and no new state is visited.

(A3) For $P \in \mathcal{C}$ in the word w_l, the interval $P' = [\mathrm{im}_l(s_P), \mathrm{im}_l(e_P))$ is a piece of U of the class $\mathcal{C}$, with the root of P and the length L'_P; at an offset $o \leq L'_P$ it carries the state that P carries at the offset o, and at $o < L'_P$ the same letter. The map $P \mapsto P'$ is a bijection from $\mathcal{C}$ onto the pieces of U of the class of ζ.

(A4) If $o \leq L_P$ and $o' \leq L'_P$ are offsets with $o = o'$, or with $o, o' \geq \alpha$ and $o \equiv o'$ modulo m, then P at o and P' at o' carry the same state, by (A3) and (P3).

For (A3): the deleted block has a length divisible by d, so P' has period d and the stated letters; the states follow from (A2) and (P3); and the breaks at both ends of P survive, because the site is at least $\beta \geq 2d$ away from them. Conversely, let R' be a piece of U of this class. If a junction, of a piece P, lies in R' or at one of its ends, then R' and P' overlap in at least d positions, because R' has at least $2d$ positions and P' extends by $2d$ positions on both sides of its junction; so $R' = P'$ by (P1). Otherwise R' and its two breaks consist of original text, so R' is the image of a piece of W of the class, which is not shortened.

A cut c' of u_l is *clean for* θ if no shortened piece P of w_l with $\mathrm{im}_l(s_P) \leq c'$ has its junction strictly between c' and $c' + \theta$. For every $P \in \mathcal{C}$ and every θ, the cut $\mathrm{im}_l(e_P) - d$ is clean for θ, and its greatest preimage is $e_P - d$: a piece Q' that begins at or before this cut and has its junction after it contains the last d positions of P', by (A1), so $Q' = P'$ by (P1), but the junction of P' lies before the cut.

Two words y, y' *agree on* θ *positions from the cuts* c, c' if $y[c + x] = y'[c' + x]$ for all $x < \theta$, these positions being defined.

▶ **Lemma D.1** (Copying pieces). *Let u_h and u_g agree on θ positions from the cuts a' and b', and let R' be a piece of U of the class $\mathcal{C}$ in u_h, between the cuts s' and e'.*

1. *If $a' < s'$ and $s' + 2d \leq a' + \theta$, then u_g has a piece P' of the class that begins at $b' + (s' - a')$, with the same root.*
2. *If $s' \leq a'$ and $a' + \ell \leq e'$ for some ℓ with $2d < \ell \leq \theta$, then u_g has a piece P' of the class that contains the positions from b' to $b' + \ell$.*
3. *In both cases, if $e' < a' + \theta$ then P' ends at $b' + (e' - a')$, and otherwise P' contains all positions from its beginning to $b' + \theta$.*

The same holds with the roles of u_h and u_g exchanged.

Proof. The copy in u_g of the part of R' inside the agreement is a factor of length at least $2d$ with period d and a primitive conjugate root, so it lies in a piece of the class. A break of R' whose two letters lie in the agreement is copied, which determines the corresponding end of P'. ◀

▶ **Lemma D.2** (Clean cuts). *Let $T = u_g[b':] \preceq u_h[a':]$ and $\theta = |T|$. If b' is clean for θ in u_g and a' is clean for θ in u_h, then $w_g[\gamma(b'):] \preceq w_h[\gamma(a'):]$.*

Proof. By induction on θ. If no junction lies strictly between a' and $a' + \theta$ in u_h or strictly between b' and $b' + \theta$ in u_g, both texts are original, by (9). Otherwise consider the shortened pieces whose junctions lie strictly inside one of the two windows. By cleanness their images begin after a', respectively b'; let $r > 0$ be the least distance of such a beginning from the beginning of its window. By Lemma D.1(1), applied in one of the two directions, there are pieces R' in u_h beginning at $a' + r$ and P' in u_g beginning at $b' + r$, with the same root; they are images of pieces R and P of $\mathcal{C}$. By the choice of r and by cleanness, no junction lies strictly between a' and $a' + r$ or at $a' + r$, and likewise in u_g; so

$$w_h[\gamma(a'):s_R] = u_h[a':a' + r] = u_g[b':b' + r] = w_g[\gamma(b'):s_P]. \tag{11}$$

By Lemma D.1(3) there are two cases.

Both pieces end inside the comparison, at the same distance, so $L'_P = L'_R$ and $u_g[\mathrm{im}_g(e_P) - d:] \preceq u_h[\mathrm{im}_h(e_R) - d:]$, a comparison of length less than θ between clean cuts. By the induction hypothesis, $w_g[e_P - d:] \preceq w_h[e_R - d:]$. In particular the followers of P and R are comparable. If P and R differed in length by at least κ, so would P' and R'; so their lengths differ by less than κ, $x_P = x_R$ and $L_P = L_R$. Two pieces with the same root and the same length carry the same text, so $w_g[s_P:] \preceq w_h[s_R:]$.

P' ends at the end of u_g, and $L'_P \le L'_R$. Then P ends at the end of w_g and has the empty follower, which is comparable with that of R. If the lengths differ by less than κ, then $x_P = x_R$; otherwise the longer piece remains the longer one. In both cases $L_P \le L_R$, and $w_g[s_P:]$, a word of period d with the root of R, is a prefix of $w_h[s_R:]$.

In both cases the claim follows with (11). ◀

▶ **Lemma D.3** (Comparisons inside pieces). *Let $P, R \in \mathcal{C}$ lie in the words w_g and w_h, and let $b' = \mathrm{im}_g(s_P) + p_T$ and $a' = \mathrm{im}_h(s_R) + p_S$ with $p_T + d \le L'_P$ and $p_S + d \le L'_R$. Suppose $u_g[b':] \preceq u_h[a':]$ and $L'_P - p_T \le L'_R - p_S$, and that P' ends at the end of u_g or $L'_P - p_T = L'_R - p_S$. Then there are offsets $o_T \le L_P$ and $o_S \le L_R$ such that $w_g[s_P + o_T:] \preceq w_h[s_R + o_S:]$ and the states of W at these two cuts are the states of U at b' and a'.*

Proof. We first show that the followers of P and R are comparable, and that $w_g[e_P:] \preceq w_h[e_R:]$ if P' does not end its word. In that case $L'_P - p_T = L'_R - p_S$, and shifting the given comparison yields $u_g[\mathrm{im}_g(e_P) - d:] \preceq u_h[\mathrm{im}_h(e_R) - d:]$ between clean cuts; Lemma D.2 gives $w_g[e_P - d:] \preceq w_h[e_R - d:]$. If P' ends its word, so does P, and its follower is empty.

It suffices to find offsets o_T, o_S that are related to p_T, p_S as in (A4) and satisfy $L_P - o_T \le L_R - o_S$, with equality if P does not end its word. Indeed, the states are then as required; the letters of w_g from $s_P + o_T$ and of w_h from $s_R + o_S$ agree on d positions, because they are the letters of u_g from b' and of u_h from a', by (A3) and since m is a multiple of d; so by periodicity they agree up to the end of P, and beyond it the followers are compared.

If the lengths of P and R differ by less than κ, then $x_P = x_R$, and $o_T = p_T$, $o_S = p_S$ work, since both lengths change by the same amount. Otherwise the lengths differ by at least κ, before and after the deletion and in the same direction, and $\kappa \ge \beta + m \ge \alpha$.

Suppose $L'_P - p_T = L'_R - p_S$. If R is the longer piece, then $p_S = p_T + L'_R - L'_P \geq \kappa$; let $o_T = p_T$ and $o_S = p_T + L_R - L_P \geq \kappa$, which differs from p_S by $m(x_R - x_P)$. If P is the longer piece, let $o_S = p_S$ and $o_T = p_S + L_P - L_R$, symmetrically.

Suppose P' ends its word and $L'_P - p_T \leq L'_R - p_S$. If P is the longer piece, then $p_T \geq p_S + \kappa \geq \beta$; let $o_T = p_T + m\,x_P$ and $o_S = p_S$, so that $L_P - o_T = L'_P - p_T \leq L'_R - p_S \leq L_R - o_S$. If R is the longer piece, let $o_T = p_T$, and let $o_S = p_S$ if $p_S < \alpha + m$ and otherwise the number in $[\alpha, \alpha + m)$ that is congruent to p_S modulo m. Then $L_R - o_S > L_R - \alpha - m \geq L_P \geq L_P - o_T$. ◀

Proof of Lemma 7.14. *Domination.* Endpoints and visited states are treated in (A2). Let $T = u_g[b':] \preceq u_h[a':]$ be a comparison of U, $\theta = |T|$. We show that the pair of states at a' and b' lies in $\mathsf{Ch}_{hg}(W)$.

Case 1: b' is not clean for θ. Then b' lies in the image P' of a shortened piece, at an offset $p_T < \beta$, and P' continues for more than 5β positions after b', inside T. By Lemma D.1(2) and (3), from u_g to u_h, a' lies in a piece R' of the class at an offset p_S, and either both pieces end inside the comparison, at the same distance, or P' ends at the end of u_g and $L'_P - p_T \leq L'_R - p_S$. Lemma D.3 applies.

Case 2: b' and a' are clean for θ. Lemma D.2 applies, with the states of (A2).

Case 3: b' is clean for θ and a' is not. Then a' lies in the image R' of a shortened piece, at an offset $p_S < \beta$, with the junction of R' strictly inside the comparison. If the comparison continues beyond the end of R', then by Lemma D.1 b' lies in a piece P' of the class that ends at the same distance, and Lemma D.3 applies. So let $\theta \leq L'_R - p_S$. If $\theta > 2d$, then by Lemma D.1 T lies in a piece P' of the class that ends at the end of u_g, and Lemma D.3 applies again. If $\theta \leq 2d < \beta$, no junction lies strictly inside the window of the target, since by cleanness it would belong to a piece that begins inside the window; so $T = w_g[\gamma(b'):]$, and T is read in w_h at the offset p_S of R, with the same state, by (A3).

First rejected cuts. Let W be a hole. By (A2) and Lemma 5.3, U is a hole with the same endpoints. The states of u_l are those of w_l at retained cuts, in the same order, and the states of w_l are rejected exactly from J_l on; so the first rejected cut of u_l is $\mathrm{im}_l(J_l)$, the image of J_l or of the right end of the site that contains J_l.

Other classes. Let R be a piece with a period $d' \leq d$, a root that is not conjugate to ζ, and a length of at least $2d \geq d + d'$. If a position of a site of $P \in \mathcal{C}$ lay in R or next to it, then by (A1) and $\beta \geq 4d$ the pieces P and R would overlap in at least $2d$ positions, contradicting (P2). So R and the letters next to it are not affected, and its image is a piece of U with the same period, root and length. ◀

D.3 The Cap

▶ **Lemma D.4** (Simple paths). *Let $P_1, \ldots, P_v$ be pairwise different pieces of $\mathcal{C}$ such that consecutive ones have comparable followers. Suppose that all but at most r of them belong to a set $\mathcal{H}$, and that every subset of $\mathcal{H}$ with pairwise comparable followers has at most h elements. Then $v \leq (r+1)\,2^h - 1$.*

Proof. Removing the pieces outside $\mathcal{H}$ leaves at most $r + 1$ paths of pieces of $\mathcal{H}$. We show that such a path has at most $2^h - 1$ vertices, by induction on h. Order its pieces by the prefix order of their followers, with ties broken by an arbitrary linear order; this is a partial order in which the elements below any element form a chain, and consecutive pieces of the path are comparable. So the path has a least element, which is comparable with all its pieces. Removing it leaves at most two paths, in each of which pairwise comparable sets have at most $h - 1$ elements, since the least element can be added to them. Hence the path has at most $1 + 2(2^{h-1} - 1)$ vertices. ◀

Proof of Lemma 7.15. We determine integers $z_P \le 0$ and put $x_P = -z_P$. Consider the following difference constraints, for all pieces $P, Q \in \mathcal{C}$ with comparable followers:

- $z_P \ge z_Q$ and $z_Q \ge z_P$, if the lengths differ by less than κ;
- $z_Q \ge z_P - \lfloor (L_Q - L_P - \kappa)/m \rfloor$, if $L_P + \kappa \le L_Q$;

together with the lower bounds $z_P \ge \underline{z}_P$, where $\underline{z}_P = -\lfloor (L_P - \phi)/m \rfloor$ if $L_P \ge \phi$ and $\underline{z}_P = 0$ otherwise. The zero vector satisfies all of them. A vector $z \le 0$ that satisfies them gives an admissible deletion: a shortened piece keeps at least ϕ letters by the lower bound, and the second kind of constraint says $L_Q + mz_Q \ge L_P + mz_P + \kappa$.

Let z be the vector whose component z_P is the largest value of $\underline{z}_Q + c$ over all pieces Q and all paths from Q to P in the graph of the constraints, c being the weight of the path. Since the zero vector is a solution, the graph has no cycle of positive weight, so the maximum exists and is attained by a path without repeated vertices; and z is a solution with $\underline{z} \le z \le 0$, the least one. Fix P and such a path $Q = P_1, \ldots, P_v = P$. By maximality, $z_{P_1} = \underline{z}_{P_1}$ and every constraint along the path is an equality for z.

Consider the retained lengths $L'_{P_u} = L_{P_u} + mz_{P_u}$. First, $L'_{P_1} < \phi + m$. Along a constraint of the first kind the retained length changes by less than κ. Along a constraint of the second kind, from P_u to P_{u+1}, equality gives $L'_{P_{u+1}} < L'_{P_u} + \kappa + m$. So $L'_P < \phi + m + (v-1)(\kappa + m)$. The pieces of the path are pairwise different and consecutive ones have comparable followers, so $v \le (r+1)2^h - 1$ by Lemma D.4, with $\mathcal{H}$ the set of the pieces that are not exceptional. ◀

D.4 The Passes

A *tail decomposition* of a hole U with the parameters (η, D, E) is a set of at most η pieces of U, the *designated pieces*, with periods at most D and lengths at least $2D$, such that at most E positions of the rejected tails of the coordinates with immortal endpoints lie in no designated piece; these are the *uncovered* positions.

▶ **Lemma D.5** (One pass). *A hole with a tail decomposition with the parameters (η, D, E), $\eta \ge 1$, has a dominating hole with a tail decomposition with the parameters $(\eta - 1, D, E')$, where*

$$E' = E + \eta \left(\phi_D + m_D + \left((2\eta + E + 1)\, 2^{k(N-1)} - 1 \right) (\kappa_D + m_D) \right).$$

Proof. Let d be the largest period of a designated piece, ζ its root and $\mathcal{C}$ the class of ζ. Let $\mathcal{H}$ consist of the pieces of $\mathcal{C}$ that lie in a word with a mortal endpoint or end at or before the first rejected cut of their word. If two pieces of $\mathcal{H}$ in one word, ending at $e < e'$, have comparable followers, then the rest after e' is a prefix of the rest after e, so the states at e and e' differ by Lemma 5.6. Hence a subset of $\mathcal{H}$ with pairwise comparable followers has at most $k(N-1)$ elements.

The other pieces of $\mathcal{C}$ lie in words u_l with immortal endpoints and end after J_l; we claim that there are at most $2\eta + E$ of them. The last position of such a piece P lies in the tail. Either it is one of the E uncovered positions, or it lies in a designated piece R. In the second case $P = R$, or P ends in the first d positions of R by (P1), if $R \in \mathcal{C}$, or P begins before R and ends in the first $2d$ positions of R by (P2), if $R \notin \mathcal{C}$. Since the ends of different pieces of $\mathcal{C}$ in one word are more than d apart, each designated piece accounts for at most two pieces.

By Lemma 7.15 there is an admissible deletion after which every piece of $\mathcal{C}$ has at most $\phi_d + m_d + ((2\eta + E + 1)2^{k(N-1)} - 1)(\kappa_d + m_d)$ letters, and this is at most the corresponding value for D. By Lemma 7.14 the result U' is a hole that dominates U, its tails consist of the images of the retained positions of the tails of U, and the designated pieces outside $\mathcal{C}$ are

pieces of U' with the same periods and lengths. We keep these as designated pieces; there are at most $\eta - 1$ of them. A position of a tail of U' that lies in none of them is the image of an uncovered position of U or lies in the image of one of the at most η designated pieces of $\mathcal{C}$. ◀

Proof of Theorems 5.7 and 7.16. By Proposition 7.12 a reader with a hole has a hole with a tail decomposition with the parameters (g, D, E_0), $g = k(\mathsf{Q}+3)$: a maximal periodic factor with a period of at most D and a length of at least $2D$ is a piece. Applying Lemma D.5 as long as there are designated pieces, at most g times, gives a hole in which all positions of the rejected tails of the coordinates with immortal endpoints are uncovered, and their number is at most E_g, because the sequence E_r is nondecreasing and its recurrence is that of Lemma D.5 with $\eta = g - r$. The closed form is proved in Appendix F.1, and $\Lambda(0, N) = 0$ by Section 7. ◀

E Proofs for Section 8: Regular and Finite Trees

E.1 Regular Trees

▶ **Lemma E.1** (Regular cones). *A tree whose two cones are regular languages has finitely many subtrees.*

Proof. Let π be a position of t and t' the subtree at π. The leaves of t' are the paths v with πv a leaf of t, with the same label and with $\mathrm{pol}(\pi v) = \mathrm{pol}(\pi) + \mathrm{pol}(v)$. So $C^s(t')$ is the residual $\{v \mid \pi v \in C^s(t)\}$ if $\mathrm{pol}(\pi) = 0$, and the residual of $C^{\overline{s}}(t)$ if $\mathrm{pol}(\pi) = 1$. A regular language has finitely many residuals, and a tree is determined by its cones (Theorem 4.2(1)). ◀

Proof of Theorem 8.3. It remains to complete the sketch in Section 8.2 with the regularity of the cones of the hull. For a signed variable x^s, the paths v with $(x^s, v) \in \mathrm{LC}$ are those with a prefix v' such that $\mathsf{D}(x^s, v')$ meets Fct (Lemma A.1), a regular language by Lemma 4.9. A seed $A_i = (p_i, w_i)$ satisfies (R1), so by Lemma 4.10 the paths v with $A_i \vdash (x^s, v)$ are the words $v'\, w_i[a{:}]\, \sigma$ with a a cut of w_i, $\mathsf{D}(x^s, v') \cap \mathsf{U}(p_i, w_i[{:}a]) \neq \varnothing$ and σ arbitrary; for each of the finitely many cuts a this is a regular language. By Lemma E.1 the assignment of the hull consists of regular trees. ◀

E.2 Finite Trees

▶ **Lemma E.2** (Observations as bounds). *For every path w there are finite trees $g_w^+, g_w^-, \hat{g}_w^+, \hat{g}_w^-$ such that for every tree t:*

$$w \in C^+(t) \iff g_w^+ \leq t, \qquad w \notin C^+(t) \iff t \leq \hat{g}_w^+,$$
$$w \in C^-(t) \iff t \leq g_w^-, \qquad w \notin C^-(t) \iff \hat{g}_w^- \leq t.$$

Proof. We define the trees by their cones (Theorem 4.2). Let E_w be the set of paths wv, and O_w the set of paths that are neither prefixes of w nor in E_w; both are extension closed and they are disjoint. Let g_w^+ have the cones (E_w, O_w), as cones of the signs $+$ and $-$, and g_w^- the cones (O_w, E_w). Their positions are the prefixes of w and the paths ui with u a proper prefix of w, so they are finite. If $w \in C^+(t)$, then $E_w \subseteq C^+(t)$, and $C^-(t)$ contains neither a prefix of w, since it is extension closed and disjoint from $C^+(t)$, nor an element of E_w; so $C^-(t) \subseteq O_w$ and $g_w^+ \leq t$ by Theorem 4.2(2). The converse is clear, and g_w^- is symmetric. Let $\hat{g}_w^+$ have the cones $(N_w, \varnothing)$, where N_w is the set of paths that are not prefixes of w, and $\hat{g}_w^-$ the cones $(\varnothing, N_w)$; their positions are the prefixes of w and their children. Then $t \leq \hat{g}_w^+$ if and only if no prefix of w lies in $C^+(t)$, that is, if and only if $w \notin C^+(t)$. ◀

The proof of Lemma 8.5 has three steps. We first prove it for a covariant constructor, from a characterization of finite satisfiability proved in [46]. We then introduce medians of trees, and treat variance through a translation into a covariant constraint on the signed variables, whose finite solutions are made symmetric by medians.

Finite satisfiability of covariant constraints. In this paragraph and the next, f is covariant. The constraints of [46] have no variable inequalities: their literals have the forms $f(\bar{a}) \leq u$, $u \leq f(\bar{b})$, $u = \bot$ and $u = \top$. For such a constraint χ, the *closure* D_χ is the least reflexive and transitive relation on its variables such that $D_\chi(a_i, b_i)$ for every $i \in [n]$ whenever $f(\bar{a}) \leq u$ and $v \leq f(\bar{b})$ are literals of χ with $D_\chi(u, v)$. For a path π, the *lower* and *upper path relations* L_π and U_π are defined by recursion on π: $L_\varepsilon(a, b)$ and $U_\varepsilon(a, b)$ hold if $D_\chi(a, b)$, and

$$L_{i\pi}(a, b) \iff \text{some literal } f(\bar{z}) \leq c \text{ of } \chi \text{ has } D_\chi(c, b) \text{ and } L_\pi(a, z_i),$$
$$U_{i\pi}(a, b) \iff \text{some literal } c \leq f(\bar{z}) \text{ of } \chi \text{ has } D_\chi(a, c) \text{ and } U_\pi(z_i, b).$$

In every solution, $D_\chi(a, b)$ forces the value of a below that of b, $L_\pi(a, b)$ forces the subtree of b at π, if it exists, above the value of a, and $U_\pi(a, b)$ forces the subtree of a at π, if it exists, below the value of b. A *cycle clash* of χ is a nonempty path π together with variables a and b such that $L_\pi(a, a)$, $D_\chi(a, b)$ and $U_\pi(b, b)$; it excludes finite solutions.

▶ **Theorem E.3** ([46, Theorems "Least shape and satisfiability" and "Covariant finite satisfiability"]). *Let f be covariant. A constraint without variable inequalities has a solution over $\mathcal{T}_{\text{fin}}$ if and only if it has a solution over $\mathcal{T}$ and no cycle clash.*

Covariant pinning. A bound $g \leq x$ or $x \leq g$ of a variable by a finite tree g is *compiled* into flat literals on fresh variables, by recursion on g. For $g \leq x$: nothing if $g = \bot$; the literal $x = \top$ if $g = \top$; and if $g = f(g_1, \ldots, g_n)$, the literal $f(z_1, \ldots, z_n) \leq x$ with fresh variables z_i, together with the compiled bounds $g_i \leq z_i$. For $x \leq g$: nothing if $g = \top$; the literal $x = \bot$ if $g = \bot$; and if $g = f(g_1, \ldots, g_n)$, the literal $x \leq f(z_1, \ldots, z_n)$ with fresh variables z_i, together with the compiled bounds $z_i \leq g_i$. By the rules (2), every solution of the compiled literals satisfies the bound, and every assignment that satisfies the bound extends to a solution of the compiled literals, with the subtrees of g as the values of the fresh variables.

▶ **Lemma E.4** (Covariant pinning). *Lemma 8.5 holds for a covariant constructor.*

Proof. For $n = 0$ every tree is finite. Let $n \geq 1$. We first remove the variable inequalities: replace every literal $u \leq v$ of Γ by the two literals $f(u, u, \ldots, u) \leq y$ and $y \leq f(v, u, \ldots, u)$, with a fresh variable y for each literal. A solution of the two literals satisfies $u \leq v$, by (2) at the first argument, and an assignment with $u \leq v$ extends to one by $y = f(u, \ldots, u)$, which is finite if the value of u is. So the new constraint satisfies the hypotheses of the lemma, and the restriction of a finite solution of it that satisfies the bounds is a finite solution of Γ that satisfies them. We may therefore assume that Γ has no variable inequalities.

Compile every bound with its own fresh variables, and let Δ be the compiled literals. The solution over $\mathcal{T}$ that satisfies the bounds extends to a solution of $\Gamma \cup \Delta$, and the restriction of a solution of $\Gamma \cup \Delta$ to the variables of Γ solves Γ and satisfies the bounds. By Theorem E.3 it suffices to show that $\Gamma \cup \Delta$ has no cycle clash. We write D, L_π and U_π for its closure and path relations, and D_Γ, L^Γ_π and U^Γ_π for those of Γ.

The variables of Γ are *old*. A fresh variable is a *source* if it comes from the compilation of a bound $g \leq x$, and a *sink* if it comes from a bound $x \leq g$. Its *rank* is its depth in the recursion, 1 for the variables z_i of the first literal of the compilation; old variables have rank

0. Every lower constructor literal $f(\bar{z}) \leq y$ of Δ has an old variable or a source y and sources z_i whose rank is that of y plus one; every upper constructor literal $y \leq f(\bar{z})$ of Δ has an old variable or a sink y and sinks z_i whose rank is that of y plus one. No source occurs in Γ or in an upper constructor literal, and no sink occurs in Γ or in a lower constructor literal. Constant literals enter neither the closure nor the path relations.

(a) *If $D(a, b)$ and b is a source, then $a = b$; if $D(a, b)$ and a is a sink, then $a = b$.* By induction on the derivation of $D(a, b)$: reflexivity is clear, a transitivity step follows from the induction hypothesis applied twice, and a decomposition step derives $D(a_i, b_i)$ from literals $f(\bar{a}) \leq u$ and $v \leq f(\bar{b})$, so b_i, a child of an upper constructor literal, is not a source, and a_i, a child of a lower one, is not a sink.

(b) *If $D(a, b)$ with a and b old, then $D_\Gamma(a, b)$.* By induction on the derivation. In a transitivity step from $D(a, c)$ and $D(c, b)$, the variable c is old: if it were a source, then $a = c$ by (a), and if it were a sink, then $c = b$. In a decomposition step from $f(\bar{a}) \leq u$, $D(u, v)$ and $v \leq f(\bar{b})$ with old a_i and b_i, the two literals belong to Γ, because the children of the literals of Δ are fresh; so u and v are old.

(c) *If $L_\pi(a, b)$ and b is a source, then a is a source whose rank is at least that of b plus $|\pi|$; if $U_\pi(a, b)$ and a is a sink, then b is a sink whose rank is at least that of a plus $|\pi|$.* By induction on π. For $\pi = \varepsilon$ this is (a). If $L_{i\pi}(a, b)$ by a literal $f(\bar{z}) \leq c$ with $D(c, b)$ and $L_\pi(a, z_i)$, then $c = b$ by (a), so the literal belongs to Δ, and z_i is a source whose rank is that of b plus one; apply the induction hypothesis to $L_\pi(a, z_i)$. The statement for U is symmetric.

(d) *If $L_\pi(a, b)$ with a and b old, then $L^\Gamma_\pi(a, b)$, and similarly for U.* By induction on π. For $\pi = \varepsilon$ this is (b). If $L_{i\pi}(a, b)$ by a literal $f(\bar{z}) \leq c$ with $D(c, b)$ and $L_\pi(a, z_i)$, the literal does not belong to Δ: otherwise z_i would be a source, and so would a, by (c). So it belongs to Γ, the variables c and z_i are old, and (b) and the induction hypothesis apply. If $U_{i\pi}(a, b)$ by a literal $c \leq f(\bar{z})$ with $D(a, c)$ and $U_\pi(z_i, b)$, a literal of Δ would make z_i a sink, and b a sink by (c).

Now let a nonempty path π and variables a and b form a cycle clash of $\Gamma \cup \Delta$. By (c), $L_\pi(a, a)$ excludes that a is a source, and $U_\pi(b, b)$ that b is a sink, since the rank would increase strictly. If a were a sink, $D(a, b)$ would give $b = a$ by (a), a sink; if b were a source, it would give $a = b$, a source. So a and b are old, and by (b) and (d) they form a cycle clash of Γ. This contradicts Theorem E.3, because Γ has a finite solution. ◀

Medians. From now on the variance is arbitrary again. We also use the covariant order on the same trees, written $\leq_0$: the order of Definition 2.2 for the variance with $c(i) = 0$ for every i. Its cones are the set $C^+_0(t)$ of the paths along which t reaches $\top$ and the set $C^-_0(t)$ of those along which it reaches $\bot$, and Theorem 4.2 applies to it. The *dual* t^* of a tree t exchanges $\bot$ and $\top$ at every leaf. It exchanges the two covariant cones, so $s \leq_0 t$ if and only if $t^* \leq_0 s^*$, and $f(t_1, \ldots, t_n)^* = f(t^*_1, \ldots, t^*_n)$. A tree is finite if and only if, for some d, every path of length d has a prefix that is a leaf: the paths without such a prefix are the positions labeled f, and they are closed under prefixes.

For trees t_1, t_2, t_3, the *median* $\mathrm{med}(t_1, t_2, t_3)$ is the tree whose covariant cone of each sign s consists of the paths that belong to $C^s_0(t_j)$ for at least two of the three indices j. These two sets of paths are extension closed, and they are disjoint, since a path in both would lie in both cones of one of the trees; so the median is well defined by Theorem 4.2(1).

▶ **Lemma E.5** (Medians). *For all trees:*

1. *the median is symmetric in its arguments, and* $\mathrm{med}(t,t,u) = t$*;*
2. *if* $t_j \leq_0 u_j$ *for* $j = 1,2,3$*, then* $\mathrm{med}(t_1,t_2,t_3) \leq_0 \mathrm{med}(u_1,u_2,u_3)$*;*
3. $\mathrm{med}(f(\bar{a}), f(\bar{b}), f(\bar{c})) = f\big(\mathrm{med}(a_1,b_1,c_1), \ldots, \mathrm{med}(a_n,b_n,c_n)\big)$*;*
4. $\mathrm{med}(t_1^*, t_2^*, t_3^*) = \mathrm{med}(t_1,t_2,t_3)^*$*;*
5. *if* t_1*,* t_2 *and* t_3 *are finite, so is their median.*

Consequently, the componentwise median of three solutions of a constraint in the covariant order is a solution, and if two assignments satisfy a bound $g \leq_0 x$ *or* $x \leq_0 g$*, then so does their median with any third assignment.*

Proof. (1) is clear. (2) and (4) follow from Theorem 4.2(2): inclusions between the cones of the arguments give inclusions between the majorities, and duality exchanges the two cones. For (3), the empty path lies in no cone of the three arguments, hence in no cone of the median, which is therefore a constructor tree, and by Theorem 4.2(3) its ith subtree has the majority cones of a_i, b_i and c_i. For (5), let d exceed the lengths of all positions of the three trees. Every path of length d lies in a cone of each t_j; two of these three cones have the same sign, so the path lies in a cone of the median. For the consequences, a literal $x \leq y$ is preserved by (2), the constructor literals by (3) and (2), and the literals $x = \bot$ and $x = \top$ by (1). If $g \leq_0 \rho_1(x)$ and $g \leq_0 \rho_2(x)$, then $g = \mathrm{med}(g,g,\rho_3(x)) \leq_0 \mathrm{med}(\rho_1(x),\rho_2(x),\rho_3(x))$ by (1) and (2), and symmetrically for upper bounds. ◀

Variance. For a tree t, let $\eta(t)$ be the tree with the same domain obtained by exchanging $\bot$ and $\top$ at the leaves of polarity 1. Then η is an involution, $\eta(\bot) = \bot$, $\eta(\top) = \top$, and

$$s \leq t \iff \eta(s) \leq_0 \eta(t),$$
$$\eta(f(t_1,\ldots,t_n)) = f(t_1',\ldots,t_n'), \quad \text{where } t_i' = \begin{cases} \eta(t_i) & \text{if } i \text{ is covariant,} \\ \eta(t_i)^* & \text{if } i \text{ is contravariant.} \end{cases} \tag{12}$$

The first statement holds because $\eta(s)$ and $\eta(t)$ have the common paths of s and t, and at the paths of polarity 1, where Definition 2.2 compares the labels in the reverse direction, exchanging $\bot$ and $\top$ reverses the order $\bot < f < \top$. The second holds because the path $i\pi$ has the polarity $c(i) + \mathrm{pol}(\pi)$. The covariant cones of $\eta(t)$ are the cones of t.

The *signed translation* $\Gamma^\pm$ of Γ is the constraint over the signed variables $V^\pm$, read in the order $\leq_0$, with the following two literals for each literal of Γ, where $x^{[i]}$ is the signed variable of Section 4.2:

Literal of Γ	Literals of $\Gamma^\pm$	
$x \leq y$	$x^+ \leq y^+$	$y^- \leq x^-$
$f(\bar{x}) \leq y$	$f(x_1^{[1]},\ldots,x_n^{[n]}) \leq y^+$	$y^- \leq f(\overline{x_1^{[1]}},\ldots,\overline{x_n^{[n]}})$
$y \leq f(\bar{x})$	$y^+ \leq f(x_1^{[1]},\ldots,x_n^{[n]})$	$f(\overline{x_1^{[1]}},\ldots,\overline{x_n^{[n]}}) \leq y^-$
$x = \bot$	$x^+ = \bot$	$x^- = \top$
$x = \top$	$x^+ = \top$	$x^- = \bot$

An assignment ρ of V gives the assignment $\rho^\pm$ of $V^\pm$ with $\rho^\pm(x^+) = \eta(\rho(x))$ and $\rho^\pm(x^-) = \eta(\rho(x))^*$; so $\rho^\pm(x^{[i]})$ is the tree t_i' of (12) for $t_i = \rho(x)$. For an assignment A of $V^\pm$, the assignment $\tilde{A}$ is defined by $\tilde{A}(x^s) = A(x^{\bar{s}})^*$, and A is *coherent* if $\tilde{A} = A$.

▶ **Lemma E.6** (Signed translation)**.** *Let* ρ *be an assignment of* V *and* A *one of* $V^\pm$*.*

1. ρ *solves* Γ *if and only if* $\rho^\pm$ *solves* $\Gamma^\pm$*;* $\rho^\pm$ *is coherent, and finite if* ρ *is.*

2. $\tilde{\tilde{A}} = A$, *and if* A *solves* $\Gamma^{\pm}$, *so does* $\tilde{A}$.
3. *If* A *is coherent, then* $A = \rho^{\pm}$ *for the assignment* $\rho(x) = \eta(A(x^+))$.
4. *Let the* pins *of a bound* $g \leq x$ *be* $\eta(g) \leq_0 x^+$ *and* $x^- \leq_0 \eta(g)^*$, *and those of a bound* $x \leq g$ *be* $x^+ \leq_0 \eta(g)$ *and* $\eta(g)^* \leq_0 x^-$. *Then* ρ *satisfies a bound if and only if* $\rho^{\pm}$ *satisfies its pins, and* A *satisfies the pins of a bound if and only if* $\tilde{A}$ *does.*

Proof. (1) By (12),

$$\rho(y) \leq f(\rho(x_1), \dots, \rho(x_n)) \iff \rho^{\pm}(y^+) \leq_0 f(\rho^{\pm}(x_1^{[1]}), \dots, \rho^{\pm}(x_n^{[n]})),$$

which is the first translated literal. Since duality reverses $\leq_0$ and commutes with f, and $\rho^{\pm}(r)^* = \rho^{\pm}(\bar{r})$, this is equivalent to the second translated literal. The other literals are similar, and coherence and finiteness are clear. (2) The first statement holds because duality is an involution. The map $A \mapsto \tilde{A}$ exchanges the two literals of each row of the table: for instance, $\tilde{A}$ satisfies $y^+ \leq_0 f(x_1^{[1]}, \dots, x_n^{[n]})$ if and only if $A(y^-)^* \leq_0 f(A(\overline{x_1^{[1]}})^*, \dots, A(\overline{x_n^{[n]}})^*)$, that is, if and only if A satisfies $f(x_1^{[1]}, \dots, \overline{x_n^{[n]}}) \leq_0 y^-$. (3) $\rho^{\pm}(x^+) = \eta(\eta(A(x^+))) = A(x^+)$, and $\rho^{\pm}(x^-) = A(x^+)^* = \tilde{A}(x^-) = A(x^-)$. (4) By (12), $g \leq \rho(x)$ if and only if $\eta(g) \leq_0 \rho^{\pm}(x^+)$, and by duality if and only if $\rho^{\pm}(x^-) \leq_0 \eta(g)^*$; similarly for $x \leq g$. Finally, $\tilde{A}$ satisfies $\eta(g) \leq_0 x^+$ if and only if A satisfies $x^- \leq_0 \eta(g)^*$, and conversely, and similarly for the pins of $x \leq g$. ◀

Proof of Lemma 8.5. Let δ be a finite solution of Γ and ρ a solution over $\mathcal{T}$ that satisfies the bounds. By Lemma E.6, $\delta^{\pm}$ is a finite coherent solution of $\Gamma^{\pm}$, and $\rho^{\pm}$ is a solution of $\Gamma^{\pm}$ that satisfies all pins, which are bounds by the finite trees $\eta(g)$ and $\eta(g)^*$. By Lemma E.4, applied to $\Gamma^{\pm}$ in the covariant order, some finite solution A of $\Gamma^{\pm}$ satisfies all pins, and by Lemma E.6 so does the finite solution $\tilde{A}$. Let $B(r) = \mathrm{med}(A(r), \tilde{A}(r), \delta^{\pm}(r))$ for every signed variable r. By Lemma E.5, B is a finite solution of $\Gamma^{\pm}$ that satisfies all pins, and it is coherent:

$$\tilde{B}(x^s) = \mathrm{med}\big(A(x^{\bar{s}}), \tilde{A}(x^{\bar{s}}), \delta^{\pm}(x^{\bar{s}})\big)^* = \mathrm{med}\big(\tilde{A}(x^s), A(x^s), \delta^{\pm}(x^s)\big) = B(x^s),$$

by Lemma E.5(4), $\tilde{\tilde{A}} = A$, the coherence of $\delta^{\pm}$ and the symmetry of the median. By Lemma E.6(3), $B = \rho'^{\pm}$ for an assignment ρ' of V, which is finite because η preserves domains, solves Γ by Lemma E.6(1), and satisfies the bounds by Lemma E.6(4). ◀

Proof of Theorem 8.4. If Γ has no finite solution, the entailment over $\mathcal{T}_{\mathrm{fin}}$ holds vacuously, and if the entailment holds over $\mathcal{T}$ it holds over $\mathcal{T}_{\mathrm{fin}}$. Conversely, let Γ have a finite solution and a countermodel ρ over $\mathcal{T}$. By Lemma 4.6 there are modes and paths such that S_ρ contains the seeds A_i and none of the exclusions B_i. By Lemma E.2 these $2k$ conditions are bounds of variables by finite trees, which ρ satisfies. By Lemma 8.5 some finite solution ρ' of Γ satisfies them, and by Lemma 4.6 it is a countermodel.

It remains to decide finite satisfiability. For $n = 0$ every tree is finite, and finite satisfiability is satisfiability. For $n \geq 1$, replace every literal $u \leq v$ by $f(\bar{a}) \leq y$ and $y \leq f(\bar{b})$, with a fresh variable y, where $a_j = b_j = u$ for $j \neq 1$, and $(a_1, b_1) = (u, v)$ if the first argument is covariant and $(a_1, b_1) = (v, u)$ if it is contravariant. By (2), the two literals imply $f(\bar{a}) \leq f(\bar{b})$ and hence $u \leq v$, and an assignment with $u \leq v$ extends to them by $y = f(\bar{a})$. So the new constraint, which has no variable inequalities, has a finite solution if and only if Γ has one. For constraints without variable inequalities, finite satisfiability is decided in [46, Lemma "Finite signed correspondence" and Theorem "Covariant finite satisfiability"], for every variance, by finite tests on the closure of a signed translation. ◀

F Proofs for Section 9: Complexity

F.1 The Closed Forms of the Bounds

We prove Theorem 9.1, which contains the closed forms of Theorems 6.9 and 7.16. All numbers are natural numbers, and $\exp_h$ is as in Section 9. We use the following inequalities, for all x, a, b, c and $h \geq 1$:

(X1) $x + 1 \leq 2^x$, and $N! \leq N^N \leq 2^{N^2}$;
(X2) $2^a + 2^b \leq 2^{a+b+1}$;
(X3) $\exp_h(a) + \exp_h(b) \leq \exp_h(a+b+1)$ and $\exp_h(a) \cdot \exp_h(b) \leq \exp_h(a+b+1)$;
(X4) $\exp_2(a)^c \leq \exp_2(a+c)$.

(X2) holds because both summands are at most 2^{a+b}. For (X3), the sum is at most $2\exp_h(a+b) \leq \exp_h(a+b+1)$, because $\exp_{h-1}(a+b+1) \geq \exp_{h-1}(a+b)+1$; the product is $2^{\exp_{h-1}(a)+\exp_{h-1}(b)}$, which is 2^{a+b} for $h = 1$ and at most $2^{\exp_{h-1}(a+b+1)}$ for $h \geq 2$, by the inequality for sums at the height $h-1$. For (X4), $\exp_2(a)^c = 2^{c2^a} \leq 2^{2^c 2^a}$.

Fresh tuples. Since $\mathsf{J}(0,N) = 0$, let $k \geq 1$; then $\mathsf{m} \leq 3kN + 5$ and $F = 2^{k(N-1)} \leq 2^{kN}$. Put $a = k(kN+1) + N^2 + 7kN + k + 12$ and $a' = 4a + 4$. First, $E \leq 2^a$ and $F \leq 2^a$: by (X1), $(k+1)^{kN+1}N^N \leq 2^{k(kN+1)}2^{N^2}$, and $2\mathsf{m} + 1 \leq 6kN + 11 \leq 2^{6kN+11}$, so $E \leq 2^{k(kN+1)+N^2+6kN+12}$ by (X2). Second, $\nu_d \leq 2^{a'3^d}$ for every d. Indeed $\nu_0 = k+1 \leq 2^k$, and if $\nu_d \leq 2^x$, then $\nu_d F \leq 2^{x+a}$, $\nu_d F + 2 \leq 2^{x+a+2}$ and

$$\nu_{d+1} = \nu_d + \nu_d F(\nu_d F + 2)E \leq 2^x + 2^{2x+3a+2} \leq 2^{2x+3a+3},$$

which is at most $2^{a'3^{d+1}}$ for $x = a'3^d$, because $3a + 3 \leq a'$. Third, for $d = kN$, $X = (\nu_d F + 2)E \leq 2^{a'3^d+a+2} \cdot 2^a \leq 2^{2a'3^d}$. Finally, with $3^d \leq 2^{2d}$, (X1) and (X2),

$$\mathsf{J}(k,N) = kX^{d+1} \leq 2^{k+2a'3^d(d+1)} \leq 2^{2^k+2^{a'+3d+1}} \leq \exp_2(k+a'+3d+2),$$

and $k + a' + 3d + 2 = 4k^2N + 4N^2 + 31kN + 9k + 54 \leq (k+N+16)^4$.

Search lengths. For the tails let $k \geq 1$ and $N \geq 1$, since $\Lambda(0,N) = 0$. Let Q, $s = \mathsf{Q} + N$, Φ, L_r and B_r be as in Appendices C.2 and C.3, for $m = k$ target words, and put $C_r = 2^{(k+N+10)(s+3)^r}$ for $r \leq k$. We show: *if* $b + 2 \leq 2^t$, *then* $\mathsf{L}_r(b) + b + 2 \leq 2^{C_r t}$. Then $t \geq 1$. For $r = 0$, $b + 1 \leq 2^t$ and $kb(b+1)/2 + 1 \leq (k+1)2^{2t}$, so $\Phi(b) \leq 2^t \cdot 2^N \cdot 2^k 2^{2t}$ and

$$\mathsf{L}_0(b) + b + 2 = \Phi(b) + 1 + b + 2 \leq 3 \cdot 2^{3t+k+N} \leq 2^{(k+N+5)t} \leq 2^{C_0 t}.$$

For the step from r to $r+1$, let $C = C_r$ and $G(b) = b + \mathsf{L}_r(b)$. By induction on j, $G^j(b) + 2 \leq 2^{C^j t}$: apply the statement for r to $G^j(b)$ and $C^j t$. So $x = G^s(b)$, the window b^* of the recursion, satisfies $x + 2 \leq 2^{C^s t}$; then $\mathsf{L}_r(x) + x + 2 \leq 2^{C^{s+1}t}$, and $y = x + \mathsf{L}_r(x)$ satisfies $\mathsf{L}_r(y) \leq 2^{C^{s+2}t}$. Since $b \leq x$, we have $2\mathsf{L}_r(x) + b + 4 \leq 2(\mathsf{L}_r(x) + x + 2) \leq 2^{C^{s+2}t}$, and

$$\mathsf{L}_{r+1}(b) + b + 2 = 2\mathsf{L}_r(x) + \mathsf{L}_r(y) + b + 4 \leq 2^{C^{s+2}t+1} \leq 2^{C^{s+3}t} = 2^{C_{r+1}t}.$$

Scales. For $l \geq -1$, the *values* of scale l are the five numbers e_l, X_l, X_{l-1}, Y_l and T_l; those of scale -1 are $1, 0, 0, 0, 0$, and by Appendix C.3 the values of scale $l-1$ determine those of scale l. Let $V = (k+N+10)(s+3)^k + 5$, so that $2^V = 32\,C_k$. We show: *if*

$H \ge k + N + \mathsf{Q} + N^2 + 20$ and the values of scale $l-1$ are at most 2^H, then the values of scale l are at most $2^{2^V H}$. First,

$$b_l + 2 = (N+1)e_{l-1} + (\mathsf{Q}+2)(X_{l-2}+1) + 2 \le 2^H 2^H + 2^H(2^H+1) + 2 \le 2^{2H+2} \le 2^{4H},$$

so $T'_l = \mathsf{L}_k(b_l) \le Z$ for $Z = 2^{4C_k H}$, by the bound on search lengths. The values of scale $l-1$, N, Q and $N! \le 2^{N^2}$ are at most $2^H \le Z$, and $e_l < T'_l \le Z$ by (L2). By the definitions,

$$T_l \le (Z+2)(3Z+3), \qquad Y_l \le 1 + (2Z+2)Z^2 + Z^2 + 3(Z+3)Z + 2Z,$$
$$X_l \le (Z+2)(2Z+1) + T_l + Y_l + Z \le 2Z^3 + 11Z^2 + 26Z + 9 \le 64Z^3.$$

Since $T_l, Y_l \le X_l$ and $e_l, X_{l-1} \le Z$, all values of scale l are at most $64Z^3 = 2^{12C_k H + 6} \le 2^{32C_k H} = 2^{2^V H}$.

Let $A = k + N + \mathsf{Q} + N^2 + kN + k(\mathsf{Q}+3) + 20$. The values of scale -1 are at most 2^A, so by induction on l, applying the last statement with $H = 2^{Vl}A \ge A$, the values of scale l are at most $2^{2^{V(l+1)}A}$. Let $\ell = 3(2k+1)$ and $H = 2^{A+V(\ell+1)}$. Since $2^{V(\ell+1)}A \le H$, the values of scale ℓ, the period bound $D = e_\ell$, and $T'_\ell \le T_\ell$ are at most 2^H.

Rejected tails. Let $Z = 2^H$. Since k and $\mathsf{Q}+3$ are at most $A \le H \le Z$, the number E_0 of Appendix C.6 satisfies

$$E_0 = k\big(T_\ell + (\mathsf{Q}+3)(3T'_\ell + 3) + 2(\mathsf{Q}+3)D\big) \le Z\big(Z + Z(3Z+3) + 2Z^2\big) \le 9Z^3 \le 2^{8H}.$$

For the passes of Theorem 7.16 let $F = 2^{k(N-1)}$, $\sigma = F(\kappa_D + m_D)$ and $c_n = \phi_D + m_D + ((2n+1)F - 1)(\kappa_D + m_D)$, so that the recurrence reads $E_{r+1} = E_r + (g-r)(c_{g-r} + \sigma E_r)$, and $c_n \le c_g$ for $n \le g$. Since $D \ge e_{-1} = 1$ and $g \ge 1$, we have $g\sigma \ge 1$, and induction on $r \le g$ gives

$$E_{r+1} + gc_g \le E_r + gc_g + g\sigma E_r + gc_g \le (1+g\sigma)(E_r + gc_g),$$

so $\Lambda(k,N) = E_g \le (1+g\sigma)^g(E_0 + gc_g)$. Let $G = 8H$. The numbers N, $N!$, F, D, g and E_0 are at most 2^G, because N^2, kN and g are at most A. With $\alpha_D = ND$ and $m_D = D \cdot N!$ at most 2^{2G}, we get $\kappa_D + m_D = 2\alpha_D + 4m_D \le 2^{2G+3} \le 2^{5G}$ and $\phi_D + m_D = 6\alpha_D + 9m_D \le 2^{2G+4} \le 2^{6G}$, hence $\sigma \le 2^{6G}$, $2g+1 \le 2^{3G}$, $c_g \le \phi_D + m_D + (2g+1)\sigma \le 2^{10G}$, $1 + g\sigma \le 2^{8G}$ and $E_0 + gc_g \le 2^{12G}$. Therefore

$$\Lambda(k,N) \le 2^{8Gg+12G} \le 2^{16(g+1)G} = 2^{128(g+1)H} \le 2^{2^{g+7}H} = \exp_2\big(A + V(\ell+1) + g + 7\big).$$

Finally let $R = k + N + 20$. Then $\mathsf{Q} \le 4 + 2kN$, so $s + 3 \le R^2$ and $\mathsf{Q} \le R^2$; hence $A + g + 7 = k + N + \mathsf{Q} + N^2 + kN + 2k(\mathsf{Q}+3) + 27 \le R^4$, $V \le (k+N+15)(s+3)^k \le R^{2k+1}$ and $\ell + 1 = 6k + 4 \le R^2$. So $A + V(\ell+1) + g + 7 \le R^4 + R^{2k+3} \le R^{2k+5}$, and $\Lambda(k,N) \le \exp_2((k+N+20)^{2k+5})$.

Small holes. $\mathsf{H}(0,N) = \mathsf{J}(0,1) = 0$. Let $k \ge 1$, $p = (k+N+20)^{2k+5}$ and $N' = k(N + \Lambda(k,N) + 1) + 1$, so that $\mathsf{H}(k,N) = \mathsf{J}(k,N')$. Since $\Lambda(k,N) \le \exp_2(p)$ and $x \le \exp_2(x)$, (X3) gives $N + \Lambda(k,N) + 1 \le \exp_2(N) + \exp_2(p) + \exp_2(0) \le \exp_2(N + p + 2)$, then $k(N + \Lambda(k,N) + 1) \le \exp_2(k + N + p + 3)$, and

$$k + N' + 16 \le \exp_2(k) + \exp_2(k + N + p + 3) + \exp_2(16) \le \exp_2(2k + N + p + 21).$$

By the bound on fresh tuples and (X4),

$$\mathsf{H}(k,N) \le \exp_2\big((k + N' + 16)^4\big) \le \exp_2\big(\exp_2(2k + N + p + 25)\big) = \exp_4(2k + N + p + 25),$$

and $2k + N + 25 \le 2R \le 2p$, so $2k + N + p + 25 \le 3p \le Rp = (k+N+20)^{2k+6}$. This proves Theorem 9.1.

F.2 Long Witnesses

This subsection gives the construction behind Example 9.4. Let f be binary and covariant, so the alphabet of the readers is $[n] = \{1, 2\}$.

From stopping readers to constraints. A *stopping reader* is a structural reader over $\{1, 2\}$; let S_0 be the set of its states other than † (it contains the starts). It has a set $S_1 \subseteq S_0$ of *final* states, such that $\delta(q, c) = \dagger$ for every $q \in S_1$ and every letter c, the rejected states are the final states, and $\mathsf{Adm} \subseteq S_0 \times S_0$. Conditions (S1) and (S2) hold vacuously. On a run that never reaches †, a final state can only occur at the last cut. Write Adm as a union of *rectangles* $A_1 \times B_1, \ldots, A_u \times B_u$, for instance of single pairs. The constraint Γ_R of a stopping reader R with k starts has the variables x_q and y_q for $q \in S_0$, a variable b and variables $z_1, \ldots, z_u$; we write $x_\dagger = y_\dagger = b$. Its literals are

$$
\begin{aligned}
&b = \bot, && \\
&x_q \le f(x_{\delta(q,1)}, x_{\delta(q,2)}) && \text{for } q \in S_0, \\
&f(y_{\delta(q,1)}, y_{\delta(q,2)}) \le y_q && \text{for } q \in S_0 \setminus S_1, \\
&x_p \le z_h \ \text{ and } \ z_h \le y_q && \text{for } p \in A_h,\ q \in B_h,\ 1 \le h \le u,
\end{aligned}
$$

and the queries are (x_{s_i}, y_{s_i}) for the starts s_i. The *witness tuples* of a countermodel ρ are the tuples W of paths such that, for some vector of modes, S_ρ contains the seeds and none of the exclusions of the queries along W (Section 4.3).

▶ **Lemma F.1** (Stopping readers). *The constraint Γ_R has a finite solution. For each of the classes $\mathcal{T}$, $\mathcal{T}_{\mathrm{reg}}$ and $\mathcal{T}_{\mathrm{fin}}$, the witness tuples of the countermodels of Γ_R and its queries in the class are exactly the holes of R.*

Proof. All x_q, z_h and b equal to $\bot$ and all y_q equal to $\top$ is a finite solution. Since f is covariant, Table 2 gives the following moves, where $q \in S_0$, $i \in \{1, 2\}$ and w is a path:

- $(x_q^+, iw) \to (x_{\delta(q,i)}^+, w)$ and $(x_{\delta(q,i)}^-, w) \to (x_q^-, iw)$;
- for $q \notin S_1$: $(y_{\delta(q,i)}^+, w) \to (y_q^+, iw)$ and $(y_q^-, iw) \to (y_{\delta(q,i)}^-, w)$;
- for $p \in A_h$ and $q \in B_h$: $(x_p^+, w) \to (z_h^+, w) \to (y_q^+, w)$ and $(y_q^-, w) \to (z_h^-, w) \to (x_p^-, w)$.

The only fact is (b^-, ε), and Neg consists of the x_q^+ for $q \in S_0$ and the y_q^- for $q \in S_0 \setminus S_1$. On the negative signed variables the moves have a simple structure: from a variable y_q^- they pop a letter, to a y^- or to b^-, or pass to a z_h^- without changing the path; from a z_h^- they pass to an x_p^-; and from an x_q^- or b^- they push a letter. So along a derivation by moves the negative variables are first of the form y^-, while letters are popped, and then of the forms z^-, x^- and b^-, while letters are pushed. In particular $\mathsf{Fct} = \{b^-\}$. For $q \in S_0$ let

$$
\begin{aligned}
P_q &= \{y_q^-\} \cup \{z_h^- \mid q \in B_h\} \cup \{x_p^- \mid (p, q) \in \mathsf{Adm}\}, \\
Q_q &= \{x_q^-\} \cup \{z_h^- \mid q \in A_h\} \cup \{y_r^- \mid (q, r) \in \mathsf{Adm}\}.
\end{aligned}
$$

Claim. Let $s \in S_0$, let v be a path and $q(0), \ldots, q(|v|)$ the run of R on v from s. If no $q(a)$ is †, then $\mathsf{U}(y_s^-, v) = P_{q(|v|)}$ and $\mathsf{D}(x_s^-, v) = Q_{q(|v|)}$. If $q(|v|) = \dagger$ and no earlier $q(a)$ is †, then $\mathsf{D}(x_s^-, v) = \{b^-\}$. Indeed, a derivation by moves from (y_s^-, v) to an atom with the empty path pops the letters of v one by one, through the atoms $(y_{q(a)}^-, v[a{:}])$, which is possible because $q(a) \notin S_1$ for $a < |v|$, and leaves the variables y^- only at the empty path, since afterwards only pushes follow; at the empty path it can pass from $y_{q(|v|)}^-$ to the z_h^- with

$q(|v|) \in B_h$ and then to the x_p^- with $p \in A_h$. A derivation from an atom with the empty path to (x_s^-, v) pops nothing, so it consists of moves at the empty path followed by pushes. Read backwards, the pushes follow the run: the atom before $(x_{q(a)}^-, v[a:])$ is $(x_{q(a+1)}^-, v[a+1:])$. If the run avoids †, they start at $(x_{q(|v|)}^-, \varepsilon)$, which can be reached at the empty path from the z_h^- with $q(|v|) \in A_h$ and from the y_r^- with $r \in B_h$. If $q(|v|) = \dagger$, they start at (b^-, ε), which is reached by moves only from atoms with a nonempty path.

Bottom modes. Let all modes be $\bot$, so that the seeds are $(y_{s_i}^-, w_i)$ and the exclusions $(x_{s_i}^-, w_i)$. By Lemma 4.9 and the claim, coordinate i of the reader $M_{\bar{\mu}}$ of Definition 4.13 is in the state $\varphi(q) = (P_q, Q_q)$ at every cut at which the run of R on w_i is in the state $q \neq \dagger$, since $b^- \notin Q_q$ and P_q has no positive variables; and it is dead at the first cut at which the run of R reaches †, because $b^- \in \mathsf{Fct}$. Moreover $\varphi(q)$ is rejected if and only if $P_q \cap \mathsf{Neg} = \varnothing$, that is, if and only if $q \in S_1$. And $(\varphi(h), \varphi(t))$ is admitted if and only if $P_t \cap (Q_h \cup \overline{P_h}) \neq \varnothing$; since $\overline{P_h}$ has only positive variables, this means $P_t \cap Q_h \neq \varnothing$, which holds if and only if $(h, t) \in \mathsf{Adm}$: through $y_t^- \in Q_h$, through $x_h^- \in P_t$, or through a z_g^- with $h \in A_g$ and $t \in B_g$. So the holes of $M_{\bar{\mu}}$ are exactly the holes of R.

Other modes. Let $\mu_i = \top$, with the seed $(x_{s_i}^+, w_i)$. From (x_q^+, iw) there is a move to $(x_{\delta(q,i)}^+, w)$. If the run of R on w_i never reaches †, these moves lead from the seed to (x_q^+, ε) for its endpoint $q \in S_0$, so the upper set at the endpoint meets Neg, and the endpoint of coordinate i is not rejected. If the run reaches † first at the cut a, they lead from $(x_{s_i}^+, w_i[:a])$ to (b^+, ε), so the upper set at the cut a contains b^+, its opposite is the fact b^-, and the coordinate is dead at the cut a or earlier. In both cases coordinate i is not eligible, and $M_{\bar{\mu}}$ has no hole.

By Theorem 4.14, the witness tuples of the countermodels over $\mathcal{T}$ are therefore the holes of R. A countermodel over $\mathcal{T}_{\mathrm{reg}}$ or $\mathcal{T}_{\mathrm{fin}}$ is one over $\mathcal{T}$. Conversely, for a hole of R there is a countermodel over $\mathcal{T}$ with this witness tuple; its $2k$ observations are bounds by finite trees (Lemma E.2), and since Γ_R has a finite solution, Lemma 8.5 gives a finite solution that satisfies them, a finite countermodel with the same witness tuple. ◀

The constraint Γ_R has $2|S_0| + 1 + u$ variables and at most $2|S_0| + 1 + \sum_h (|A_h| + |B_h|)$ literals.

The modular reader. Let $r \geq 1$ and $p_1, \ldots, p_r \geq 1$, with least common multiple L. The first component has *boundary* states $b_1, \ldots, b_r$, *cycle* states $c_{j,z}$ for $1 \leq j \leq r$ and $0 \leq z < p_j$, and a final state f_1, with the transitions

$$\delta(b_j, 1) = c_{j,1 \bmod p_j}, \quad \delta(c_{j,z}, 1) = c_{j,(z+1) \bmod p_j},$$

$$\delta(b_j, 2) = b_1, \qquad \delta(c_{j,z}, 2) = \begin{cases} b_{j+1} & \text{if } z = 0 \text{ and } j < r, \\ \mathsf{f}_1 & \text{if } z = 0 \text{ and } j = r, \\ b_1 & \text{if } z \neq 0. \end{cases}$$

So after $e \geq 1$ letters 1 from b_j the state is $c_{j,e \bmod p_j}$. The second component is a disjoint copy of this construction for the single modulus 1, with the boundary state b_1', the cycle state $c_{1,0}'$ and the final state f_2. The reader R_{mod} has the states of both components and †, the starts b_1 and b_1', the final states f_1 and f_2, whose transitions lead to †, and

$$\mathsf{Adm} = (\mathrm{Bnd} \times \mathrm{Cyc}) \cup (\mathrm{Cyc} \times \mathrm{Bnd}),$$

where Bnd and Cyc are the sets of all boundary and of all cycle states of both components. It is a stopping reader with $r + p_1 + \cdots + p_r + 5$ states, and Adm consists of two rectangles.

▶ **Lemma F.2** (The modular reader). *The holes of R_{mod} are exactly the tuples $W_t = ((1^{tL}2)^r, 1^{tL}2)$ with $t \geq 1$.*

Proof. Let (w_1, w_2) be a hole. The run of w_1 avoids † and ends in the final state f_1, which can only occur at the last cut and is entered only by the letter 2 from $c_{r,0}$. Write $w_1 = 1^{a_1}2\,1^{a_2}2 \cdots 1^{a_h}2$ with $h \geq 1$ and $a_j \geq 0$. Every letter 2 before the last one leads to a boundary state, so the cuts at the beginnings of the blocks $1^{a_j}2$ carry boundary states, and the cuts inside a block after at least one letter 1 carry cycle states. The last block has $\ell = a_h \geq 1$, because its letter 2 is read in a cycle state.

All blocks have the length of the last one. If $a_j < \ell$ for some $j < h$, the cut at the beginning of block j carries a boundary state and its rest begins with $1^{a_j}2$, and the cut of the last block after $\ell - a_j \geq 1$ letters 1 carries a cycle state and has the rest $1^{a_j}2$; this gives a pair of Bnd × Cyc on the channel from 1 to 1, also for $a_j = 0$. If $a_j > \ell$, the cut of block j after $a_j - \ell \geq 1$ letters 1 carries a cycle state and its rest begins with $1^{\ell}2$, and the beginning of the last block carries a boundary state and has the rest $1^{\ell}2$; this gives a pair of Cyc × Bnd. Both are excluded, so $a_j = \ell$ for all j.

A block $1^{\ell}2$ leads from b_j to b_{j+1}, or to f_1 for $j = r$, if p_j divides ℓ, and back to b_1 otherwise. If some p_j does not divide ℓ and j_0 is the least such index, the blocks lead from b_1 through $b_2, \ldots, b_{j_0}$ and back to b_1, so the run never reaches f_1. Hence every p_j divides ℓ, every block advances one stage, and since f_1 occurs only at the last cut, $h = r$: $w_1 = (1^{\ell}2)^r$ and L divides ℓ. In the same way $w_2 = 1^{\ell'}2$ with $\ell' \geq 1$.

If $\ell < \ell'$, the cut 0 of w_1 carries the boundary state b_1 and its rest begins with $1^{\ell}2$, and the cut of w_2 after $\ell' - \ell$ letters 1 carries the cycle state $c'_{1,0}$ and has the rest $1^{\ell}2$: a pair of Bnd × Cyc on the channel from 1 to 2. If $\ell > \ell'$, the cut of w_1 after $\ell - \ell'$ letters 1 carries a cycle state and its rest begins with $1^{\ell'}2$, and the cut 0 of w_2 carries b'_1 and has the rest $1^{\ell'}2$: a pair of Cyc × Bnd. So $\ell = \ell'$, and $\ell = tL$ with $t \geq 1$.

Conversely, in W_t both runs avoid † and end in final states. A cut at the beginning of a block carries a boundary state, and its rest begins with $1^{tL}2$; a cut inside a block after d letters 1, $1 \leq d \leq tL$, carries a cycle state, and its rest begins with $1^{tL-d}2$; the last cuts carry final states. An admitted pair needs a target in a boundary or a cycle state, whose rest begins with $1^{e'}2$, and a host in the other kind of state, whose rest begins with $1^{e}2$, with $e \neq e'$. Then the two rests differ at the position $\min(e, e')$, so the rest of the target is not a prefix of the rest of the host. So W_t is a hole. ◀

So the shortest hole of R_{mod} has total length $(r+1)(L+1)$. By Lemma F.1, the constraint $\Gamma_{R_{\mathrm{mod}}}$, with two variables z_1 and z_2 for the two rectangles, has $2(r + p_1 + \cdots + p_r + 4) + 3$ variables and $O(p_1 + \cdots + p_r)$ literals, has finite solutions, and the witness tuples of its countermodels over each of the three classes are exactly the tuples W_t. This proves the claims of Example 9.4; with the first r primes as moduli, the sum of the moduli is $\Theta(r^2 \log r)$ and L, their product, is $2^{\Theta(r \log r)}$, by the prime number theorem.

G The Lean Development

The development is a Lean 4 library over Mathlib, with the root namespace `DeciSigma1`; all names below are relative to it. Table 3 lists its parts, Tables 4 and 5 relate the definitions and results of the paper to declarations, and Appendix G.1 lists the differences between the two presentations.

Table 3 The parts of the development.

Part	Contents
`Main.lean`	the public statements, with a map of the proof
`Encodings.lean`	satisfiability and entailment as sentences
`Semantics`	trees over both signatures, the order with variance, finite and regular trees
`Constraints`	flat literals, closure, the signed translation, compiled bounds by finite trees
`Satisfiability`	satisfiability and finite satisfiability
`Automata`	runs of transition functions, the run automata of constraints
`Words`	periods, primitive words, rays, deletion of positions, suffix chains
`Arithmetic`	difference constraints, simple paths, maximal repetitions, iterated exponentials
`Readers`	tables, structural readers, holes, domination
`Holes`	the small-hole theorem: `Fresh`, `Periodic`, `Tail`, and the bounds
`Full`	the reduction to holes over $\{\bot, f, \top\}$, regular and finite countermodels, deciders
`Entailment`	the signature $\{\bot, f\}$ and its reduction by a guard
`Logic`	sentences, normal forms, the decision procedure for sentences
`Examples`, `Audit.lean`	examples; the check of axioms and of computability

Table 4 Definitions.

Paper	Lean
trees, the order (Definitions 2.1 and 2.2)	`Semantics.Tree`, `Semantics.Tree.Le`, `Semantics.polarity`
trees over $\{\bot, f\}$	`Semantics.PrefixSet`, `Semantics.IsRegularTree`, `Semantics.IsFiniteTree`
classes of trees	`Full.Logic.TreeClass`, `Full.Models.RegularTop`, `Full.Models.FiniteTop`
sentences (Section 2.3)	`Logic.QuantifierFreeFormula`, `Logic.Sigma1`, `Full.Logic.BooleanSentence`, `HoldsIn`
flat constraints, entailment (Definitions 2.4 and 2.5)	`Constraints.LitT`, `Constraints.SatT`, `Full.DisjEntailsIn`, `Full.HasCountermodelIn`
cones, atoms, rules (Definition 4.1 and Table 2)	`Full.Closure.leaf`, `Config`, `Move`, `Clause`, `Fact`, `Neg`, `IsSolution`, `leastClosure`
seeds, exclusions (Section 4.3)	`Full.seed`, `Full.exclusion`
runs (Definition 4.8)	`Automata.upperRun`, `Automata.lowerRun`, `upperStep`, `lowerStep`
readers, channels, holes (Definitions 4.11 and 4.12)	`Readers.StructuralReader`, `Readers.Table.channel`, `StructuralReader.Eligible`, `IsHole`, `firstRejection`
the reader of a mode (Definition 4.13)	`Full.Reduction.compile`
domination, fresh, immortal (Definitions 5.2, 5.4 and 5.5)	`Readers.Dominates`, `FreshDominates`, `IsMinimalFresh`, `StructuralReader.Immortal`
the cover (Definition 5.9)	`Holes.cover`, `Holes.coverProjection`
markers, anchors (Definition 6.1)	`Holes.Fresh.level`, `marker`, `active`, `contextAnchors`, `anchors`, `topLevel`
parses, trie, block deletion (Definitions 6.5 and 6.7)	`upperOccurrences`, `upperParse`, `trieNodes`, `Edge`, `Edge.TraversedBy`, `label`, `deletionMargin`, `BlockDeletion`
rays, ray system (Definitions 7.1 and 7.3)	`Words.Ray`, `Ray.onset`, `Ray.period`, `Holes.Tail.raySystem`

Paper	Lean
suffix nodes, closed factors (Definition 7.5)	`suffixDictionary`, `suffixNode`, `NodeClosedAt`, `NoSuffixOccurrenceEndsBetween`, `pumpFactor`
suffix chains	`Words.suffixReturnGap`, `Words.SuffixChain`, `ChainBoundFrom`
scales (Definition 7.10 and Appendix C.3)	`LevelBounds`, `levelStep`, `levelBounds`, `CoordinateCover`, `LevelCoverProperty`
forbidden overlaps (Definition C.7)	`PrefixOverlap`, `PeriodicTailOverlap`, `ReversePrefixOverlap`
pieces, admissible deletions (Definition 7.13)	`Holes.Periodic.Piece`, `FollowerEdge`, `AdmissibleDeletion`, `editedTuple`
deletion of positions	`Words.deletePositions`, `retainedCount`, `greatestPreimage`

Table 5 Results.

Paper	Lean
Theorem 1.1	`decideBooleanSentence`, `decideBooleanSentence_correct`, `booleanSentence_regular_iff`, `decideBooleanSentenceTopFree`, `booleanSentenceTopFree_regular_iff`
Lemma 3.1 and Proposition 3.3	`Logic.QuantifierFreeFormula.holds_iff_dnf`, `Logic.ConjunctiveClause.realizable_iff_not_disjEntailsIn`, `Logic.decideBooleanSentence`, `Logic.BooleanSentence.holdsIn_iff_of_disjEntailsIn_iff`
the table of Section 2.4	`Full.Logic.holdsIn_satisfiabilitySentence_iff`, `Full.Logic.holdsIn_entailmentSentence_iff`
Example 2.6, (3)	`Examples`: `unary_comparable`, `not_unary_comparable_single`, `binary_not_comparable`, `arrow_least`, `not_arrow_least_single`
Theorem 4.2 and Lemma 4.3	`Full.Closure.treeLe_iff_leaf`, `leaf_decode`, `sat_iff_isSolution`, `sat_iff_leastClosure`
Lemmas 4.4 and 4.6	`Full.Closure.moves_dual`, `Full.failures_iff_exists_leaf_witnesses`
Theorem 4.7	`Full.Reduction.SeedConsistency`, `hasCountermodelIn_iff_exists_seedConsistency`, `seedConsistency_iff_terminalRootConditions`
Lemmas 4.9 and 4.10, Lemma A.1	`Automata.lowest_point`, `Full.Reduction.reach_iff_event`, `Full.Closure.leastClosure_membership`
Theorem 4.14	`Full.Reduction.hole_iff_terminalRootConditions`, `not_disjEntails_iff_hole`
Lemmas 5.3 and 5.6	`StructuralReader.IsHole.of_dominates`, `runPrefix_ne_of_mortal`, `runPrefix_ne_of_le_firstRejection`
Lemma 5.10 and Theorem 5.1	`Holes.cover_fresh`, `cover_project_of_dom`, `Holes.exists_small_hole`, `smallHoleBound`, `Holes.decideHole`
Theorems 5.8 and 6.9	`Holes.Fresh.totalLength_le_freshTupleBound`, `exists_freshDominates_totalLength_le`, `freshTupleBound_le_double_exp2`
Lemmas 6.3 and 6.4, Lemma B.1	`marker_eq_of_infix`, `marker_occurrences_eq_of_nested`, `contextAnchors_marker_succ`, `topLevel_le`, `card_anchors_Ioc_marker_le`, `card_prefixOccurrenceCuts_le`
Lemmas B.2 and B.3	`relativeAnchors_eq`, `BlockDeletion.traversedBy_of_text_eq`
Theorem 6.8	`contextAnchors_apply`, `BlockDeletion.transfer`, `BlockDeletion.channel`, `BlockDeletion.freshDominates`
Lemma B.6	`label_local`, `exists_blockDeletion`
Theorems 5.7 and 7.16	`Holes.Tail.exists_hole_rejectedTailLength_le`, `tailLengthBound`, `tailLengthBound_le_double_exp2`
Lemma 7.4	`ray_prefix_hole`, `Arithmetic.DifferenceSystem.exists_long_solution`

Paper	Lean
Lemmas 7.7 and 7.9	`comparison_of_pumpFactor`, `pumpFactor_eligible`, `exists_closed_factor_free_of_suffix_occurrences`, `Words.chain_stop_from`
Lemmas C.3 and C.4	`levelCoverProperty_succ`, `exists_stable_level`, `levelCoverProperty_chainBound`, `levelCoverProperty_chainBoundFrom`, `Words.chainBound_mortal`
Lemmas C.5 and C.6	`exists_terminal_ray`, `exists_many_rayCandidates`, `exists_compatible_rayCandidate`
Lemma 7.11	`card_prefixOverlap_le`, `card_periodicTailOverlap_le`, `card_reversePrefixOverlap_le`, `not_mem_closed_walk_of_noForbiddenRayOverlap`
Proposition 7.12	`exists_rayFamily`, `exists_hole_with_run_cover`
Lemma 7.14	`Holes.Periodic.editedTuple_dominates`, `editedTuple_firstRejection`, `exists_preserved_piece_of_nonconjugate_root`
Lemma 7.15, Lemmas D.4 and D.5	`cap_exists`, `simplePath_le_pow_split`, `Arithmetic.simplePath_le_pow_prefix`, `exists_hole_compressed_step`
Theorem 9.1	`smallHoleBound_le`
Theorem 8.1 and Lemma 8.2	`decideDisjEntails`, `Full.Closure.decideSat`
Theorem 8.3	`Full.Reduction.disjEntails_regular_iff`, `SeedConsistency.regular_realizes`, `Full.Closure.regular_of_leaf`
Theorem 8.4 and Lemma 8.5	`Full.Reduction.disjEntails_finite_iff`, `Full.Models.finiteBoundPinning`, `observation_bound`, `Constraints.Ranked.Extension.satFin_iff`
Lemma 8.6 and Theorem 8.7	`Semantics.Tree.isTopFree_of_le_of_le_node`, `Entailment.disjEntailsIn_iff_guarded` and its variants for regular and finite trees

G.1 Differences between the Paper and the Development

The paper and the development follow the same route. They differ in the following points of representation and arrangement.

1. *Representation.* Argument positions are numbered from 0, the variance is a Boolean function with *true* for contravariant positions, and the signs $+, -$ are the Booleans *false* and *true*. Constraints are lists of literals over an arbitrary finite type of variables with decidable equality. The alphabet and the state set of a reader are arbitrary finite types, and the rejected states and admitted pairs are Boolean functions.
2. *Runs.* The development defines the upper and lower runs for covariant constraints and applies them to the *signed translation* of a constraint with variance, a covariant constraint on two copies of each variable; the paper defines them directly on atoms. The reader of a mode tests rejection against the set of signed variables from which a variable of Neg is reachable by moves that keep the path empty, which is equivalent on upper runs because they are closed under such moves.
3. *Realization.* The development states Theorem 4.7 with a condition for each seed (no forbidden atom, the opposite not in the least closure, no derivation of its own opposite) and conditions for pairs of different seeds, and proves the equivalence with the form used by the readers; the paper states the second form directly.
4. *Labels.* A block deletion requires equal labels in every traversing occurrence, as in Definition 6.7; the pigeonhole uses the pair of the marker texts and the map g_d of Lemma B.6 as a joint label.

5. *Closed factors.* Closed factors for escaping coordinates are found among cuts that are $B+1$ apart, which makes the factor longer than its window and yields the hypothesis of the count for the overlap (O2). An alternative is to add to the suffix node, for each terminal ray, the longest suffix of the text that is a factor of the periodic part of the ray, with a capped length; the development and the paper use the spaced cuts.
6. *Chains.* The chain budget of a fixed word is a bound on the number of members of a suffix chain that starts at a long window (Lemma C.4). An alternative bounds the number of maximal stretches of a chain with a constant gap and the length of each stretch; counting members is simpler and gives the same form of the recursion.
7. *State periods.* The development uses the period $d \cdot N!$ of the states inside a piece of period d, as the paper does; the least common multiple of $1, \dots, N$ in place of $N!$ would also do.
8. *Scale functions.* The development packages the inequalities (L1)–(L5), and a few monotonicity statements, in a structure, proves the tail bound from the structure, and then instantiates it with the recursion of Appendix C.3.
9. *Finite trees.* Lemma 8.5 is proved in the development as in Appendix E.2: bounds by finite trees are compiled into one-sided constructor literals on fresh variables, such an extension creates no cycle clash, and a finite solution of the signed translation is made symmetric under the exchange of the two copies by a median. The development proves the characterization of finite satisfiability by cycle clashes itself, for constraints with variable inequalities, whose closure contains these inequalities; the paper cites it from [46], where it is stated without variable inequalities, and removes them first.
10. *Degenerate cases.* The development covers arity 0, the case of no queries, empty words and readers with a single state without separate treatment, with truncated subtraction in the numerical definitions; the paper treats $k = 0$ separately (Section 7) and assumes $k \geq 1$ and $N \geq 2$ in Appendices C and D, which holds for every reader with at least one start and a hole.
11. *Complexity.* The development proves the numerical bound of Theorem 9.1. It does not contain Corollary 9.2, Proposition 9.3 or the construction of Appendix F.2.

H Notation

Table 6 lists the notation of the paper in the order of its introduction. A few letters are used with different meanings in different sections, mostly with different decorations: σ is a path or a word, σ_l^t a marker (Section 6); $\tau_r(l, c)$ is a marker text (Section 6) and τ the repeated factor of a ray (Section 7); D is a lower run, $\mathcal{D}_B$ a dictionary and D a bound on periods; E bounds the length of an edge (Section 6), and E_0, E_r count uncovered positions (Section 7); J_i is a first rejected cut and J the fresh-tuple bound; m is the margin of block deletions and m_d the period of the states in a piece. Without a change of decoration, $\mathcal{C}$ is a class of trees and, in Appendix D, the class of the pieces with a given root, and F is the number $2^{k(N-1)}$ of Section 6.5 and, in Appendix B.4, the deleted text.

Table 6 Notation.

Symbol	Meaning	Introduced in
Words, trees and sentences		
$\lvert w\rvert$, $w[{:}a]$, $w[a{:}]$, $w[a{:}b]$	length; prefix up to the cut a; rest after a; factor between two cuts	Section 2.1
$u \preceq v$	u is a prefix of v	Section 2.1

Symbol	Meaning	Introduced in
n, $[n]$, c, pol	arity, argument positions, variance, polarity of a path	Section 2.2
$\mathcal{T}$, $\mathcal{T}_{\mathrm{reg}}$, $\mathcal{T}_{\mathrm{fin}}$, $\mathcal{C}$	all, regular, finite trees; a class of trees	Section 2.2
f^{ω}	the tree with f at every path	Section 2.2
Γ, V, ρ	constraint, its variables, an assignment	Definition 2.4
(x_i, y_i), k	queries and their number	Definition 2.5
$\Gamma \models_{\mathcal{C}} \dots$	disjunctive entailment over $\mathcal{C}$	Definition 2.5
Γ_K	flat constraint of a clause K	Section 3.2
The reduction to holes		
$C^s(t)$, $\overline{s}$	cone of sign s of the tree t; opposite sign	Definition 4.1
$V^{\pm}$, x^s, (p, w)	signed variables; an atom	Section 4.2
$x^{[i]}$	x^+ for a covariant, x^- for a contravariant position i	Section 4.2
S_ρ	set of atoms of an assignment	Section 4.2
$a \to b$, $a \vdash b$, $a \Rightarrow b$	move or extension; derivation by rules; derivation by moves	Section 4.2
LC, Neg, Fct	least closure; signed variables with a forbidden atom; signed variables reached from facts	Section 4.2, (4)
μ_i, $\bar{\mu}$	mode of a query, vector of modes	Lemma 4.6
$A_i = (p_i, w_i)$, $B_i = (p_i', w_i)$	seed and exclusion of query i	(5)
$\mathsf{U}(p, w)$, $\mathsf{D}(p, w)$, $\delta^{\uparrow}$, $\delta^{\downarrow}$	upper and lower run, their transition functions	Definition 4.8 and Lemma 4.9
A, S, δ, s_i, †, Rej, Adm, N	alphabet, states, transitions, starts, dead state, rejected states, admitted pairs and size of a reader	Definition 4.11
$W = (w_1, \dots, w_k)$, $\|W\|$	tuple of words, its total length	Section 4.6
$q_i(a)$	state of coordinate i at the cut a	Section 4.6
$\mathsf{Ch}_{ij}(W)$	channel from the host i to the target j	Definition 4.12
J_i	first rejected cut of coordinate i	Section 4.6
$M_{\bar{\mu}}$, (P, Q)	reader of a mode vector; its states	Definition 4.13
Small holes		
$\mathsf{Vis}_i(W)$, $U \sqsubseteq W$	visited states; domination	Definition 5.2
H, Λ, J	bounds on holes, rejected tails, fresh tuples	Theorems 5.1, 5.7 and 5.8
$\widehat{M}$, (i, s), (i, m), o, pr	cover of a hole, its states, its projection	Definition 5.9
ℓ_i, L	length of a tracked tail and its bound	Section 5.3
$\exp_h$	iterated exponential	Section 1
Fresh tuples		
σ_l^t, Act_t	marker of level t of coordinate l; active coordinates	Definition 6.1
$\mathrm{lcs}(Z, Y)$, (Z, a, b)	common suffix of two parses, in lower blocks; an edge	Definition 6.5
$\tau_r(l, c)$, $\omega(l, c)$, $\mathrm{lab}(l, c)$	marker text, longest marker length and label of a cut	Section 6.4
m	margin of block deletions	Section 6.4
E, F, ν_d, X	bounds on edges, on parses per text, on texts, on lower blocks per upper block	Section 6.5
im, γ	image of a retained cut; greatest preimage of a cut	Appendix B.4

Symbol	Meaning	Introduced in
Rejected tails		
Ω, $\mathrm{per}(\Omega)$, $\mathrm{on}(\Omega)$, L_i	ray, its period and onset; length of a prefix	Definition 7.1
$t_1, \ldots, t_m$, B, $v_B(t)$	target words; a window; suffix of length B	Definition 7.5
$\mathcal{D}_B$, $\mathrm{node}_B(e)$	suffix dictionary; suffix node of a cut	Definition 7.5
$\mathrm{gap}_t(b)$	gap of the window b in a suffix chain of t	Section 7.3
L_m, B_m, Φ	search length, window bound, bound on closed factors	Lemma 7.9
e_l, T_l, Y_l, b_l	numbers of scale l: periods, exceptional positions, long suffixes, initial window	Section 7.4
T'_l, B_l, X_l	search stretch, window bound and chain bound of scale l	Appendix C.3
Q	$4 + 2(k-1)(N-1)$; $\mathsf{Q}+1$ periodic factors are allowed	Section 7.4
F_l, K_l	almost periodic tail; and no long periodic suffix	Definition 7.10
X_u, A_u, Π_u	stretch, onset and periodic part of the ray of u	Section 7.5, Appendix C.5
d, ζ, α_d, m_d	period and root of a piece; transient and period of its states	Section 7.6
β_d, κ_d, ϕ_d, x_P	offset of the deleted blocks, length threshold, retained length; number of deleted blocks	Section 7.6, Definition 7.13
D, E_0, E_r, g	bound on periods; uncovered positions; number of passes	Proposition 7.12 and Theorem 7.16
Decision procedures		
Γ^g	a constraint over $\{\bot, f\}$ with the guard variable u	Section 8.4